# Online Appendix for:

## Explaining the Historical Increase and Recent Decline in Social Security Disability Insurance Enrollment


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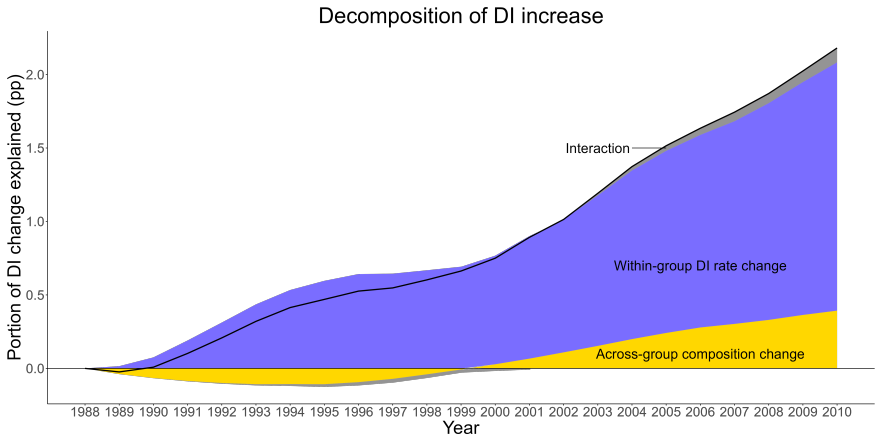
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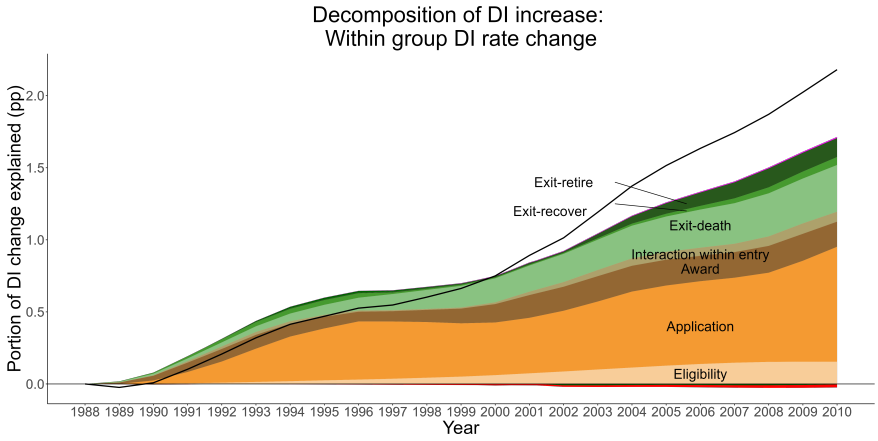
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# A Appendix Figures and Tables

Appendix Figure A.1: SSDI increase (1988–2010) attributable to higher application/award, lower exit



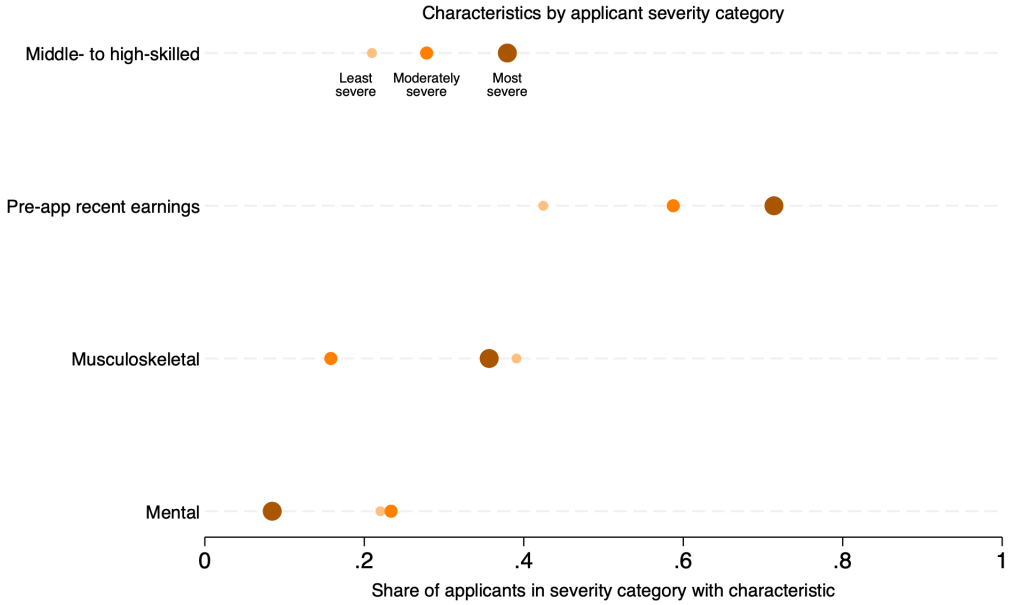

(a) Across-group versus within-group changes




(b) Within-group changes in behavior

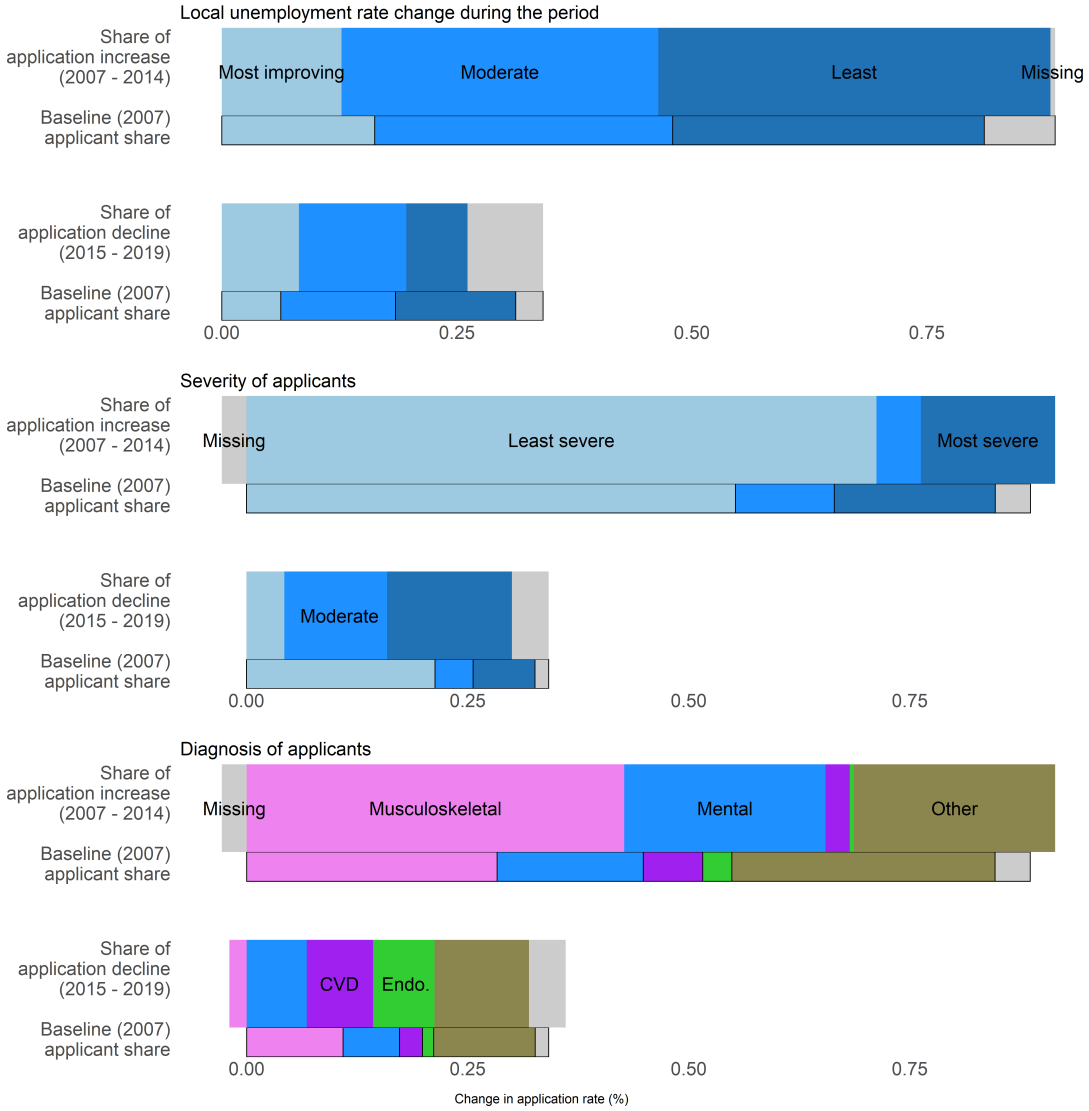
Notes: Top figure plots the decomposition of the increase in SSDI enrollment rates from 1988 to 2010 into across-group composition change, within-group change in SSDI enrollment rate, and the interaction between the two, as given by equation (5). Bottom figure plots the decomposition of the within-group change in SSDI enrollment rate into specific components of SSDI entry, SSDI exit, friction in the population, and population discrepancy. The specific components of SSDI entry are partial eligibility (i.e., the part that can only be attributed to eligibility and no other component), partial application, partial award, and the sum of interactions between the three entry components (eligibility, application, and award), as given by equation (9). The specific components of SSDI exit are partial death, partial retirement (defined as aging into the retirement program), partial recovery (defined as being found no longer eligible due to continuing disability review), and partial other (any remaining exit after removing death, retirement, and recovery), as given in equation (8). "Friction in population" and "Population discrepancy" are described in footnotes 13 and 15. The point estimates we report are the ones in which flows (e.g., $B_t$, $A_t$, $E_t$, $X_t$), when fixed, are held fixed at their counterfactual values. See Section 2.1 for details on the decomposition method. See Table 1 for exact estimates.

Appendix Figure A.2: Higher-severity applicants are higher-skilled and have higher labor force attachment



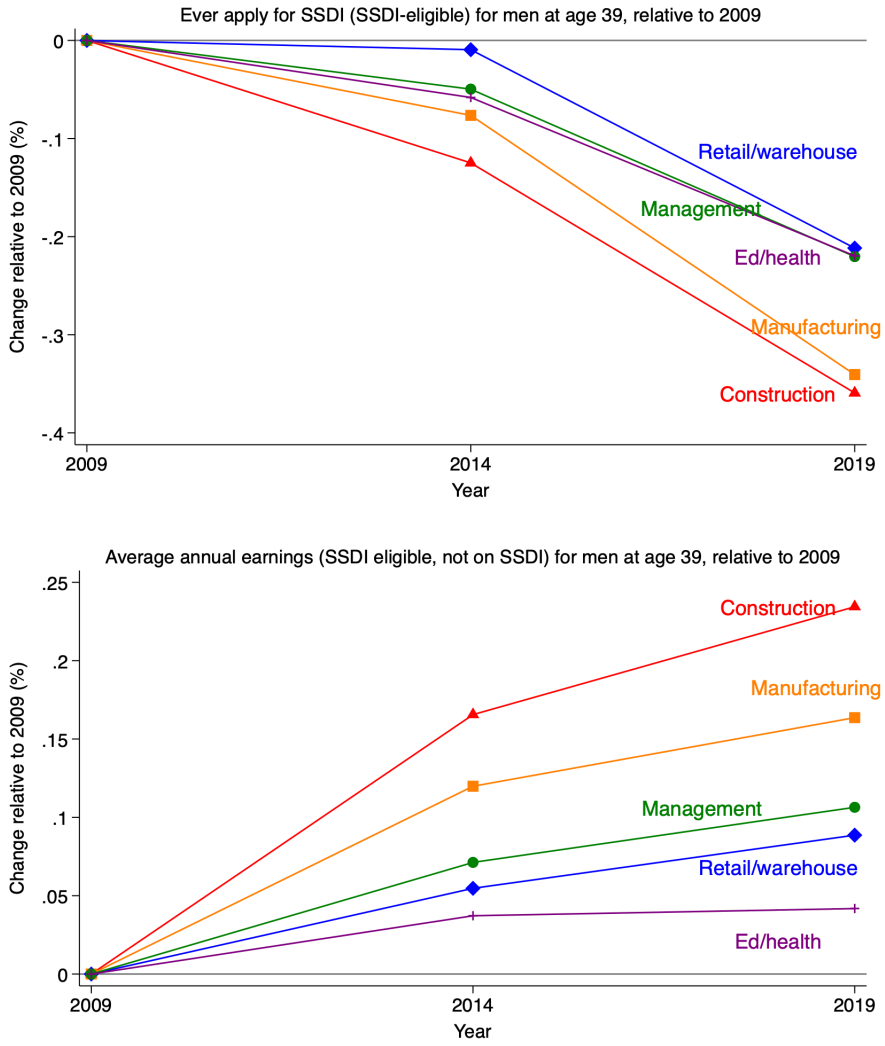

Notes: Figure shows share of SSDI applicants in 2019 in a given severity category who have a particular characteristic. "Middle- to high-skilled" means in top two earnings quintile in one's birth cohort between ages 25 to 35. "Pre-app recent earnings" means earning at least $15,000 in at least three of the five years preceding application. "Least severe" means that the applicant was rejected at the disability examiner level (but could have been allowed on appeal). "Moderately severe" means that the disability examiner assigned a diary reason of "medical improvement expected" or "medical improvement possible." "Most severe" means that the disability examiner assigned a diary reason of "medical improvement not expected." The sample is constructed from the F831 file.

Appendix Figure A.3: Marginal (non-)applicants during Great Recession (2007-2014) and post-recession period (2014-2019) are different



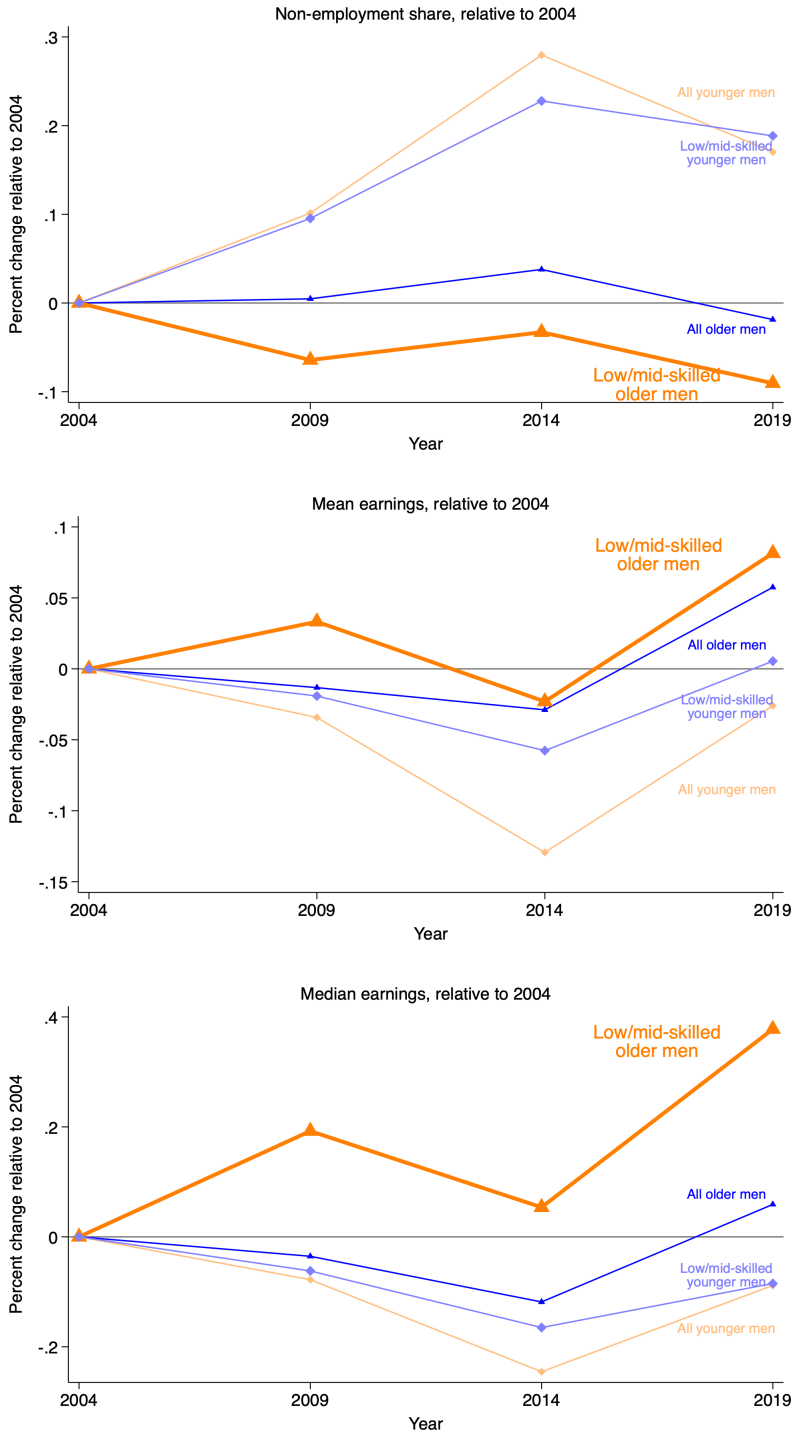

Notes: Figure characterizes the change in applicants by characteristics during 2007-2014 ("Great Recession," when there was an increase in applications) and during 2014-2019 ("post-recession," when there was a decrease in applications). Each bar consists of a main bar and a sub-bar. The main bar gives the fraction of the decline from that period represented by that characteristic, using equation (15). The sub-bar gives the hypothetical fractions that would have been proportional to applicant shares in 2007. We link ZIP code to Bureau of Labor Statistics Local Area Unemployment Statistics. "Most improving labor market" means that the change in the unemployment rate from 2007-14 (or 2014-19) is in the top tercile. "Moderately improving" means in the middle tercile. "Least improving" means the bottom tercile. "Least severe" means that the applicant was rejected at the disability examiner level (but could have been allowed on appeal). "Moderately severe" means that the disability examiner assigned a diary reason of "medical improvement expected" or "medical improvement possible." "Most severe" means that the disability examiner assigned a diary reason of "medical improvement not expected." Diagnosis is recorded by the disability examiner assigned to the case. See Appendix Section H.3 for exact estimates.

Appendix Figure A.4: Men working in construction and manufacturing have largest SSDI application declines and largest earnings gains



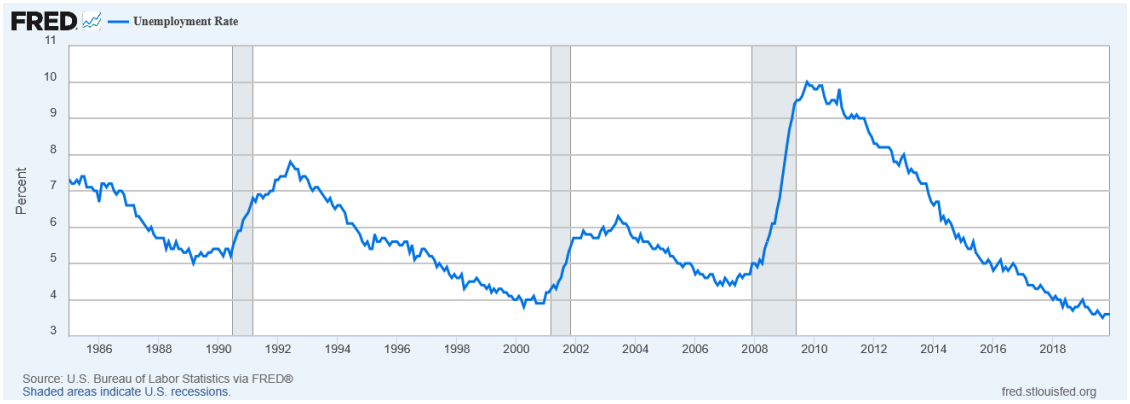

Notes: Top figure plots the likelihood of ever applying for SSDI, relative to 2009. Bottom figure plots average annual earnings relative to 2009. Sample is men working in that industry between ages 25 to 34 years with average annual earnings of $1-$30,000 during that time. The sample is derived from the Continuous Work History Sample linked to the Numident, F831 file, Payment History Update System, and Master Earnings File.

Appendix Figure A.5: Non-employment has risen for younger men, but declined for older low/mid-skilled men



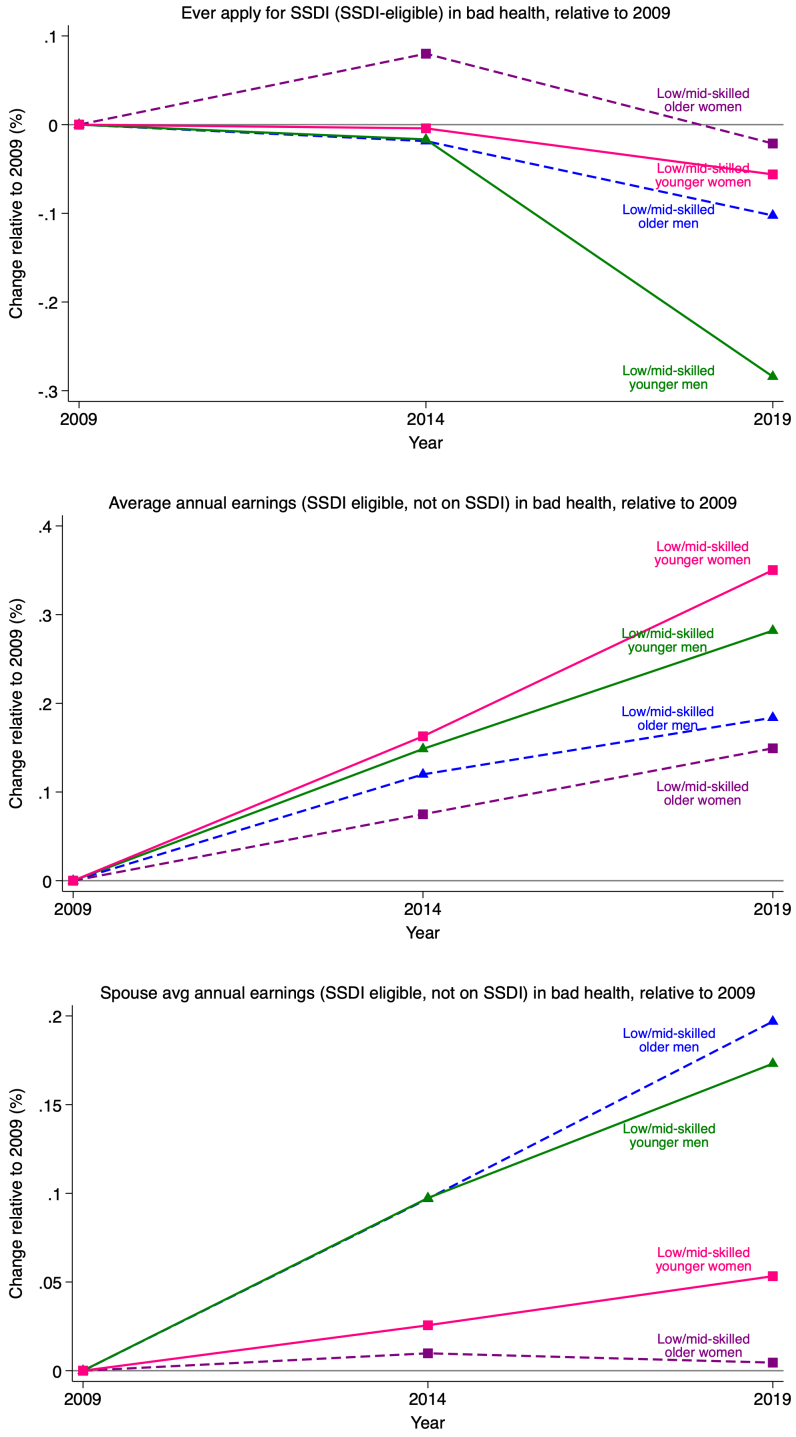

Notes: Figures plot non-employment likelihood, mean earnings, and median earnings for low-to-middle-skilled men and all men at ages 40 and 55, over time. Low-to-middle-skilled men are defined as having average earnings between $1 and $30,000 between ages 25 to 34 years old. The sample is derived from the Continuous Work History Sample linked to the Numident and Master Earnings File.

Appendix Figure A.6: Prolonged economic recovery following Great Recession



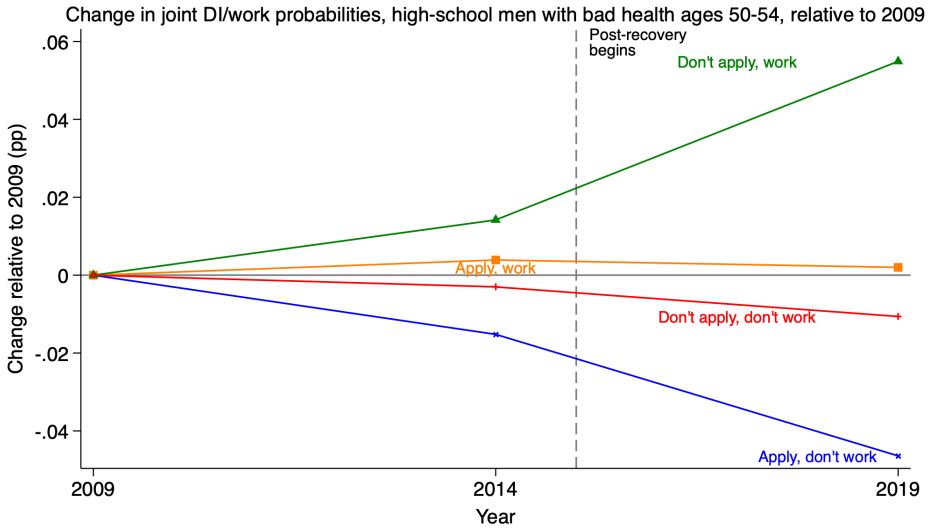

Notes: Figure plots US unemployment rate over time. Gray areas indicate recession periods as determined by the National Bureau of Economic Statistics. Source: U.S. Bureau of Labor Statistics, Unemployment Rate [UNRATE], retrieved from FRED, Federal Reserve Bank of St. Louis; https://fred.stlouisfed.org/series/UNRATE, July 18, 2025.

Appendix Figure A.7: Among low-to-middle-skilled people *with poor health*, falling SSDI application/receipt and rising earnings



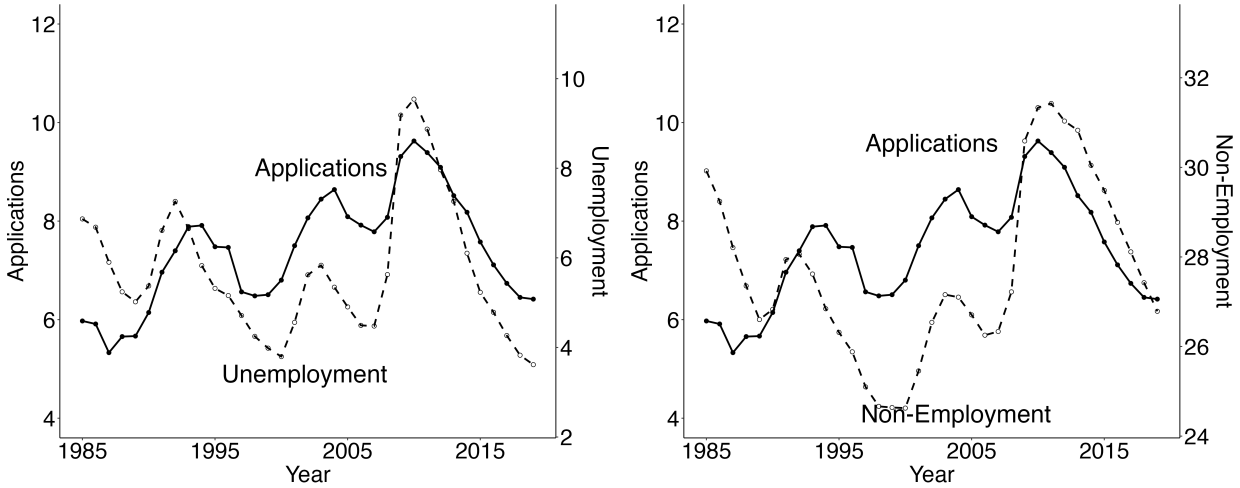

Notes: Figure plots the following outcomes over time, relative to 2009, for individuals in poor health: ever applied for SSDI, average annual earnings, and average annual spouse earnings conditional on having a spouse. The sample is derived from the 1996, 2001, 2004, 2008, and 2014 panels of the SIPP linked to the Numident, F831 file, Payment History Update System, and Master Earnings File. "Bad health" is defined as self-reported "fair" or "poor" health. Low/mid-skilled is defined as having a high school degree. See Appendix Section B.1 for more details on the SIPP analysis.

Appendix Figure A.8: Low/mid-skilled men with health conditions have become less likely to apply, more likely to work (and not apply)




Notes: Figure plots the joint probabilities of working and applying for SSDI benefits for men with a high school degree or less over time. Figure uses men from the SIPP who have a high school degree or less and report being in "fair" or "poor" health; the sample is derived from the 1996, 2001, 2004, 2008, and 2014 panels of the SIPP linked to the Numident, F831 file, Payment History Update System, and Master Earnings File.

Appendix Figure A.9: SSDI application, unemployment, and non-employment rates for all adults age 18-66

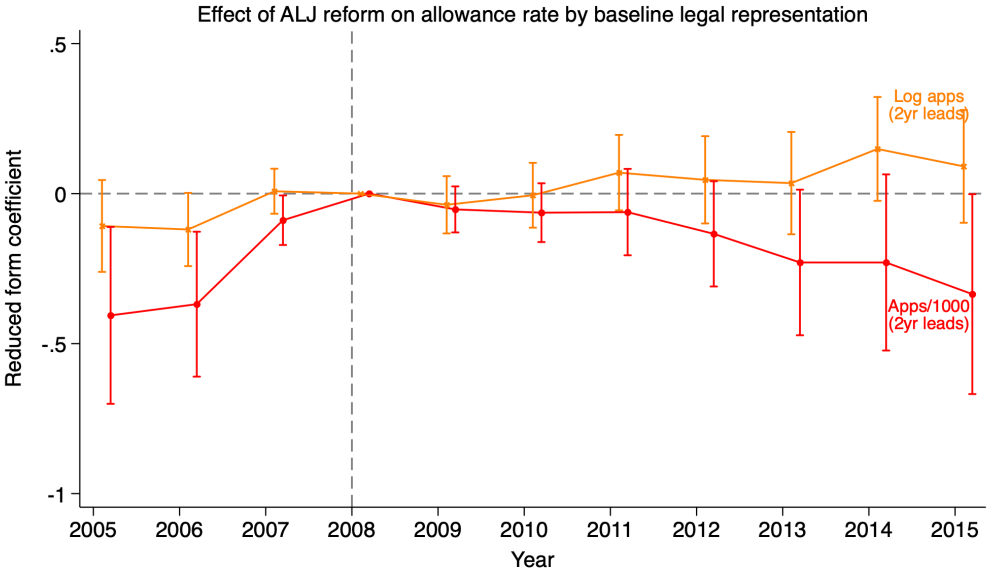



(a) Applications and unemployment, all adults ages 18-66

(b) Applications and non-employment, all adults ages 18-66

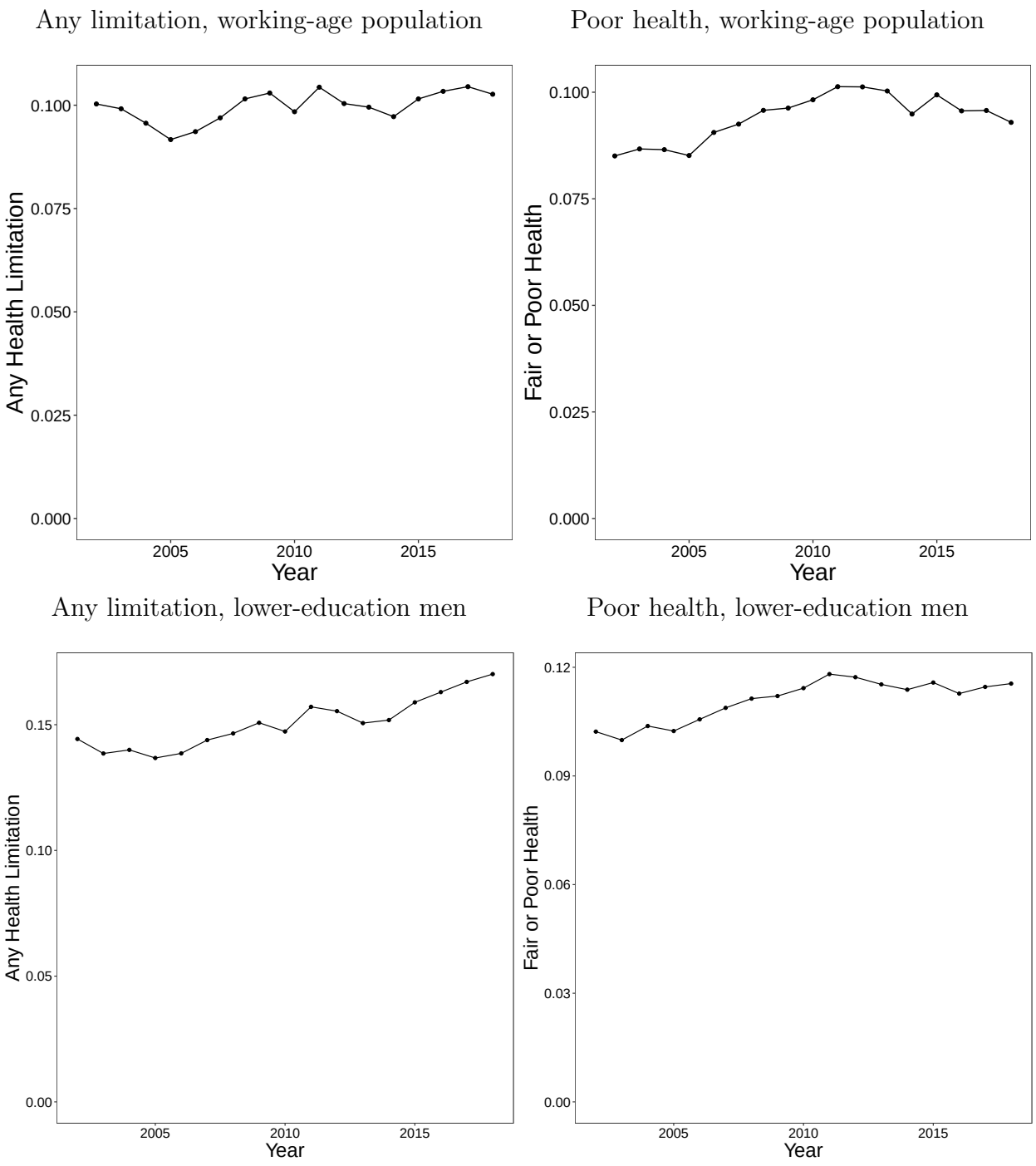
Notes: Figure plots SSDI applications, unemployment rates, and non-employment rates over time for all adults ages 18-66. SSDI applications are per thousand working-age adults. Employment measures (dashed lines) are plotted against the secondary axis in panel. Non-employment is reported as a percent of the working-age population, and unemployment as a percent of the labor force. Both are measured using monthly CPS data from IPUMS for individuals age 18 to 66. Overall time series correlations of applications with non-employment and unemployment are 0.45 and 0.58, respectively. The correlation between non-employment and unemployment is 0.81.

Appendix Figure A.10: No differential effect of ALJ reform on future applications by baseline legal representation



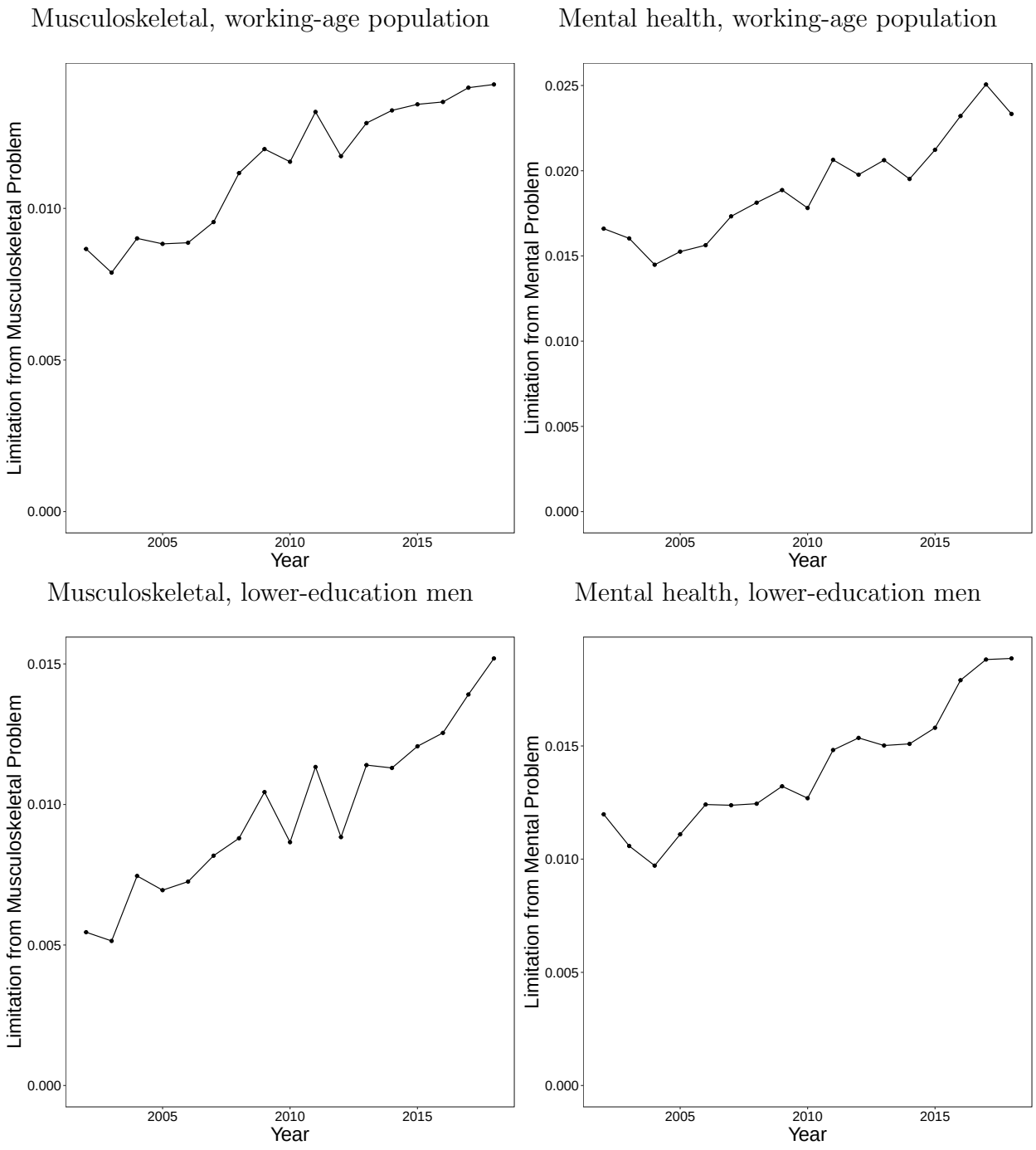

Notes: Figure plots estimates of the effect of a county having a higher baseline SSDI legal representation rate in 2007 (i.e., $\beta_t$ from equation (22), where baseline ALJ award rate is replaced by baseline SSDI legal representation rate) on the number of SSDI applications two years later (divided by 100) in the county (red) and log SSDI applications two years later in the county (orange). The sample is constructed from the F831 file linked to the MEDIB.

Appendix Figure A.11: Population health did not improve substantially from 2010–2018



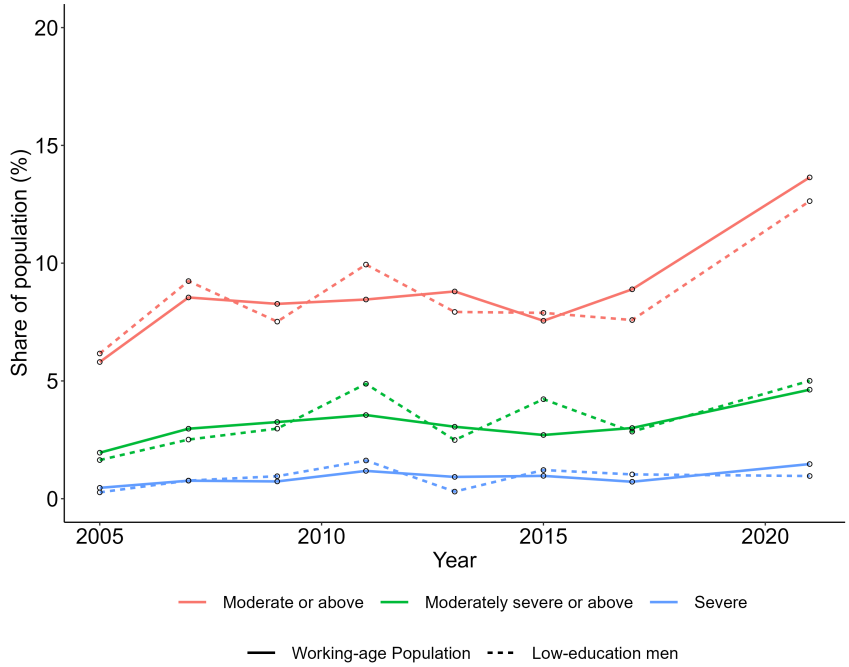

Notes: Figures plot self-reported health in the National Health Interview Survey among the overall working-age population (top) and lower-education men (without a college degree; bottom). "Any limitation" means individuals who report being "limited in any way in any activities because of physical, mental, or emotional problems." "Poor health" means those reporting "fair" or "poor" overall health (as opposed to "good," "very good," or "excellent.")

Appendix Figure A.12: Mental and musculoskeletal health did not improve substantially from 2010–2018



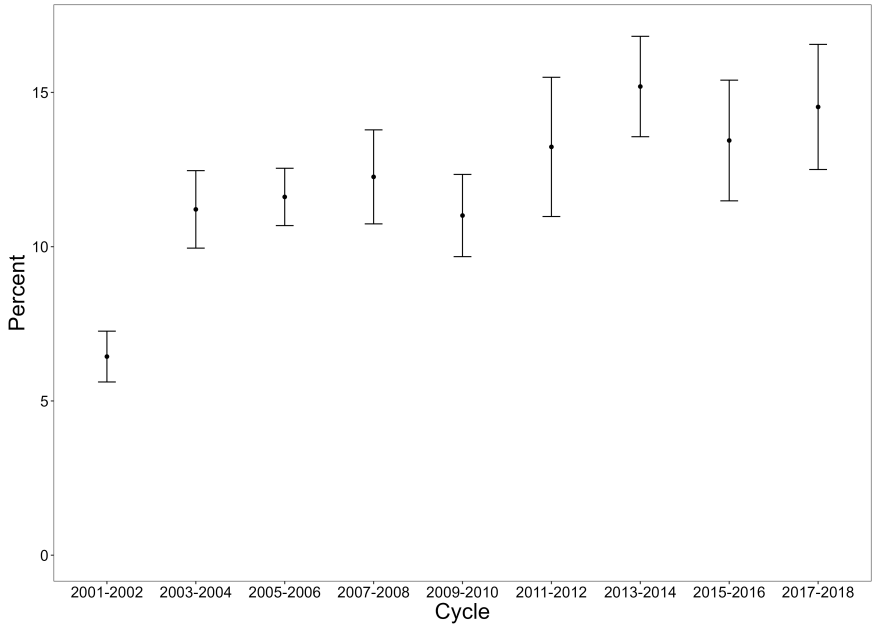

Notes: Figures plot self-reported health in the National Health Interview Survey among the overall working-age population (top) and lower-education men (without a college degree; bottom). Each graph reports the share of respondents who indicate having a work or functional limitation caused by musculoskeletal issues (left panels) or depression, anxiety, and emotional problems (right panels).

Appendix Figure A.13: Severe depression rates did not fall from 2010–2019



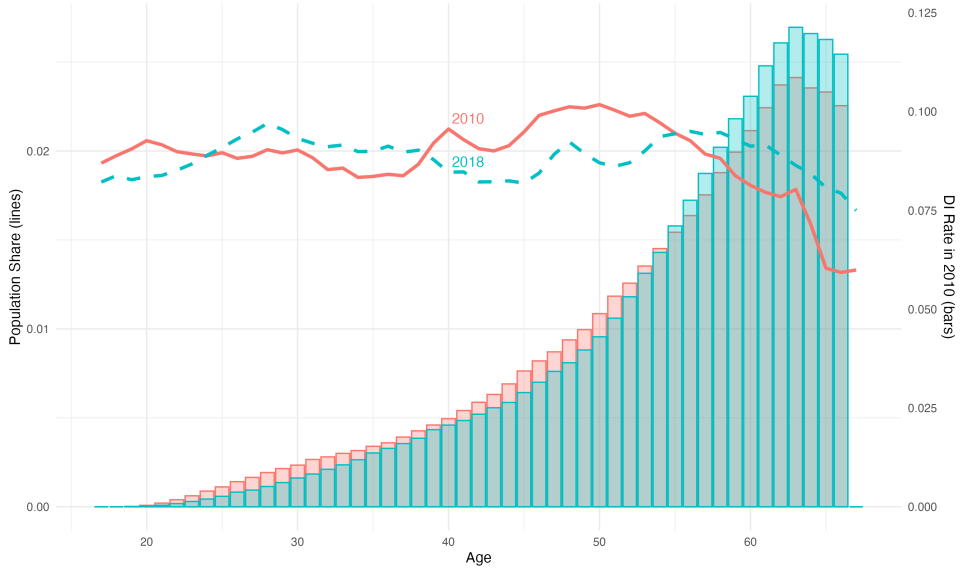

Notes: Figure plots depression rates among the working-age population (solid lines) and low-education men (dashed lines) estimated from the PHQ-9 depression screener in the National Health and Nutrition Examination Survey (NHANES). We use the standard PHQ-9 thresholds for "moderate," "moderately severe," and "severe" depression. Markers indicate the years in which the NHANES was administered; there is no NHANES data available between 2017 and 2021.

Appendix Figure A.14: Use of antidepressants and antipsychotics increased around 2011




Notes: Figure plots fraction of US adults using antidepressants or antipsychotics. Source: National Center for Health Statistics, National Health and Nutrition Examination Survey 2001–2018.

Appendix Figure A.15: From 2010 to 2018, working-age population became older on average, and older working-age people have higher SSDI rates




Notes: Figure plots age distribution of working-age population in 2010 and 2018 (left axis), plus SSDI enrollment rates by age in the same years (right axis). The sample is constructed from the Continuous Work History Sample (CWHS) linked to the Payment History Update System (PHUS) and the Numident.

Appendix Table A.1: SSDI increase decomposition results: Liebman (2015) vs this paper

| | Liebman (2015) 1985–2007 | This paper 1988–2010 |
|---|---|---|
| Increase in SSDI enrollment to be explained | 1.9 | 2.2 |
| *Percent explained by:* | | |
| Eligibility | 12% | 7% |
| Application + award | 51% | 44% |
| Population aging | 20% | 22% |
| Exit (mortality) | 5% | 15% |
| Exit (recovery) | −1% | 2% |
| Interaction terms (age x eligible) | 13% | |
| Other components | | 8% |
| *Percent explained by:* | | |
| Men | 41% | 40% |
| Women | 59% | 61% |

Notes: Table presents decomposition results from Liebman (2015) and analogous estimates from Table 1 of this paper. Liebman (2015) uses years 1985-2007 and ages 20-64 years. This paper uses years 1988-2010 and ages 25-66.

Appendix Table A.2: Decomposition results for decrease in SSDI enrollment: 2013 to 2019

| | Absolute contribution (pp) | | As a share of total (%) | |
|---|---|---|---|---|
| | Minimum | Maximum | Minimum | Maximum |
| **Across-group composition** | **-0.05** | **-0.06** | **9.2** | **11.9** |
| Sex | 0.00 | -0.01 | -0.4 | 2.1 |
| Age | 0.08 | -0.01 | -14.6 | 1.3 |
| Earnings | 0.01 | -0.07 | -2.3 | 12.3 |
| Immigrant status | -0.04 | -0.11 | 7.9 | 19.7 |
| **Within-group SSDI rates** | **-0.48** | **-0.50** | **88.1** | **90.8** |
| Entry | -0.42 | -0.43 | 76.3 | 79.6 |
| *Eligibility* | *-0.01* | *-0.02* | *2.5* | *4.0* |
| *Application* | *-0.31* | *-0.34* | *57.6* | *62.1* |
| *Award* | *-0.07* | *-0.09* | *12.9* | *17.0* |
| Exit | -0.06 | -0.06 | 10.4 | 11.7 |
| *Death* | *0.01* | *0.01* | *-1.2* | *-1.0* |
| *Recovery* | *-0.05* | *-0.06* | *9.3* | *10.3* |
| *Retirement* | *0.01* | *0.01* | *-1.3* | *-1.1* |
| *Other* | *-0.02* | *-0.02* | *3.3* | *3.7* |
| Friction in population | -0.00 | -0.00 | 0.6 | 0.6 |
| Population discrepancies | 0.00 | 0.00 | -0.1 | -0.0 |
| **Within-across interaction** | | | | |
| Total to be explained | -0.55 | -0.55 | 100.0 | 100.0 |

Notes: Table presents decomposition of the decrease in SSDI enrollment rates from 2013 to 2019 into various components. Across-group composition change, within-group change in SSDI enrollment rate, and the interaction between the two are calculated using equation (5). Across-group composition change is decomposed into specific characteristics using equation (11). Within-group SSDI rate is decomposed into entry, exit, friction in population, and population discrepancies using equation (6) and footnotes 13 and 15. Entry is decomposed into partial eligibility (i.e., the part that can only be attributed to eligibility and no other component), partial application, partial award, and the sum of interactions between the three entry components (eligibility, application, and award) using equation (9). Exit is decomposed into death, retirement (defined as aging into the retirement program), recovery (defined as being found no longer eligible due to continuing disability review), and "other" (the residual) using equation (8). "Minimum" means the minimum bound, and "Maximum" means the maximum bound. See Section 2.1 for details on the decomposition method, and Appendix Section F for details on the construction of bounds.

Appendix Table A.3: Decomposition results for decrease in SSDI enrollment: 2010 to 2021

| | Absolute contribution (pp) | | As a share of total (%) | |
|---|---|---|---|---|
| | Minimum | Maximum | Minimum | Maximum |
| **Across-group composition** | **-0.02** | **-0.11** | **1.2** | **7.3** |
| Sex | 0.00 | -0.03 | -0.1 | 1.9 |
| Age | 0.20 | -0.02 | -13.6 | 1.2 |
| Earnings | 0.03 | -0.14 | -2.2 | 9.3 |
| Immigrant status | -0.07 | -0.20 | 4.4 | 13.7 |
| **Within-group SSDI rates** | **-1.37** | **-1.46** | **92.7** | **98.8** |
| Entry | -1.31 | -1.42 | 88.7 | 95.9 |
| *Eligibility* | *-0.05* | *-0.10* | *3.2* | *6.7* |
| *Application* | *-0.64* | *-0.86* | *43.4* | *58.3* |
| *Award* | *-0.43* | *-0.68* | *28.8* | *46.2* |
| Exit | -0.02 | -0.07 | 1.2 | 4.7 |
| *Death* | *0.01* | *0.00* | *-0.8* | *-0.3* |
| *Recovery* | *-0.07* | *-0.10* | *4.7* | *6.6* |
| *Retirement* | *0.06* | *0.04* | *-3.9* | *-2.6* |
| *Other* | *-0.01* | *-0.02* | *0.8* | *1.5* |
| Friction in population | -0.01 | -0.01 | 0.6 | 0.9 |
| Population discrepancies | 0.00 | 0.00 | -0.3 | -0.1 |
| **Within-across interaction** | | | | |
| Total to be explained | -1.48 | -1.48 | 100.0 | 100.0 |

Notes: Table presents decomposition of the decrease in SSDI enrollment rates from 2010 to 2021 into various components. Across-group composition change, within-group change in SSDI enrollment rate, and the interaction between the two are calculated using equation (5). Across-group composition change is decomposed into specific characteristics using equation (11). Within-group SSDI rate is decomposed into entry, exit, friction in population, and population discrepancies using equation (6) and footnotes 13 and 15. Entry is decomposed into partial eligibility (i.e., the part that can only be attributed to eligibility and no other component), partial application, partial award, and the sum of interactions between the three entry components (eligibility, application, and award) using equation (9). Exit is decomposed into death, retirement (defined as aging into the retirement program), recovery (defined as being found no longer eligible due to continuing disability review), and "other" (the residual) using equation (8). "Minimum" means the minimum bound, and "Maximum" means the maximum bound. See Section 2.1 for details on the decomposition method, and Appendix Section F for details on the construction of bounds.

Appendix Table A.4: Decomposition results for the decrease in SSDI enrollment from 2010 to 2019: adjustment for change in applicant severity composition

| | | Absolute contribution (pp) | | | As a share of total (%) | | |
|---|---|---|---|---|---|---|---|
| | | Minimum | Maximum | Used in graph | Minimum | Maximum | Used in graph |
| Original | | | | | | | |
| | **Entry** | **-0.94** | **-1.00** | **-0.99** | **90.3** | **96.9** | **95.2** |
| | *Eligibility* | -0.05 | -0.09 | -0.08 | 4.5 | 8.4 | 8.1 |
| | *Application* | -0.48 | -0.59 | -0.58 | 46.1 | 56.8 | 55.6 |
| | *Award* | -0.31 | -0.45 | -0.44 | 29.9 | 43.0 | 42.0 |
| Adjusted | | | | | | | |
| | **Entry** | **-0.94** | **-1.00** | **-0.99** | **90.3** | **96.9** | **95.2** |
| | *Eligibility* | -0.05 | -0.09 | -0.08 | 4.5 | 8.4 | 8.1 |
| | *Application + Applicant share* | -0.59 | -0.72 | -0.61 | 57.1 | 69.1 | 59.2 |
| | *Application* | -0.48 | -0.59 | -0.58 | 46.2 | 56.7 | 55.6 |
| | *Applicant share* | -0.10 | -0.15 | -0.15 | 9.4 | 14.7 | 14.4 |
| | *Award* | -0.19 | -0.32 | -0.31 | 18.6 | 31.0 | 30.3 |

Notes: Table presents decomposition of the decrease in SSDI enrollment rates from 2010 to 2019 into the components of Entry. "Original" reproduces the results from Table 2. "Adjusted" presents results using equation (18) to adjust for changes in applicant severity. Since applicants became less severe over this period, the award shares in Table 2 reflect a mechanical effect of lower award rates due to less-severe applicants. The "adjusted" numbers correct for this issue. In the adjusted numbers, "Award" represents the first term in equation (18) and "Applicant share" represents the second term in equation (18). "Minimum" means the minimum bound, "Maximum" means the maximum bound, and "Used in graph" means the point estimate used in Figure 3. See Section 2.1 for details on the decomposition method, and Appendix Section F for details on the construction of bounds.

Appendix Table A.5: Robustness Checks for Sensitivity of Applications to ALJ Reform

| | Specification of Controls | | | | | |
|---|---|---|---|---|---|---|
| **Outcome Specification** | (1) | (2) | (3) | (4) | (5) | (6) |
| Level Applications | 4.3 | -9.7 | 19.6 | 9.8 | 42.3 | -37.3 |
| | (45.4) | (44.2) | (47.8) | (48.4) | (51.8) | (59.2) |
| | *Pre-Reform Mean Applications:* | | | | *986* | |
| Log Applications | -0.09 | -0.10 | -0.06 | -0.04 | -0.01 | -0.10 |
| | (0.06) | (0.05) | (0.05) | (0.06) | (0.05) | (0.06) |
| **Additional Controls** | | | | | | |
| Unemp. Rate | - | X | X | X | X | X |
| Initial (DDS) Allowance Rate | - | - | X | - | X | X |
| SSDI Application Rate | - | - | - | X | X | X |
| 2006 Avg Applicant Earnings | - | - | - | - | - | X |
| Mental/Musculoskeletal Applicant Share | - | - | - | - | - | X |

Notes: Table reports estimates of the effect of the ALJ reform on SSDI applications using a pre-post version of equation (22): i.e., $\beta_t$ from $Y_{it} = \alpha_i + \gamma_t + \beta_t(\text{BaseAppealAllow}_i \times \text{Post2008}_t) + \delta X_{it} + \epsilon_{it}$. $Y_{it}$ is either level applications or log applications. See Section 5.2.2 for details. Columns denote alternative model specifications, using different sets of controls in the regression models.

# B Details of empirical analysis

## B.1 SIPP analysis

We use the 1996, 2001, 2004, 2008, and 2014 panels of the Survey of Income and Program Participation linked to SSA administrative data, including the Numident, F831 files, Payment History Update System, and Summary Earnings Record. For the 1996, 2001, 2004, and 2008, we use the core files and the topical module on adult disability, which includes questions about health conditions, self-reported health status, and activities of daily living and other questions about daily function. For the reengineered 2014 panel, we use the SSA supplement which asks many of the health questions previously asked in the adult disability topical modules.

We append all five panels together. In order to study the subgroup that drives the decline in SSDI applications, we limit sample to men with a high-school degree or less. We classify individuals by birth cohort in 5-year intervals (1940–44, 1945–1950, 1950–54, 1955–1960, 1960–64, 1965–1970). We also create groups for different measures of health in the SIPP; our main classification is reporting "fair" or "poor" health (EHSTAT in the 1996, 2001, 2004, and 2008 panels; EHLTSTAT in the 2014 SSA supplement). We identify spouses using the SIPP family relationship variables.

For each year, we create variables for cumulative SSDI application (ever apply up to that year), cumulative SSDI receipt (ever receive SSDI up to that year), own earnings and employment (average annual), and spouse earnings and spouse employment (average annual, conditional on having a spouse and unconditional on having a spouse). Earnings variables are inflation-adjusted using 2000 as the base year (as in our CWHS analysis). We condition the earnings and employment variables on being financially eligible for SSDI (calculated using individual earnings history) and not receiving SSDI at that age.

We use SIPP survey weights. The SIPP survey weights are consistent across panels for the 1996, 2001, 2004, and 2008 panels. However, the scale of the weights changes by several orders of magnitude for the reengineered 2014 panel, so we normalize the average 2014 panel weight to equal the average 2008 panel weight.

## B.2 Evidence on potential explanations for decline in SSDI applications

### B.2.1 Details on Social Security field office closing calculations

Deshpande and Li (2019) estimate large declines in applications and awards resulting from

Social Security field office closings. We use estimates from Deshpande and Li (2019) to determine how much of the decline in SSDI applications can be explained by Social Security field office closings.

In ZIP codes where the closest office closes, SSDI applications fall on average by 6.8 percent and SSDI awards by 14.8 percent. In ZIP codes where a second-closest or third-closest office closes, SSDI applications fall by 2.8 percent and SSDI awards by 9.0 percent. In level terms, these application declines imply 306 fewer applications in total per office closing.

Assuming that effects of office closings are persistent, we use this estimate to construct the counterfactual DI application rate presented in Section 6. It implies that office closings can explain 4.4 percent of the cumulative decline in DI applications from 2010 to 2019.

### B.2.2 Details on pollution abatement analysis

We estimate the effect of local pollution on SSDI enrollment using variation from the Clean Air Act as in Isen et al. (2017). Following Isen et al. (2017), we construct a crosswalk between Numident place of birth and counties, as pollution data is measured at the county level. This is a challenge because place of birth information in the registry data (a "city of birth" and a "state of birth") is hand-coded and recorded at varying levels of geographic granularity.

We follow the matching procedure developed by Isen et al. (2017). We construct a list of known geographic units that are linked to counties by stacking four databases: the Census Bureau's database of Census 1) places, 2) counties, and 3) minor civil divisions, as well as 4) the United States Geological Survey's Geographic Names Information System file. Each of these files provides a list of strings identifying local geographic units and the counties to which they belong. We then merge county information into the registry data using the following two procedures: First, we exactly match individual city and state of birth in the registry file as-is to the constructed county information file. Second, we extract the first word from an individual's place of birth and match this to the first word of location names in the county file, keeping any successful matches which identify a unique county within the state. Taking individuals from the Numident file who were working-age adults between 2010 and 2019, we can match place of birth for nearly one-half of individuals.

For the individuals with successful matches on place of birth, we then assign Clean Air Act non-attainment status at the county-level using classifications from Isen et al. (2017) (data provided by the authors). We determine if the decline in applications is disproportionately concentrated among individuals born in non-attainment counties, in which Clean Air Act regulations imposed improvements in air quality over time. We report our results

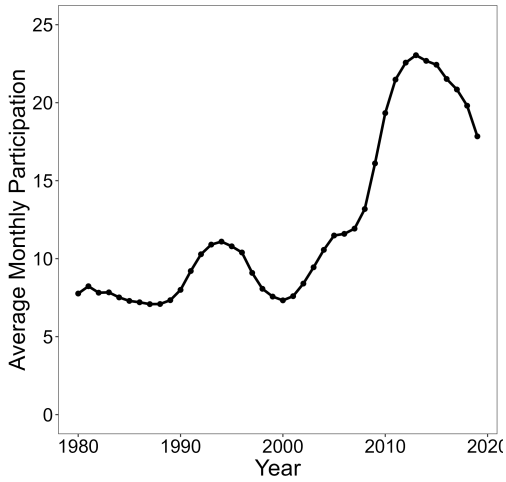
in Appendix Section G.3. We find that the SSDI application decline is disproportionately concentrated in these counties.

### B.2.3 Generosity of other social welfare benefits

The explanation of higher generosity of other social welfare programs is inconsistent with the patterns we observe in SSDI applications and employment. Still, we consider how the enrollment and generosity of other social safety programs have changed since 2010.

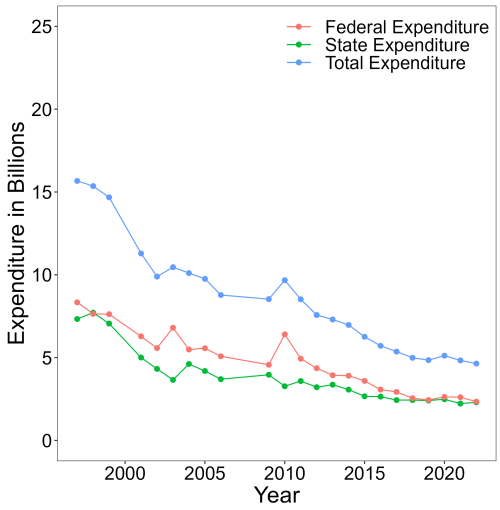
**SNAP, TANF, and UI.** There was no increase in SNAP, TANF, or UI benefit amount or enrollment between 2010 and 2019 (see Appendix Figures B.1, B.2, and B.3).

Appendix Figure B.1: SNAP program participation did not increase from 2010–2019



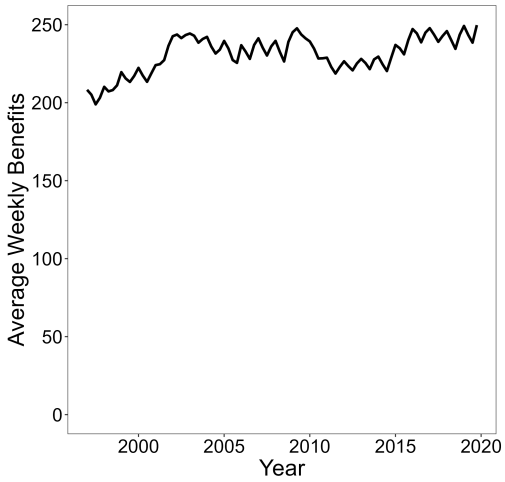

Notes: Figure plots average monthly SNAP participation from 1980 to 2020, in millions. Source: United States Department of Agriculture, "Persons, Households, Benefits, and Average Monthly Benefit per Person & Household."

Appendix Figure B.2: TANF spending did not increase from 2010–2019




Notes: Figure plots federal, state, and total expenditures on TANF (all direct assistance expenditures) over time, in 2000 dollars. Source: United States Department of Health and Human Services, Office of Family Assistance. FY 1997-2022 Federal TANF & State MOE Financial Data.

Appendix Figure B.3: UI benefit amount did not increase substantially from 2010–2019




Notes: Figure plots average weekly UI benefits in each quarter averaged across states and adjusted for inflation to 2000 dollars. Source: United States Department of Labor, Unemployment Insurance Data Base (`https://oui.doleta.gov/unemploy/data_summary/DataSum.asp`).

**Health insurance.** The Affordable Care Act of 2010 expanded Medicaid to more low-income adults and subsidized health insurance for lower- and middle-income adults through the exchanges. If some potential SSDI applicants would have applied primarily to get health insurance coverage, these health insurance expansions could have contributed to the decline in SSDI applicants.

We investigate this hypothesis in several ways. First, we determine whether states that experienced larger increases in health care coverage over this period have larger SSDI application declines. Appendix Figure B.4 shows this is not the case; if anything, states with greater enrollment in Medicaid and health insurance exchanges (left graph) or changes in health insurance coverage in general (right graph) had smaller declines in SSDI application rates.

Second, we study whether the decline in SSDI applications is concentrated in states that expanded Medicaid coverage (Appendix Table H.13). We find no heterogeneity in the SSDI application decline by state Medicaid expansion status.

Finally, we review the large existing literature on the effect of Medicaid expansion on SSDI and SSI applications. Appendix Table B.1 summarizes the population, outcome, and estimated effect for each of six papers (Baicker et al., 2014; Maestas et al., 2014; Burns and Dague, 2017; Anand et al., 2019; Schmidt et al., 2020).[36] It then presents our calculation extrapolating the effect size to SSDI applications from all adults.

Two papers (Staiger et al., 2024; Ne'eman and Maestas, 2023a) estimate the effects of Medicaid expansion on SSDI enrollment. Both find expansion increased enrollment. We infer effects on SSDI applications by assuming Medicaid expansion has no effect on rejected applications, and then converting the excess pool of SSDI recipients over the study period into annualized excess SSDI applications.[37]

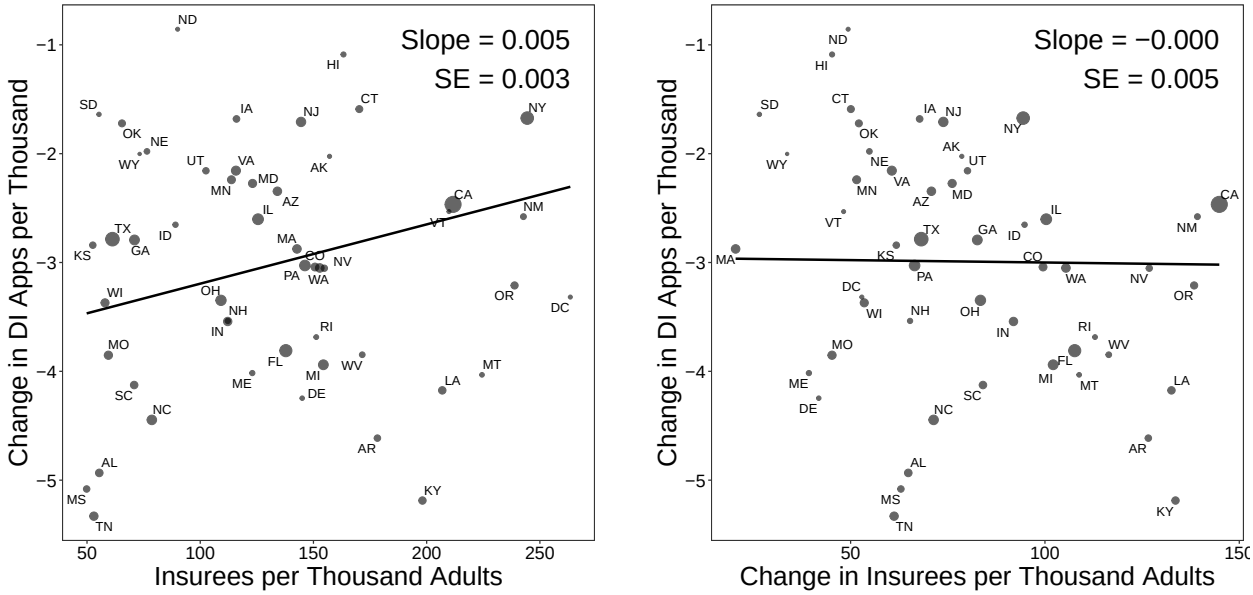
Two other papers (Maestas et al., 2014; Baicker et al., 2014) estimate effects for specific states (Massachusetts and Oregon respectively) from pre-expansion variation. For these, we assume that these state-specific effects are representative of effects in Medicaid-expanding states more generally. For Maestas et al. (2014), we also scale their quarterly estimates into annual ones.

Finally, two papers (Anand et al., 2019; Schmidt et al., 2020) directly estimate the impact of Medicaid expansion on SSDI applications. We report their estimates as-is.

[36] Burns and Dague (2017) and Soni et al. (2017) additionally examine effects on SSI applications. (Heim et al., 2021) tests for effects of Medicaid expansion by looking for changes in beneficiary composition.

[37] Staiger et al. (2024) specifically estimates effects on enrollment for disabled individuals, who they estimate to represent 11.4 percent of the total working-age population. For this estimate, we further assume no effect of Medicaid expansion on SSDI applications from non-disabled individuals.

Appendix Figure B.4: State-level changes in SSDI application rates vs changes in health insurance coverage




(a) Expanded Medicaid and Exchanges

(b) All Health Insurance

Notes: Panel (a) plots, for each state, the number of insurees enrolled in either expanded Medicaid or the exchanges through the Affordable Care Act in 2019 (relative to the adult population in 2019) (x-axis) against the change in SSDI application rates from 2010 to 2019 (y-axis). On the x-axis, we count insurees from Medicaid Group 8 (adult enrollees eligible for Medicaid under Affordable Care Act expansions) and state health insurance exchanges. Data on Medicaid Group 8 enrollment comes from the Medicaid Budget and Expenditure System. Data on enrollment in exchanges and basic health plans is taken from the Kaiser Family Foundation, originally published in Health Insurance Marketplace Open Enrollment Reports by the Office of the Assistant Secretary for Planning and Evaluation (2014-2016) and the Marketplace Open Enrollment Period Public Use Files by the Centers for Medicare and Medicaid Services (2017-2019). Panel (b) repeats the exercise using growth in *all* health insurance coverage, as measured in the American Community Survey.

| | Staiger et al. (2024) | Ne'eman and Maestas (2023) | Maestas et al. (2014) | Baicker et al. (2014) | Anand et al. (2019) | Schmidt et al. (2020) |
|---|---|---|---|---|---|---|
| Population: | Disabled Adults | All Adults | Adults in MA | Adults in OR | All adults | All adults |
| Outcome: | Enrollment Stock | Enrollment Stock | Quarterly Apps | Applications | Applications | Applications |
| Effect Estimate (per Thousand): | 20.20 | 8.00 | 0.05 | 0.07 | 0.04 | 0.00 |
| **Extrapolating effect estimate to:** | | | | | | |
| All Adults: | 2.30 | 8.00 | 0.05 | 0.07 | 0.04 | 0.00 |
| **Translating effect estimate into:** | | | | | | |
| Annual Apps from All Adults: | 0.77 | 2.67 | 0.19 | 0.07 | 0.04 | 0.00 |

Notes: Table reports our calculations of the effects of Medicaid expansion on annual SSDI application rates implied by each of the six papers presented. The second to last row reports the effect actually appearing in the given paper (which, per the row before it, is not necessarily the effect of Medicaid expansion on annual SSDI application rates). The last row converts this estimate into an effect on annual application rates. For papers reporting effects on enrollment, we assume Medicaid expansion has *no* effect on rejected applications, and then translate the excess pool of SSDI beneficiaries over the study period into annualized excess SSDI applications. For Ne'eman and Maestas (2023a), we further scale by the share of the working-age population that reports a disability (11.23 percent).

Appendix Table B.1: Estimating the effect of Medicaid expansion on SSDI applications

**Health care access.** Anstreicher (2021) and Anstreicher and Strand (2025) find that community health center (CHC) expansions starting in 2000 reduced SSDI applications in surrounding areas. However, CHC expansions begin well before the decline in SSDI enrollment, and Anstreicher and Strand (2025) find that they can only account for about 3% of the total decline in SSDI applications.

**Veterans Disability Compensation.** We consider two major changes to Veterans Disability Compensation that could have affected SSDI application rates, with complete results in Appendix Section G.2. First, enrollment in Veterans Disability Compensation (VDC) has increased from 3.2 million in 2010 to 5.7 million in 2023, and the average benefit amount has increased from $11,365 to $23,500 (in nominal dollars) (U.S. Department of Veterans Affairs, 2010, 2023). While there is no restriction on receiving both VDC and SSDI benefits,[38] it is possible that higher VDC receipt could reduce SSDI application and enrollment through an income effect. To evaluate this hypothesis, we identify veterans in our data based on military EINs. We find that veterans do not disproportionately contribute to the decline in SSDI applications; in fact, veterans have a disproportionately small decline in SSDI applications relative to their size (see Appendix Table G.26).

Second, from 2000 to 2020, the fraction of men ages 45 to 64 who report in the CPS that they are veterans fell from 34 percent to 12 percent (Chan et al., 2021). To the extent that VDC enrollment increases SSDI enrollment, a decline in the population share of veterans could contribute to the decline in SSDI enrollment, especially among men. To investigate this hypothesis, we redo the decomposition exercise in Table 2 including veteran status (and thus the interaction between veteran status and age). The results are shown in Appendix Table G.23. Including veteran status as a characteristic of the decomposition does not change the main results: the application contribution is similar at 46–58% (vs 46–57%), the award contribution is identical at 30–43%, and the composition (including veteran status) contribution is similar at -1 to 6% (vs 1–5%). These results indicate that the declining veteran share among older men is not a large enough phenomenon to explain a large part of the SSDI enrollment or application decline.

[38] See, e.g., `https://www.ssa.gov/pubs/EN-64-125.pdf`

# C Details of application counterfactuals

To take to data our model of application behaviors from Section 5, we let $p_t(v) \equiv \mathbb{P}(A_t|v_t = v)$ denote the application rate among those of type $v$ regardless of eligibility status (i.e., the application-to-population ratio). This aligns with the measures used in our empirical evidence in Section 5 and Appendix Section B, and it is necessary because determinants of one's type (e.g., education for our exercise fixing labor market conditions) can only be observed for applicants in our administrative data sources. We find no meaningful trends in eligibility rates, so we expect trends in the application-to-population ratio to roughly map into trends in application rates among the eligible.

Except where otherwise specified, we define one's type $v$ by a combination of age (older than 40 versus 40 and younger), educational attainment, and sex. Around 5% of SSDI applications are missing information on the applicant's education. To address this, we assume education is missing at random conditional on the other determinants of type and the year of applying for SSDI.

**Homogeneity assumption.** For our baseline results, we assume homogeneity across types in utility parameters. This means that, for example, the type-specific condition pinning down the labor market shifter, $\beta_k^w$, using the following equality:

$$\underbrace{\frac{\partial \mathbb{P}(A_t = 1 \mid \nu_t = v)}{\partial z_{kt}^w}}_{\text{Identified from (quasi)-experiment}} = \underbrace{h(m_t(v))}_{\text{Identified from application data}} \times \underbrace{\beta_k^w(v)}_{\text{Unobserved}},$$

implies a condition that must also hold in expectation across types and time periods:

$$\mathbb{E}\left[\frac{\partial \mathbb{P}(A_t = 1|\nu_t = v)}{\partial z_{kt}^w}\right] = \mathbb{E}[h(m_t(v))]\beta_k^w \tag{25}$$

To actually estimate $\beta_k^w$, we take an estimate of the quasi-experimental effect of the shifter $z_k^w$ on application behaviors to be an estimate of the left-hand side term. We divide this by an estimate for $\mathbb{E}[h(m_t(v))]$, which we get by averaging over type and period-specific densities at the margin of applying, as implied by their application rates. Our estimates for parameters governing the utility of applying for SSDI follow from this homogeneity assumption in a similar way.

## C.1 Counterfactuals fixing less-skilled labor market conditions

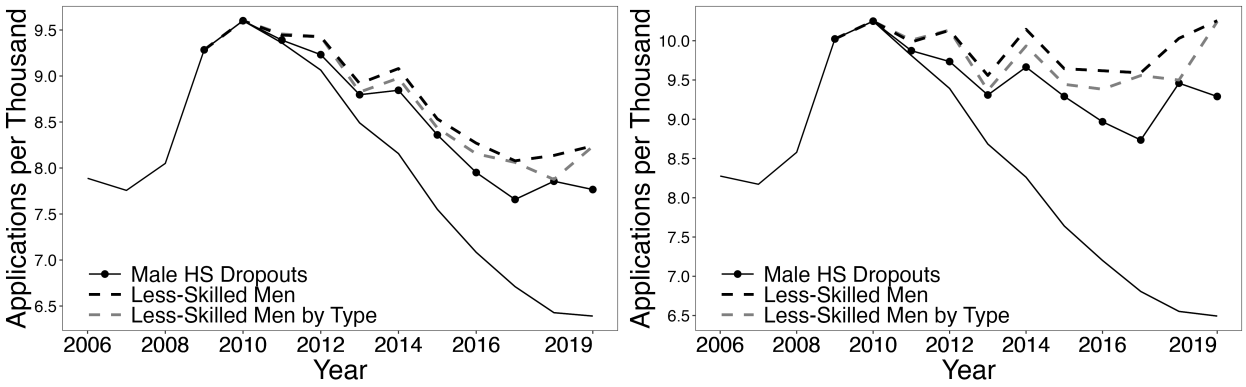
Appendix Figure C.1 presents alternative counterfactual application rates, fixing labor market conditions for less-skilled men (i.e., men with no college education). Panel (a) reports overall application rates, averaging counterfactuals for less-skilled men with application rates for other (unaffected) subpopulations, and panel (b) reports application rates specifically among men. These panels both report three counterfactuals which point to the same conclusion: fixing labor market conditions for less-skilled men can explain a substantial share of the recent decline in overall application rates because they can entirely explain the recent decline for men in particular.

The first counterfactual we report (the dashed line) is derived from the causal estimate in the first panel of Appendix Table C.1. This is the counterfactual appearing also in Figure 12. We next relax the homogeneity assumption underlying that counterfactual, using type-specific causal estimates from the second panel of Appendix Table C.1 to estimate type-specific sensitivities to labor market conditions. This can only be done for types with an existing first stage, so in this counterfactual (the dotted line), we do not fix labor market conditions for the one type that has a low first stage F-statistic (older male high school dropouts). In our last counterfactual, we fix labor market conditions for male high school dropouts in particular (echoing the choice of Autor and Duggan (2003) to focus on dropouts). This counterfactual (the connected line) is based on a sensitivity estimate for male dropouts, derived from causal estimates for that group: an estimate of -0.21 (standard error 0.42) with a first stage of 1.20 (standard error 1.10).

A potential concern with each of these counterfactuals, though, is that they are derived from effect estimates with relatively wide standard errors. Panel (a) of Appendix Figure C.2 shows that, indeed, the counterfactual confidence interval based on the first panel of Appendix Table C.1 contains the actual path of overall application rates.

In Section 5.2.1, we estimated two alternative specifications using additional years of data. The resulting estimates then became larger and significant. Panels (b) and (c) of Appendix Figure C.2 show that even the lower end of the 95-percent confidence interval of the alternative estimates suggests that changes in employment rates are a key driver of the decline in SSDI applications from 2010 to 2019.

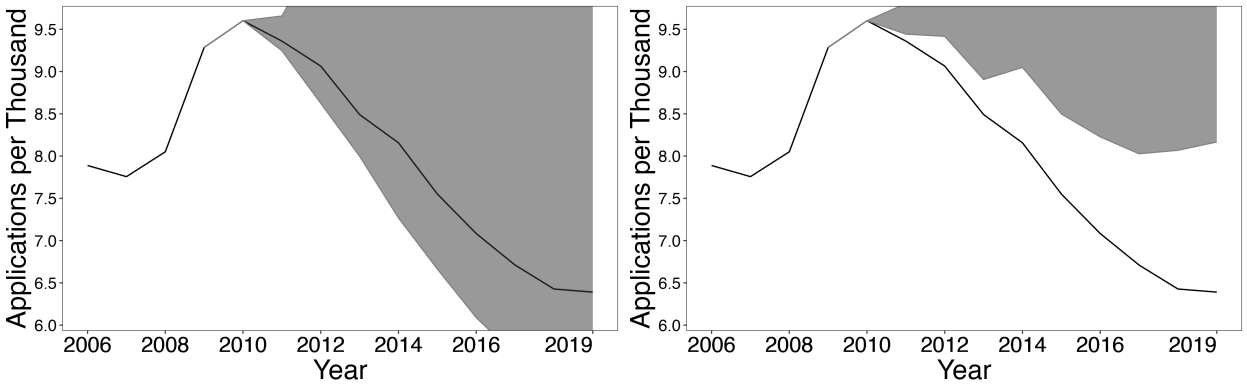
Appendix Figure C.1: Application counterfactuals fixing labor market conditions




(a) Application rate among working-age adults (b) Application rate among working-age men

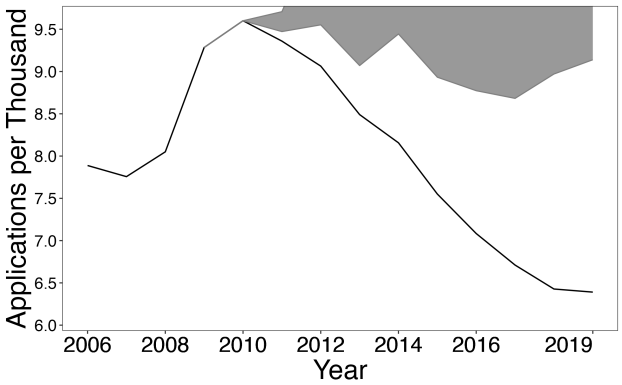
Notes: Graphs present counterfactuals fixing labor market conditions for less-skilled men. "Less-Skilled Men" uses the effect estimates from the first panel of Appendix Table C.1. "Less-Skilled Men by Type" uses estimates from the second panel of Appendix Table C.1, for the three types with a strong first stage. "Male HS Dropouts" uses the TSLS estimate for high school dropouts (-0.21 standard error: 0.42) (variance-weighted average of "HS dropout" rows in Appendix Table C.1).

Appendix Figure C.2: Inference on application counterfactuals fixing labor market conditions




(a) Estimated with data from 2013 to 2019

(b) Estimated with data from 1992 to 2019




(c) Estimated with specification of Autor and Duggan (2003) and data from 1984 to 2019

Notes: The shaded region in each plot indicates the 95 percent confidence interval on application counterfactuals for working-age adults–fixing labor market conditions for less-skilled men–implied by the stated 2SLS estimate. Panel (a) uses the effect estimates from the first panel of Appendix Table C.1. Panel (b) uses the same estimator but fit on non-overlapping intervals from 1992 to 2019 instead of 2013 to 2019 (point estimate -0.43 and standard error 0.15). Panel (c) uses the estimate based on data from the original model specification of Autor and Duggan (2003) applied to three-year intervals from 1984 to 2019 (point estimate -0.37 and standard error 0.09).

Appendix Table C.1: Estimates of the effects of labor market conditions on SSDI applications

| | | First Stage | | 2SLS | |
|---|---|---|---|---|---|
| | | Coef | F | Coef | S.E. |
| *(a) Overall Estimates for Less-Skilled Men* | | | | | |
| | | 1.659 | 15.024 | -0.130 | 0.158 |
| *(b) Less-Skilled Men, by Type (Education × Age)* | | | | | |
| Education | Age | | | | |
| HS Dropouts | 41-66 | 0.425 | 0.069 | -0.906 | 1.788 |
| HS Dropouts | 18-40 | 1.611 | 2.398 | -0.146 | 0.167 |
| Only HS | 41-66 | 1.441 | 6.219 | -0.329 | 0.146 |
| Only HS | 18-40 | 1.889 | 32.401 | -0.049 | 0.065 |

Notes: These models include the controls in equation (20): time dummies and linear controls for annualized changes in population shares by age and gender, age being split into 3 bins: 18-39, 40-54, and 55-66.

Appendix Table C.2: Estimates of the effects of labor market conditions on SSDI award rates

| | | First Stage | | 2SLS | |
|---|---|---|---|---|---|
| | | Coef | F | Coef | S.E. |
| *(a) Overall Estimates for Less-Skilled Men* | | | | | |
| | | 1.659 | 15.024 | -0.036 | 0.076 |
| *(b) Less-Skilled Men, by Type (Education × Age)* | | | | | |
| Education | Age | | | | |
| HS Dropouts | 41-66 | 0.425 | 0.069 | -0.361 | 0.906 |
| HS Dropouts | 18-40 | 1.611 | 2.398 | -0.015 | 0.016 |
| Only High School | 41-66 | 1.441 | 6.219 | -0.164 | 0.100 |
| Only High School | 18-40 | 1.889 | 32.401 | -0.004 | 0.016 |

Notes: This table reports 2SLS effects of labor market conditions on award rates among applicants–a check on the exclusion restriction of the instrument. These models include the controls in equation (20): time dummies and linear controls for annualized changes in population shares by age and gender, age being split into 3 bins: 18-39, 40-54, and 55-66.

## C.2 Counterfactuals fixing award rates

Appendix Table A.5 reports the estimated effects of ALJ reform on SSDI applications, which we will translate into effects of *award rates* on application rates for the purpose of simulating counterfactuals. To do so, we scale the reform's estimated effects on log applications by the application-to-population ratio to convert them into effects on application probabilities. We then divide this by the reform's "first stage" effect on award rates, consistent with the

following identity:

$$\frac{\partial \mathbb{P}(A_t = 1)}{\partial \mathbb{P}(\text{Award}_t = 1 | A_t = 1)} = \left[ \frac{\partial \log(\text{Apps}_t)}{\partial z_t^b} \Big/ \frac{\partial \mathbb{P}(\text{Award}_t = 1 | A_t = 1)}{\partial z_t^b} \right] \times \frac{\text{Apps}_t}{\text{Pop}_t}$$

We then translate effects of award rates on application probabilities into effects on the utility cost of applying by adjusting for the density at the margin of applying and the probability of award, based on average application and award rates over the period surrounding ALJ reforms (2009-2015).

We report in Figure 12 the most extreme counterfactual from taking the largest effect estimate reported in Appendix Table A.5.

## C.3 Counterfactuals fixing office closures

The quasi-experimental effect of office closures on SSDI applications comes from Deshpande and Li (2019) and is reported in Section B.2.1. We take the estimates which imply 306 fewer SSDI applications per office closing and translate it into an effect on application rates by dividing by population counts each year. Averaging this and densities at the margin of applying over calendar years 2000 to 2019 provides the inputs to solve for $\beta_k^b$ using a version of equation (25).

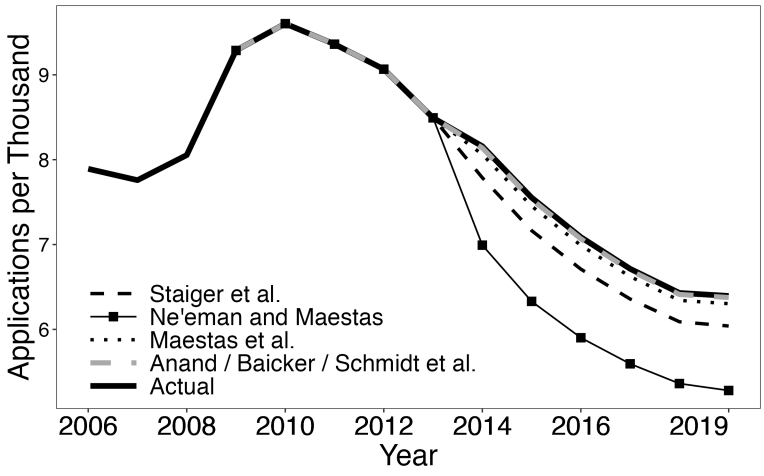
## C.4 Counterfactuals fixing Medicaid access

Appendix Figure C.3 plots counterfactual SSDI rates in the absence of the ACA Medicaid expansions, based on the alternative effect estimates reported in Appendix Table B.1. For this exercise, we take a person's type ($v$) to be state of residence, since some states expanded Medicaid and others did not. The results of Appendix Figure C.3 come from averaging the counterfactuals for Medicaid-expanding states with (unaffected) states that never expanded Medicaid. Because application responses to Medicaid are all positive or close to zero, these counterfactual application rates lie weakly below the actual application rates observed in the data. In Figure 12, we plot the counterfactual implied by estimates from Schmidt et al. (2020), which is indistinguishable from actual application rates.

## C.5 Counterfactuals fixing the age distribution

When counterfactually fixing the age distribution, we do not rely on a model of application behaviors. Instead, we hold fixed the population density across ages while allowing age-specific application rates to freely vary over time per the data. Letting type $\nu$ be defined

Appendix Figure C.3: Application counterfactuals fixing Medicaid access




Notes: Counterfactuals use effect estimates reported in the final row of Appendix Table B.1. The counterfactual line using Baicker et al. (2014); Anand et al. (2019); Schmidt et al. (2020) is on top of the "actual" line.

only by age alone, the counterfactual in $t_1$ having fixed the age distribution at its year-$t$ value is:

$$P(A_{t_1}^{\text{aging}}) = \int P(A_{t_1}|\nu_{t_1} = v) f_{\nu_t}(v) dv$$

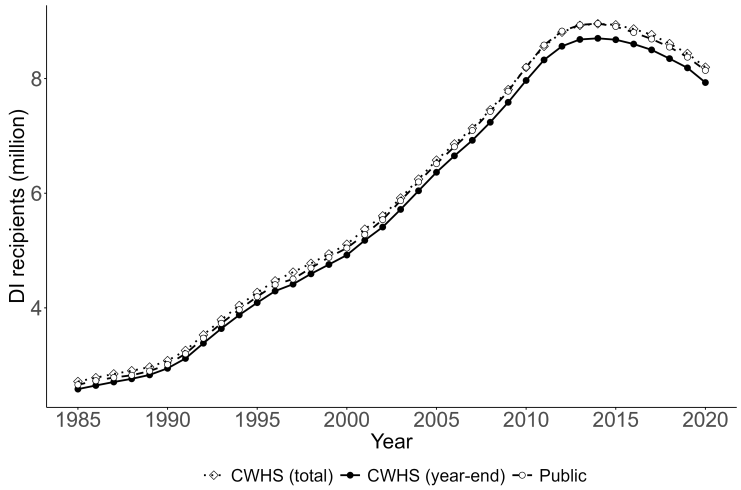
## C.6 Counterfactuals fixing veteran shares and application rates

To gauge how much veterans contribute to the decline, we define type by veteran status alone and counterfactually fix two objects. First, we fix the density of veteran status in the population, in the same way we fixed the age distribution above. Second, we also fully shut down changes over time in the application rates of veterans. This leaves only one channel driving counterfactual declines in overall application rates: changes in the application behaviors of non-veterans.

# D Data Appendix

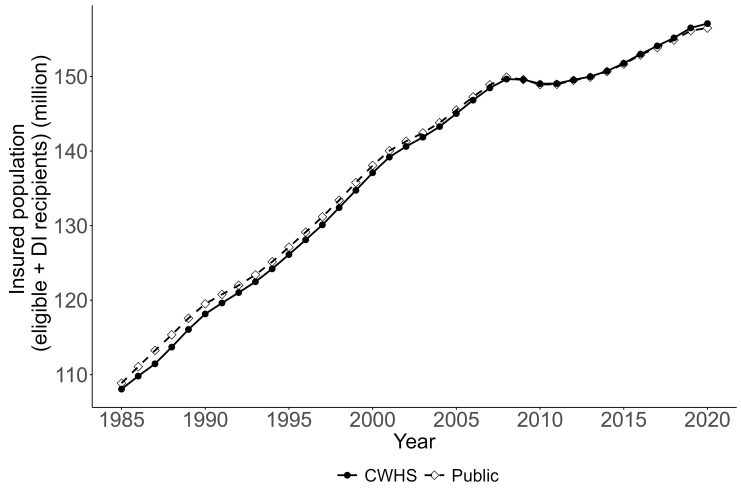
## D.1 Matching public data on SSDI

Appendix Figure D.1: CWHS vs public data: SSDI recipients



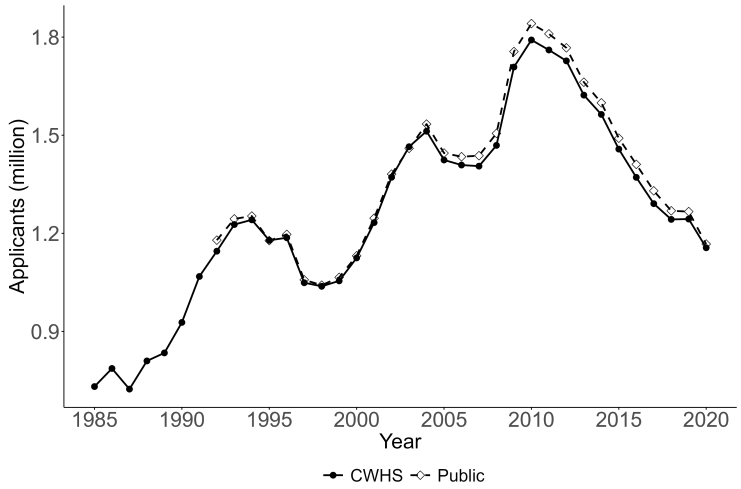

Notes: Figure plots the number of SSDI recipients over time from different data sources: authors' tabulations from the Continuous Work History Sample (CWHS) and publicly available data from the Social Security Administration. We identify SSDI recipients in the CWHS by determining whether they receive transfers in the Payment History Update System (PHUS) before retirement age. The solid line plots CWHS year-end SSDI recipients, which excludes SSDI recipients that died or terminated their status at each calendar year-end. The dotted line plots all recipients in the CWHS who were on SSDI in the calendar year. The dashed line plots the SSDI recipients reported in the first column of Table 49 in Social Security Administration (2024).

Appendix Figure D.2: CWHS vs public data: SSDI-insured population



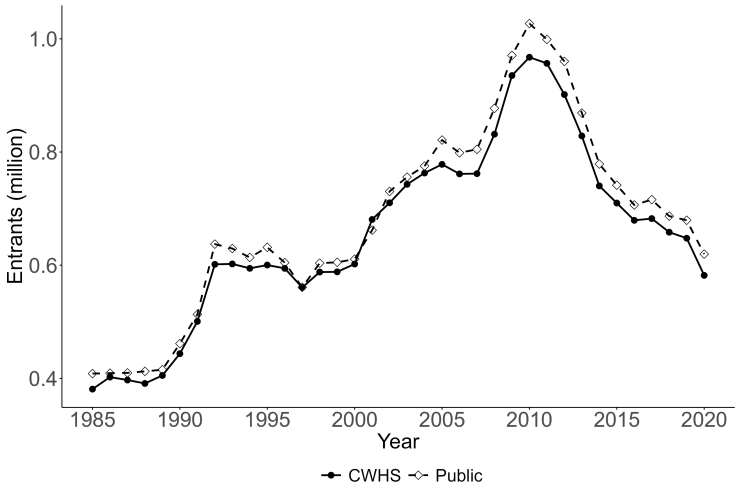

Notes: Figure plots the number of SSDI "insured" individuals—i.e., those who are financially eligible for SSDI—over time from different data sources: authors' tabulations from the Continuous Work History Sample (CWHS) and publicly available data from the Social Security Administration. We identify the insured population not on SSDI in the CWHS by calculating financial eligibility from the Master Earnings File; we then add this number to the number of SSDI recipients to get the total insured population. The solid line plots insured population in the CWHS. The dashed line plots SSDI recipients reported by the Social Security Administration in the 2024 Trustees Report (based in part on the Continuous Work History Sample, 1 percent sample, `https://www.ssa.gov/oact/STATS/table4c2DI.html`).

Appendix Figure D.3: CWHS vs public data: SSDI applicants



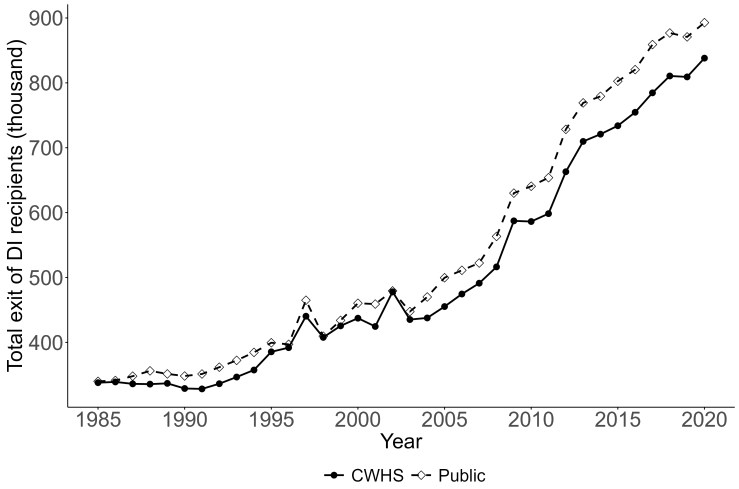

Notes: Figure plots number of SSDI applicants over time from different data sources: authors' tabulations from the Continuous Work History Sample (CWHS) and publicly available data from the Social Security Administration. We identify applicants in the CWHS by linking them to the F831 file. The solid line plots number of applicants based on the CWHS. The dashed line plots number of applicants in Table 60 from the 2023 Annual Statistical Report (Social Security Administration, 2024), where we sum the number of pending applications and all four columns in medical decisions and exclude technical denials.

Appendix Figure D.4: CWHS vs public data: SSDI entrants



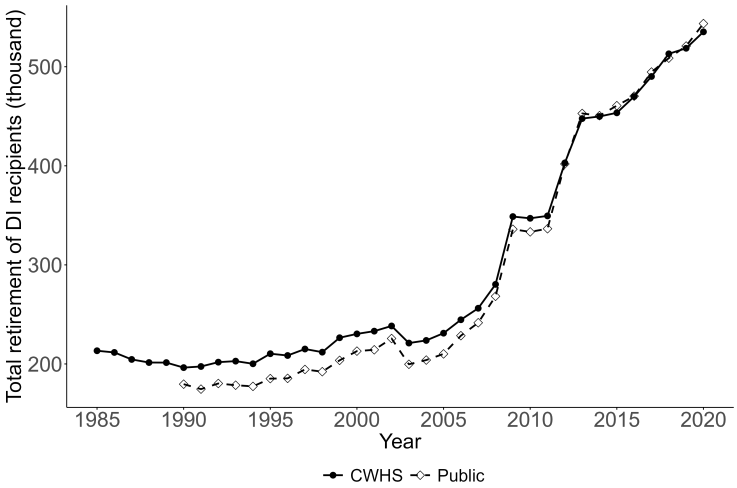

Notes: Figure plots number of SSDI entrants over time from different data sources: authors' tabulations from the Continuous Work History Sample (CWHS) and publicly available data from the Social Security Administration. We identify entrants in the CWHS by linking them to the Payment History Update System (PHUS) and the Master Beneficiary Record. The solid line plots number of entrants based on the CWHS. The dashed line plots number of entrants in Table 39 from the 2023 Annual Statistical Report (Social Security Administration, 2024).

Appendix Figure D.5: CWHS vs public data: SSDI exits



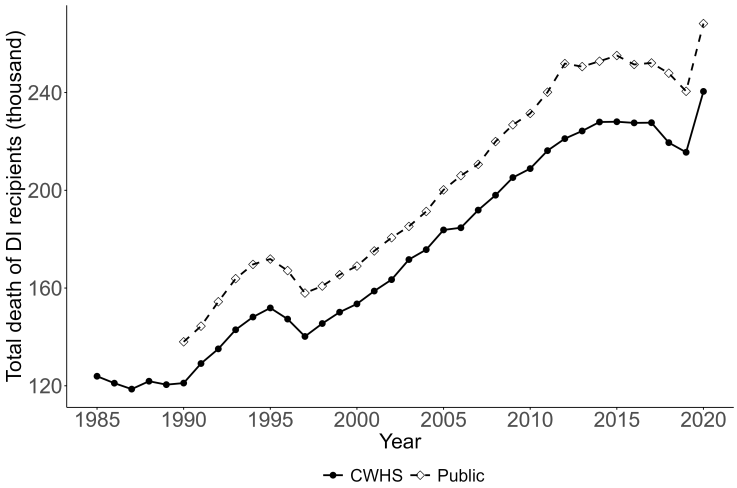

Notes: Figure plots number of SSDI exits over time from different data sources: authors' tabulations from the Continuous Work History Sample (CWHS) and publicly available data from the Social Security Administration. We identify exits in the CWHS by linking them to the Payment History Update System (PHUS) and the Master Beneficiary Record, or by determining from the Numident that they had either died. The solid line plots the number of exits based on the CWHS. The dashed line plots number of exits from the SSA Annual Statistical Report in each year.

Appendix Figure D.6: CWHS vs public data: SSDI exits from retirement



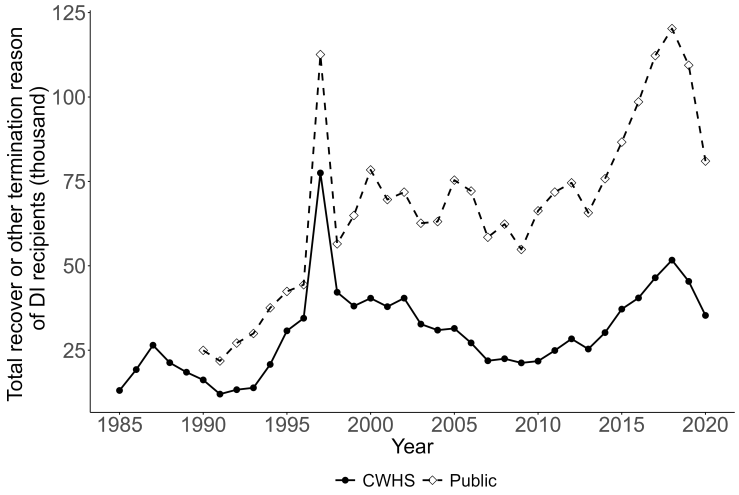

Notes: Figure plots number of SSDI exits due to retirement over time from different data sources: authors' tabulations from the Continuous Work History Sample (CWHS) and publicly available data from the Social Security Administration. We identify exits due to retirement in the CWHS by linking them to the Payment History Update System (PHUS) and the Master Beneficiary Record. The solid line plots the number of exits due to retirement based on the CWHS. The dashed line plots the number of exits due to retirement from the SSA Annual Statistical Report in each year.

Appendix Figure D.7: CWHS vs public data: SSDI exits from death



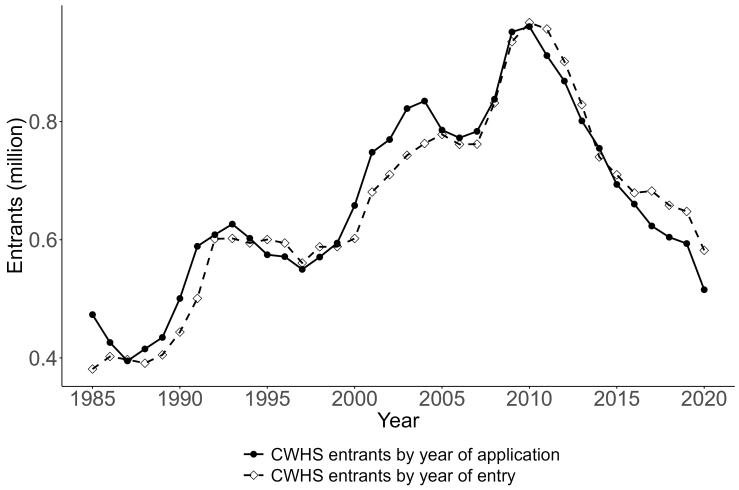

Notes: Figure plots number of SSDI exits due to death over time from different data sources: authors' tabulations from the Continuous Work History Sample (CWHS) and publicly available data from the Social Security Administration. We identify exits due to death in the CWHS by linking them to date of death in the Numident. The solid line plots number of exits due to death based on the CWHS. The dashed line plots number of exits due to death from the SSA Annual Statistical Report in each year.

Appendix Figure D.8: CWHS vs public data: SSDI exits from recovery



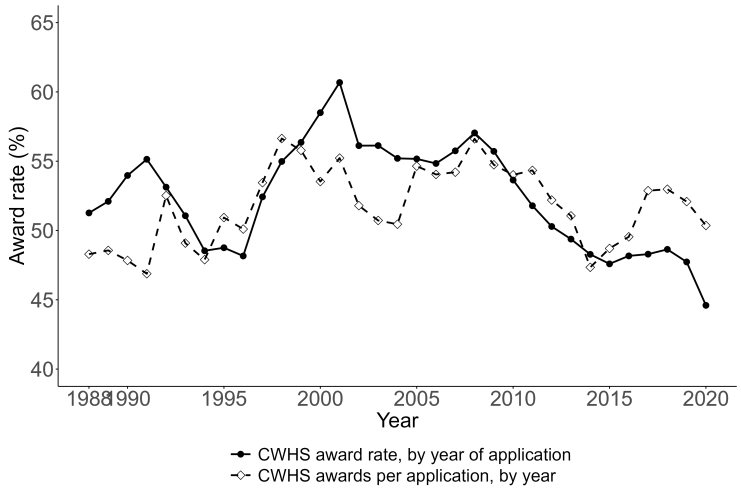

Notes: Figure plots number of SSDI exits due to recovery over time from different data sources: authors' tabulations from the Continuous Work History Sample (CWHS) and publicly available data from the Social Security Administration. We identify exits due to death in the CWHS by linking them by linking them to the Payment History Update System (PHUS) and the Master Beneficiary Record. The solid line plots the number of exits due to medical recovery or other reasons based on the CWHS. The dashed line plots the number of exits due to all non-retirement and death reason from the SSA Annual Statistical Report in each year.

Appendix Figure D.9: CWHS SSDI entrants, by year of entry vs year of application




Notes: The dashed line plots the number of people who enter SSDI in each year, regardless of when the application was submitted. The solid line plots the number of SSDI applications submitted in each year that ultimately lead to SSDI receipt. These differ because applications may not receive an allowance decision in the same calendar year in which they are submitted. Both series are calculated using the Continuous Work History Sample (CWHS) linked to the F831, PHUS, and MBR.

Appendix Figure D.10: CWHS SSDI annual awards per application vs. annual award rates




Notes: The dashed line plots the number of SSDI awards granted in each year (regardless of when the application was submitted), divided by the number of applications received in that year. The solid line plots the eventual award rate for applications submitted in each year. These differ because applications may not receive an allowance decision in the same calendar year in which they are submitted. Both series are calculated using the Continuous Work History Sample (CWHS) linked to the F831, PHUS, and MBR.

# E Decomposition details

In this section, we walk through the decomposition exercise in more detail. The exact specifications outlined here form the basis of the point estimates reported in the "Used in graph" column in Tables 1 and 2 and Figures A.1 and 3.

## Recursive structure of SSDI enrollment rates

DI enrollment rate ("on DI") $O$ is defined to be SSDI stock $S$ divided by population $P$.

$$O_{t,g} = \frac{S_{t,g}}{P_{t,g}}$$

DI stock is a recursive process that depends on past SSDI stock, past population, entry rates ($N$), and exit rates ($X$) and SSDI entrants that are immigrants ($\alpha$):

$$S_{t,g} = S_{t-1,g} + \alpha_{t,g} + N_{t,g}(P_{t-1,g} - S_{t-1,g}) - X_{t,g}S_{t-1,g} \tag{26}$$

Population is also a recursive process that depends on past population, past SSDI stock, immigration ($\alpha$ SSDI-entering immigrants; $\beta$ non-SSDI-entering immigrants), and death rates ($\gamma$, for non-DI beneficiaries; $\delta$, for SSDI beneficiaries).[39]

$$P_{t,g} = P_{t-1,g} + (\alpha_{t,g} + \beta_{t,g}) - \underbrace{\gamma_t(P_{t-1,g} - S_{t-1,g})}_{\text{death of non-DI beneficiaries}} - \underbrace{\delta_{t,g}S_{t-1,g}}_{\text{death of SSDI beneficiaries}} \tag{27}$$

We compute the actual SSDI stock using the recursive accounting equation (26) and (27) with contemporaneous entry and exit rates. The counterfactual SSDI stock is obtained by applying the base-year entry and exit rates to the same equations.[40] For both series, we initialize the recursion by setting $S^c_{0,g} = S^r_{0,g} =$ SSDI stock in the base year and $P^c_{0,g} = P^r_{0,g} =$ base-year population.

$$S^r_{t,g} = S^r_{t-1,g} + \alpha^r_{t,g} + N^r_{t,g}(P^r_{t-1,g} - S^r_{t-1,g}) - X^r_{t,g}S^r_{t-1,g}$$
$$S^c_{t,g} = S^c_{t-1,g} + \alpha^r_{t,g} + N^c_{t,g}(P^c_{t-1,g} - S^c_{t-1,g}) - X^c_{t,g}S^c_{t-1,g}$$

[39] In equation (8), we refer to $\delta_{t,g}$ as Mortality Rate$_{t,g}$. We change notation here for brevity.

[40] We set $\alpha$ to be the contemporaneous value in both series to simplify the calculation. The actual value of $\alpha$ is minimal.

We compute the actual population by using contemporaneous immigration and death rates. For the counterfactual population series, since population dynamics are not the focus of this analysis, we hold fixed only the SSDI stock $S$ and the death rate for SSDI beneficiaries $\delta$ at their base-year levels while allowing all other components to follow their actual contemporaneous values.[41] Since the number of SSDI beneficiary deaths is minimal, $P^c_{t,g} \approx P^r_{t,g}$.

$$P^r_{t,g} = P^r_{t-1,g} + \underbrace{(\alpha^r_{t,g} + \beta^r_{t,g})}_{\text{Immigrants}} - \underbrace{\gamma^r_t(P^r_{t-1,g} - S^r_{t-1,g})}_{\text{death of non-DI beneficiaries}} - \underbrace{\delta^r_{t,g} S^r_{t-1,g}}_{\text{death of SSDI beneficiaries}}$$
$$P^c_{t,g} = P^c_{t-1,g} + \underbrace{(\alpha^r_{t,g} + \beta^r_{t,g})}_{\text{Immigrants}} - \underbrace{\gamma^r_t(P^c_{t-1,g} - S^c_{t-1,g})}_{\text{death of non-DI beneficiaries}} - \underbrace{\delta^c_{t,g} S^c_{t-1,g}}_{\text{death of SSDI beneficiaries}}$$

## Step 1: Breaking down overall SSDI enrollment rate

Overall SSDI rate $O_t$ can be written as

$$O_t = \frac{S_t}{P_t} = \frac{\sum_g S_{t,g}}{P_t} = \sum_g \underbrace{\frac{S_{t,g}}{P_{t,g}}}_{O_{t,g}} \underbrace{\frac{P_{t,g}}{P_t}}_{w_{t,g}} = \sum_g O_{t,g} w_{t,g}$$

We decompose the difference between the actual and counterfactual overall SSDI rate into three components: the change in characteristics, the change in conditional SSDI rates, and their interaction.

$$\begin{aligned} O^r_t - O^c_t = & \underbrace{\sum_g O^c_{t,g}(w^r_{t,g} - w^c_{t,g})}_{\text{change in characteristics}} + \underbrace{\sum_g (O^r_{t,g} - O^c_{t,g}) w^c_{t,g}}_{\text{change from conditional SSDI rates}} \\ & + \underbrace{\sum_g (O^r_{t,g} - O^c_{t,g})(w^r_{t,g} - w^c_{t,g})}_{\text{characteristics} \times \text{SSDI rate interaction}} \end{aligned} \tag{28}$$

[41] For both series, we initialize the recursion by setting $S^c_{0,g} = S^r_{0,g}$ = SSDI stock in the base year and $P^c_{0,g} = P^r_{0,g}$ = base-year population.

## Step 2: Breaking down group-specific SSDI enrollment rate

For the first period,

$$
\begin{aligned}
O^r_{1,g} - O^c_{1,g} &= \frac{S^r_{1,g}}{P^r_{1,g}} - \frac{S^c_{1,g}}{P^c_{1,g}} \\
&= \frac{S^r_{1,g} - S^c_{1,g}}{P^c_{1,g}} + \underbrace{S^r_{1,g}(\frac{1}{P^r_{1,g}} - \frac{1}{P^c_{1,g}})}_{\text{Friction in population}} \\
&= \underbrace{\frac{S^r_{0,g} - S^c_{0,g}}{P^c_{1,g}}}_{0} + \underbrace{(N^r_{1,g} - N^c_{1,g})\frac{(P_{0,g} - S_{0,g})}{P^c_{1,g}}}_{\text{change from conditional entry rate}} \underbrace{-(X^r_{1,g} - X^c_{1,g})\frac{S_{0,g}}{P^c_{1,g}}}_{\text{change from conditional exit rate}} + \underbrace{S^r_{1,g}(\frac{1}{P^r_{1,g}} - \frac{1}{P^c_{1,g}})}_{\text{Friction in population}} \\
&\approx \underbrace{(N^r_{1,g} - N^c_{1,g})\frac{(P_{0,g} - S_{0,g})}{P^c_{1,g}}}_{\text{change from conditional entry rate}} \underbrace{-(X^r_{1,g} - X^c_{1,g})\frac{S_{0,g}}{P^c_{1,g}}}_{\text{change from conditional exit rate}}
\end{aligned}
$$

For other periods,

$$
\begin{aligned}
O^r_{t,g} - O^c_{t,g} =& \frac{S^r_{t,g}}{P^r_{t,g}} - \frac{S^c_{t,g}}{P^c_{t,g}} \\
=& \frac{S^r_{t,g} - S^c_{t,g}}{P^c_{t,g}} + \underbrace{S^r_{t,g}(\frac{1}{P^r_{t,g}} - \frac{1}{P^c_{t,g}})}_{\text{Friction in population}}
\end{aligned} \tag{29}
$$

Where

$$
\begin{aligned}
S^r_{t,g} - S^c_{t,g} =& (S^r_{t-1,g} + \alpha^r_{t,g} + N^r_{t,g}(P^r_{t-1,g} - S^r_{t-1,g}) - X^r_{t,g}S^r_{t-1,g}) \\
&- (S^c_{t-1,g} + \alpha^r_{t,g} + N^c_{t,g}(P^c_{t-1,g} - S^c_{t-1,g}) - X^c_{t,g}S^c_{t-1,g}) \\
=& (S^r_{t-1,g} - S^c_{t-1,g}) \\
&+ N^c_{t,g}[(P^r_{t-1,g} - S^r_{t-1,g}) - (P^c_{t-1,g} - S^c_{t-1,g})] + [N^r_{t,g} - N^c_{t,g}](P^c_{t-1,g} - S^c_{t-1,g}) \\
&+ [N^r_{t,g} - N^c_{t,g}][(P^r_{t-1,g} - S^r_{t-1,g}) - (P^c_{t-1,g} - S^c_{t-1,g})] \\
&- X^c_{t,g}[S^r_{t-1,g} - S^c_{t-1,g}] - [X^r_{t,g} - X^c_{t,g}]S^c_{t-1,g} - [X^r_{t,g} - X^c_{t,g}][S^r_{t-1,g} - S^c_{t-1,g}] \\
\approx& \underbrace{(N^r_{t,g} - N^c_{t,g})(P^c_{t,g} - S^c_{t,g})}_{a_{t,g}\text{: change in conditional entry}} \quad \underbrace{-(X^r_{t,g} - X^c_{t,g})S^c_{t,g}}_{b_{t,g}\text{: change in conditional exit}} \\
&+ \underbrace{(1 - N^c_{t,g} - X^c_{t,g})(S^r_{t,g} - S^c_{t,g})}_{c_{t,g}\text{: change in past conditional SSDI rate}} \\
&+ \underbrace{(N^c_{t,g} - N^r_{t,g})(S^r_{t,g} - S^c_{t,g})}_{d_{t,g}\text{: entry} \times \text{past SSDI rate interaction}} \quad \underbrace{-(X^r_{t,g} - X^c_{t,g})(S^r_{t,g} - S^c_{t,g})}_{e_{t,g}\text{: exit} \times \text{past SSDI rate interaction}}
\end{aligned}
$$

The approximation comes from the fact that $P^c_{t,g} \approx P^r_{t,g}$. If we ignore the approximation and aggregate all of the interaction terms together, we get:

$$
\begin{aligned}
S^r_{t+1,g} - S^c_{t+1,g} = & \underbrace{(N^r_{t,g} - N^c_{t,g})(P^c_{t-1,g} - S^c_{t-1,g})}_{\text{change from conditional entry rate}} \\
& \underbrace{-(X^r_{t,g} - X^c_{t,g})S^c_{t-1,g}}_{\text{change from conditional exit rate}} \\
& \underbrace{+N^r_{t,g}(P^r_{t-1,g} - P^c_{t-1,g})}_{\text{change from population discrepancy}} \\
& \underbrace{+(1 - N^r_{t,g} - X^r_{t,g})}_{\equiv K^r_{g,t}}(S^c_{t-1,g} - S^c_{t-1,g}) \\
= & \underbrace{\sum_{i=0}^{t}(N^r_{t-i,g} - N^c_{t-i,g})(P^c_{t-i,g} - S^c_{t-i,g})K^r_{g,i,t}}_{\text{cumulative change from conditional entry rate}} \\
& \underbrace{-\sum_{i=0}^{t}(X^r_{t-i,g} - X^c_{t-i,g})S^c_{t-i,g}K^r_{g,i,t}}_{\text{cumulative change from conditional exit rate}} \\
& \underbrace{+\sum_{i=0}^{t}\left[N^r_{t-i,g}(P^r_{t-i,g} - P^c_{t-i,g})\right]K^r_{g,i,t}}_{\text{cumulative change from pop. discrepancies} \approx 0}
\end{aligned}
\tag{30}
$$

## Step 3: Further decompose entry and exit

The entry rate ($N$) is the product of award rate ($B$), application rate ($A$) and eligibility rate ($E$). Thus we can decompose the difference betweeen the actual and counterfactual entry rate:

$$
\begin{aligned}
N^r_{t,g} - N^c_{t,g} =& B^r_{t,g}A^r_{t,g}E^r_{t,g} - B^c_{t,g}A^c_{t,g}E^c_{t,g} \\
=& \underbrace{(B^r_{t,g} - B^c_{t,g})A^c_{t,g}E^c_{t,g}}_{\text{partial award}} + \underbrace{B^c_{t,g}(A^r_{t,g} - A^c_{t,g})E^c_{t,g}}_{\text{partial apply}} + \underbrace{B^c_{t,g}A^c_{t,g}(E^r_{t,g} - E^c_{t,g})}_{\text{partial elig}} \\
&+ \underbrace{B^c_{t,g}(A^r_{t,g} - A^c_{t,g})(E^r_{t,g} - E^c_{t,g})}_{\text{apply} \times \text{elig}} \\
&+ \underbrace{(B^r_{t,g} - B^c_{t,g})A^c_{t,g}(E^r_{t,g} - E^c_{t,g})}_{\text{award} \times \text{elig}} \\
&+ \underbrace{(B^r_{t,g} - B^c_{t,g})(A^r_{t,g} - A^c_{t,g})E^c_{t,g}}_{\text{award} \times \text{apply}} \\
&+ \underbrace{(B^r_{t,g} - B^c_{t,g})(A^r_{t,g} - A^c_{t,g})(E^r_{t,g} - E^c_{t,g})}_{\text{award} \times \text{apply} \times \text{elig}}
\end{aligned} \tag{31}
$$

where the last four terms sum up to the interaction within entry. The exit rate ($X$) is the aggregation of exit rates arising from different reasons (retirement, death, recovery, and other exits). Therefore, these components are additively separable.

Substituting equation (31) into (30), (29) and (28), we can calculate the absolute contribution we report in the "Used in graph" column in Tables 1 and 2 and Figures A.1 and 3. For instance, the contribution from applications is defined to be:

$$
w^c_{g,t}\left[\frac{1}{P^c_{t,g}}\sum_{i=0}^{t} B^c_{t-i,g}(A^r_{t-i,g} - A^c_{t-i,g})E^c_{t-i,g}(P^c_{t-i,g} - S^c_{t-i,g})K^r_{g,i,t}\right] \tag{32}
$$

Similarly, the contribution from award is:

$$
w^c_{g,t}\left[\frac{1}{P^c_{t,g}}\sum_{i=0}^{t} (B^r_{t-i,g} - B^c_{t-i,g})A^c_{t-i,g}E^c_{t-i,g}(P^c_{t-i,g} - S^c_{t-i,g})K^r_{g,i,t}\right] \tag{33}
$$

The contribution from eligibility is:

$$
w^c_{g,t}\left[\frac{1}{P^c_{t,g}}\sum_{i=0}^{t} B^c_{t-i,g}A^c_{t-i,g}(E^r_{t-i,g} - E^c_{t-i,g})(P^c_{t-i,g} - S^c_{t-i,g})K^r_{g,i,t}\right] \tag{34}
$$

The contribution from entry is:

$$w_{g,t}^{c}\left[\frac{1}{P_{t,g}^{c}}\sum_{i=0}^{t}(N_{t-i,g}^{r}-N_{t-i,g}^{c})(P_{t-i,g}^{c}-S_{t-i,g}^{c})K_{g,i,t}^{r}\right] \tag{35}$$

The contribution from exit due to death is:

$$w_{g,t}^{c}\left[\frac{1}{P_{t,g}^{c}}\sum_{i=0}^{t}(\delta_{t-i,g}^{r}-\delta_{t-i,g}^{c})S_{t-i,g}^{c}K_{g,i,t}^{r}\right] \tag{36}$$

# F Bounds for the decomposition estimates

## Choice in $O^r_{t,g} - O^c_{t,g}$

As we break down $O^r_{t,g} - O^c_{t,g}$, we make several choices about how to allocate interaction terms. These choices shape the definition of our "partial effects" of interest (e.g., the cumulative impact of changes in entry on SSDI rates), both directly and indirectly through the way stocks are allowed to accumulate over time (as determined by $K$). To account for potential differences arising from these choices, we construct bounds that encompass all possible formulations of partial effects and discounting. The resulting bounds are reported in Tables 1 and 2.

### Bounds on Components of Entry

Recall we defined in equation (10) the cumulative impact of changes in application rates, according to a specific variant of our decomposition that fixes everything else at its counterfactual (c). The general expression for the cumulative impacts of awards, applications, and entry–accounting for how interaction terms can be differently allocated–are as follows:

$$\Delta O_{B_{t,g}}(\Theta) = \frac{1}{P^{s_P}} \sum_{i=0}^{t} \underbrace{(B^r_{t-i,g} - B^c_{t-i,g}) A^{s^A_B}_{t-i,g} E^{s^E_B}_{t-i,g}}_{\Delta B_{t-i,g}(\Theta)} (P^{s^B_B}_{t-i,g} - S^{s^B_B}_{t-i,g}) K_{g,i,t}(\Theta)$$

$$\Delta O_{A_{t,g}}(\Theta) = \frac{1}{P^{s_P}} \sum_{i=0}^{t} \underbrace{B^{s^B_A}_{t-i,g} (A^r_{t-i,g} - A^c_{t-i,g}) E^{s^E_A}_{t-i,g}}_{\Delta A_{t-i,g}(\Theta)} (P^{s^A_A}_{t-i,g} - S^{s^A_A}_{t-i,g}) K_{g,i,t}(\Theta)$$

$$\Delta O_{E_{t,g}}(\Theta) = \frac{1}{P^{s_P}} \sum_{i=0}^{t} \underbrace{B^{s^B_E}_{t-i,g} A^{s^A_E}_{t-i,g} (E^r_{t-i,g} - E^c_{t-i,g})}_{\Delta E_{t-i,g}(\Theta)} (P^{s^E_E}_{t-i,g} - S^{s^E_E}_{t-i,g}) K_{g,i,t}(\Theta)$$

where $\Theta$ denotes a vector containing a total of 15 elements describing how different components of the stock process are held fixed (i.e., how interaction terms are allocated). Each of these elements can take on counterfactual ($c$) or actually observed ($r$) values, implying a total of $2^{15}$ different ways to define these cumulative impacts.

The first element of $\Theta$ ($s_P$) represents how we fix the population in the denominator of the enrollment rate (e.g., in equation (29)). The next 9 elements ($s^y_x$ for $x$ and $y \in \{B, A, E\}$) represent how the other determinants of entry are being fixed when measuring the impact of changes in eligibility, applications, and awards. For each measure, there are three such determinants: the SSDI stock and the remaining two steps of the entry process (e.g., to

measure the impact of changes in application rates, we must decide how to fix existing SSDI stocks, eligibility rates, and award rates).[42] The 11th element ($s_I$) represents how we fix SSDI stocks and population counts when measuring the impact of interactions between the three steps of the entry process . The final 4 elements represent how we fix SSDI stocks when measuring the impact of the 4 alternative causes of exits (retirement, mortality, recovery, and other exits).

Because entry is a non-linear function of awards, applications, and eligibility, aggregating up to measure the impact of changing *entry* requires summing up these three terms along with the *interaction term* which describes the impact of their joint changes. This interaction term can be expressed as:

$$\begin{aligned}\Delta O_{(B\times A\times E)_{t,g}}(\Theta) =& \frac{1}{P^{s_P}}\sum_{i=0}^{t}\left[(N^r_{t-i,g}-N^c_{t-i,g})\right](P^{s_I}_{t-i,g}-S^{s_I}_{t-i,g})K_{g,i,t}(\Theta)\\ &-\frac{1}{P^{s_P}}\sum_{i=0}^{t}\left[\Delta B_{t-i,g}(\Theta)+\Delta A_{t-i,g}(\Theta)+\Delta E_{t-i,g}(\Theta)\right](P^{s_I}_{t-i,g}-S^{s_I}_{t-i,g})K_{g,i,t}(\Theta)\end{aligned} \tag{37}$$

For a given choice of $\Theta$, aggregating up $\Delta O_{(B\times A\times E)_{t,g}}+\Delta O_{B_{t,g}}+\Delta O_{A_{t,g}}+\Delta O_{E_{t,g}}$ gives a measure of the cumulative impact of SSDI entry. With total exits being a sum of cause-specific exits per equation (8), expressions for the cumulative impacts of cause-specific exits are:

$$\Delta O_{\text{Ret. Rate}_{t,g}}(\Theta) = \frac{1}{P^{s_P}}\sum_{i=0}^{t}(\text{Ret. Rate}^r_{t,g}-\text{Ret. Rate}^c_{t,g})S^{s_R}_{t-i,g}K_{g,i,t}(\Theta)$$

$$\Delta O_{\text{Mort. Rate}_{t,g}}(\Theta) = \frac{1}{P^{s_P}}\sum_{i=0}^{t}(\text{Mort. Rate}^r_{t,g}-\text{Mort. Rate}^c_{t,g})S^{s_M}_{t-i,g}K_{g,i,t}(\Theta)$$

$$\Delta O_{\text{Recov. Rate}_{t,g}}(\Theta) = \frac{1}{P^{s_P}}\sum_{i=0}^{t}(\text{Recov. Rate}^r_{t,g}-\text{Recov. Rate}^c_{t,g})S^{s_H}_{t-i,g}K_{g,i,t}(\Theta)$$

$$\Delta O_{\text{Other Rate}_{t,g}}(\Theta) = \frac{1}{P^{s_P}}\sum_{i=0}^{t}(\text{Other Rate}^r_{t,g}-\text{Other Rate}^c_{t,g})S^{s_O}_{t-i,g}K_{g,i,t}(\Theta)$$

Each of the 15 binary choices available when deciding how to fix determinants of the

[42]The elements $s^y_x$ for which $x=y$ control how the SSDI stock and population count are fixed when measuring the cumulative impact of changes in $x$.

stock accounting equation will affect the specification of $K_{g,i,t}$, which can we can write as:

$$K_{g,i,t}(\Theta) \equiv \begin{cases} \prod_{j=1}^{i} k_{t-j+1,g}(\Theta) & \text{if } i \geq 1 \\ 1 & \text{if } i = 0 \end{cases}$$

where $k_{t,g}(\Theta)$ is the value which makes the following accounting identity hold:

$$\begin{aligned} S^r_{t+1,g} - S^c_{t+1,g} =& \Delta B_{t,g}(\Theta)(P^{s^B_B}_{t,g} - S^{s^B_B}_{t,g}) + \Delta A_{t,g}(\Theta)(P^{s^A_A}_{t,g} - S^{s^A_A}_{t,g}) + \Delta E_{t,g}(\Theta)(P^{s^E_E}_{t,g} - S^{s^E_E}_{t,g}) \\ &+[(N^r_{t,g} - N^c_{t,g}) - \Delta B_{t,g}(\Theta) - \Delta A_{t,g}(\Theta) - \Delta E_{t,g}(\Theta)](P^{s_I}_{t,g} - S^{s_I}_{t,g}) \\ &-\Delta\text{Ret. Rate}_{t,g}(\Theta)S^{s_R}_{t,g} - \Delta\text{Mort. Rate}_{t,g}(\Theta)S^{s_M}_{t,g} \\ &-\Delta\text{Recov. Rate}_{t,g}(\Theta)S^{s_H}_{t,g} - \Delta\text{Other Rate}_{t,g}(\Theta)S^{s_O}_{t,g} \\ &+\text{Population Discrepancy}_{t,g}(\Theta) \\ &+k_{t,g}(\Theta)(S^r_{t,g} - S^c_{t,g}) \end{aligned} \tag{38}$$

where Population Discrepancy$_{t,g}(\Theta)$ is the gap between actually observed and counterfactual application counts not accounted for by changes we attribute to awards, applications, eligibility, or the interactions within entry:

$$\begin{aligned} N^r_{t,g}P^r_{t,g} - N^c_{t,g}P^c_{t,g} =& \Delta B_{t,g}(\Theta)P^{s^B_B}_{t,g} + \Delta A_{t,g}(\Theta)P^{s^A_A}_{t,g} + \Delta E_{t,g}(\Theta)P^{s^E_E}_{t,g} \\ &+[(N^r_{t,g} - N^c_{t,g}) - \Delta B_{t,g}(\Theta) - A_{t,g}(\Theta) - E_{t,g}(\Theta)]P^{s_I}_{t,g} \\ &+\text{Population Discrepancy}_{t,g}(\Theta) \end{aligned}$$

To illustrate the exact expression for $K$ and the population discrepancy and how they depend on the choice of $\Theta$, consider the examples at the end of this section. Note that we add some notation to our labels for the "partial effects" in these decompositions, to make clearer how determinants of the decomposition are being held fixed in each example.

**Example 1: Our baseline specification, used for reporting point estimates, fixing everything at counterfactual ($c$) values.**

$$
\begin{aligned}
S^r_{g,t+1} - S^c_{g,t+1} = & \underbrace{(N^r_{g,t} - N^c_{g,t})(P^c_{g,t} - S^c_{g,t})}_{\text{Partial Entry}} \underbrace{-(X^r_{g,t} - X^c_{g,t})S^c_{g,t}}_{\text{Partial Exit}} \\
& + \underbrace{N^r_{g,t}(P^r_{g,t} - P^c_{g,t})}_{\text{Population Discrepancy}} + \underbrace{(1 - N^r_{g,t} - X^r_{g,t})}_{k^r}(S^r_{g,t} - S^c_{g,t}) \\
& = \underbrace{(B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t}(P^c_{g,t} - S^c_{g,t})}_{\text{Partial Award (c,c,c)}} \\
& + \underbrace{B^c_{g,t}(A^r_{g,t} - A^c_{g,t})E^c_{g,t}(P^c_{g,t} - S^c_{g,t})}_{\text{Partial Apply (c,c,c)}} \\
& + \underbrace{B^c_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t})(P^c_{g,t} - S^c_{g,t})}_{\text{Partial Elig (c,c,c)}} \\
& + \underbrace{((N^r_{g,t} - N^c_{g,t}) - (B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t} - B^c_{g,t}(A^r_{g,t} - A^c_{g,t})E^c_{g,t} - B^c_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t}))(P^c_{g,t} - S^c_{g,t})}_{\text{Interaction (c)}} \\
& \underbrace{-(X^r_{g,t} - X^c_{g,t})S^c_{g,t}}_{\text{Partial Exit (c,c,c,c)}} \\
& + \underbrace{N^r_{g,t}(P^r_{g,t} - P^c_{g,t})}_{\text{Population Discrepancy}} + \underbrace{(1 - N^r_{g,t} - X^r_{g,t})}_{k^r}(S^r_{g,t} - S^c_{g,t})
\end{aligned}
$$

**Example 2: Fixing SSDI stocks at actual ($r$) values when measuring impacts of entry or exit.** Relative to our baseline specification, fixing SSDI stocks (in blue text) at actual values requires that $K$ fix entry and exit rates at *counterfactual* values.

$$
\begin{aligned}
S^r_{g,t+1} - S^c_{g,t+1} = & \underbrace{(N^r_{g,t} - N^c_{g,t})(P^r_{g,t} - S^r_{g,t})}_{\text{Partial Entry}} \underbrace{-(X^r_{g,t} - X^c_{g,t})S^r_{g,t}}_{\text{Partial Exit}} \\
& + \underbrace{N^c_{g,t}(P^r_{g,t} - P^c_{g,t})}_{\text{Population Discrepancy}} + \underbrace{(1 - N^c_{g,t} - X^c_{g,t})}_{k^c}(S^r_{g,t} - S^c_{g,t}) \\
& = \underbrace{(B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t}(P^r_{g,t} - S^r_{g,t})}_{\text{Partial Award (c,c,r)}} \\
& + \underbrace{B^c_{g,t}(A^r_{g,t} - A^c_{g,t})E^c_{g,t}(P^r_{g,t} - S^r_{g,t})}_{\text{Partial Apply (c,c,r)}} \\
& + \underbrace{B^c_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t})(P^r_{g,t} - S^r_{g,t})}_{\text{Partial Elig (c,c,r)}} \\
& + \underbrace{((N^r_{g,t} - N^c_{g,t}) - (B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t} - B^c_{g,t}(A^r_{g,t} - A^c_{g,t})E^c_{g,t} - B^c_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t}))(P^r_{g,t} - S^r_{g,t})}_{\text{Interaction (r)}} \\
& \underbrace{-(X^r_{g,t} - X^c_{g,t})S^r_{g,t}}_{\text{Partial Exit (c,c,c,r)}} \\
& + \underbrace{N^c_{g,t}(P^r_{g,t} - P^c_{g,t})}_{\text{Population Discrepancy}} + \underbrace{(1 - N^c_{g,t} - X^c_{g,t})}_{k^c}(S^r_{g,t} - S^c_{g,t})
\end{aligned}
$$

**Example 3: Fixing SSDI stocks at counterfactual ($c$) values when measuring impacts of exit, actual values ($r$) when measuring impacts of entry.** Relative to the baseline specification, fixing SSDI stocks at actual values for measuring the partial effect

of entry rates implies that $K'$ must fix entry rates at *counterfactual* values.

$$
\begin{aligned}
S^r_{g,t+1} - S^c_{g,t+1} = & \underbrace{(N^r_{g,t} - N^c_{g,t})(P^r_{g,t} - S^r_{g,t})}_{\text{Partial Entry}} \underbrace{-(X^r_{g,t} - X^c_{g,t})S^r_{g,t}}_{\text{Partial Exit}} \\
& + \underbrace{N^c_{g,t}(P^r_{g,t} - P^c_{g,t})}_{\text{Population Discrepancy}} + \underbrace{(1 - N^c_{g,t} - X^c_{g,t})}_{k^c}(S^r_{g,t} - S^c_{g,t}) \\
& = \underbrace{(B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t}(P^r_{g,t} - S^r_{g,t})}_{\text{Partial Award (c,c,r)}} \\
& + \underbrace{B^c_{g,t}(A^r_{g,t} - A^c_{g,t})E^c_{g,t}(P^r_{g,t} - S^r_{g,t})}_{\text{Partial Apply (c,c,r)}} \\
& + \underbrace{B^c_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t})(P^r_{g,t} - S^r_{g,t})}_{\text{Partial Elig (c,c,r)}} \\
& + \underbrace{((N^r_{g,t} - N^c_{g,t}) - (B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t} - B^c_{g,t}(A^r_{g,t} - A^c_{g,t})E^c_{g,t} - B^c_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t}))(P^r_{g,t} - S^r_{g,t})}_{\text{Interaction (r)}} \\
& \underbrace{-(X^r_{g,t} - X^c_{g,t})S^c_{g,t}}_{\text{Partial Exit (c,c,c,c)}} \\
& + \underbrace{N^c_{g,t}(P^r_{g,t} - P^c_{g,t})}_{\text{Population Discrepancy}} + \underbrace{(1 - N^c_{g,t} - X^r_{g,t})}_{k'}(S^r_{g,t} - S^c_{g,t})
\end{aligned}
$$

**Example 4: Fixing SSDI stocks at counterfactual ($c$) values when measuring impacts of exit, apply, and interaction within entry, actual values ($r$) when measuring impacts of award and eligibility.** Using equation (38), allowing these differences

in defining partial effects within entry leads to some interaction terms appearing in $k$.

$$
\begin{aligned}
S^r_{g,t+1} - S^c_{g,t+1} = & \underbrace{(N^r_{g,t} - N^c_{g,t})(P^r_{g,t} - S^r_{g,t})}_{\text{Partial Entry}} \underbrace{-(X^r_{g,t} - X^c_{g,t})S^r_{g,t}}_{\text{Partial Exit}} \\
& + \underbrace{N^c_{g,t}(P^r_{g,t} - P^c_{g,t})}_{\text{Population Discrepancy}} + \underbrace{(1 - N^c_{g,t} - X^c_{g,t})}_{k^c}(S^r_{g,t} - S^c_{g,t}) \\
& = \underbrace{(B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t}(P^r_{g,t} - S^r_{g,t})}_{\text{Partial Award (c,c,r)}} \\
& + \underbrace{B^c_{g,t}(A^r_{g,t} - A^c_{g,t})E^c_{g,t}(P^c_{g,t} - S^c_{g,t})}_{\text{Partial Apply (c,c,c)}} \\
& + \underbrace{B^c_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t})(P^r_{g,t} - S^r_{g,t})}_{\text{Partial Elig (c,c,r)}} \\
& + \underbrace{((N^r_{g,t} - N^c_{g,t}) - (B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t} - B^c_{g,t}(A^r_{g,t} - A^c_{g,t})E^c_{g,t} - B^c_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t}))(P^c_{g,t} - S^c_{g,t})}_{\text{Interaction (c)}} \\
& \underbrace{-(X^r_{g,t} - X^c_{g,t})S^c_{g,t}}_{\text{Partial Exit (c,c,c,c)}} \\
& + \underbrace{N^r_{g,t}(P^r_{g,t} - P^c_{g,t})}_{\text{Population Discrepancy}} + \underbrace{(1 - N^r_{g,t} - X^r_{g,t})}_{k^r}(S^r_{g,t} - S^c_{g,t}) \\
& + \underbrace{(B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t}[(P^c_{g,t} - S^c_{g,t}) - (P^r_{g,t} - S^r_{g,t})]}_{\text{Partial Award (c,c,c) - Partial Award (c,c,r)}} \\
& + \underbrace{B^c_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t})[(P^c_{g,t} - S^c_{g,t}) - (P^r_{g,t} - S^r_{g,t})]}_{\text{Partial Elig (c,c,c) - Partial Elig (c,c,r)}} \\
& = \underbrace{(B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t}(P^r_{g,t} - S^r_{g,t})}_{\text{Partial Award (c,c,r)}} \\
& + \underbrace{B^c_{g,t}(A^r_{g,t} - A^c_{g,t})E^c_{g,t}(P^c_{g,t} - S^c_{g,t})}_{\text{Partial Apply (c,c,c)}} \\
& + \underbrace{B^c_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t})(P^r_{g,t} - S^r_{g,t})}_{\text{Partial Elig (c,c,r)}} \\
& + \underbrace{((N^r_{g,t} - N^c_{g,t}) - (B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t} - B^c_{g,t}(A^r_{g,t} - A^c_{g,t})E^c_{g,t} - B^c_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t}))(P^c_{g,t} - S^c_{g,t})}_{\text{Interaction (c)}} \\
& \underbrace{-(X^r_{g,t} - X^c_{g,t})S^c_{g,t}}_{\text{Partial Exit (c,c,c,c)}} \\
& + \underbrace{[N^r_{g,t} - ((B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t}) - (B^c_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t}))](P^r_{g,t} - P^c_{g,t})}_{\text{Population Discrepancy}} \\
& + \underbrace{[(1 - N^r_{g,t} - X^r_{g,t}) + ((B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t}) + (B^c_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t}))]}_{k''}(S^r_{g,t} - S^c_{g,t})
\end{aligned}
$$

**Example 5: When measuring impacts of components of entry, fixing some remaining components at counterfactual ($c$) values and others at actual ($r$) values.** Using equation (37), this affects how we define the impact of interactions within the entry

process.

$$
\begin{aligned}
S^r_{g,t+1} - S^c_{g,t+1} = & \underbrace{(N^r_{g,t} - N^c_{g,t})(P^c_{g,t} - S^c_{g,t})}_{\text{Partial Entry}} \underbrace{-(X^r_{g,t} - X^c_{g,t})S^c_{g,t}}_{\text{Partial Exit}} \\
& + \underbrace{N^r_{g,t}(P^r_{g,t} - P^c_{g,t})}_{\text{Population Discrepancy}} + \underbrace{(1 - N^r_{g,t} - X^r_{g,t})}_{k^r}(S^r_{g,t} - S^c_{g,t}) \\
& = \underbrace{(B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t}(P^c_{g,t} - S^c_{g,t})}_{\text{Partial Award (c,c,c)}} \\
& + \underbrace{B^r_{g,t}(A^r_{g,t} - A^c_{g,t})E^r_{g,t}(P^c_{g,t} - S^c_{g,t})}_{\text{Partial Apply (r,r,c)}} \\
& + \underbrace{B^r_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t})(P^c_{g,t} - S^c_{g,t})}_{\text{Partial Elig (r,c,c)}} \\
& + \underbrace{((N^r_{g,t} - N^c_{g,t}) - (B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t} - B^r_{g,t}(A^r_{g,t} - A^c_{g,t})E^r_{g,t} - B^r_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t}))(P^c_{g,t} - S^c_{g,t})}_{\text{Interaction (c)}} \\
& \underbrace{-(X^r_{g,t} - X^c_{g,t})S^c_{g,t}}_{\text{Partial Exit (c,c,c,c)}} \\
& + \underbrace{N^r_{g,t}(P^r_{g,t} - P^c_{g,t})}_{\text{Population Discrepancy}} + \underbrace{(1 - N^r_{g,t} - X^r_{g,t})}_{k^r}(S^r_{g,t} - S^c_{g,t})
\end{aligned}
$$

**Example 6: Fixing non-beneficiary populations at actual ($r$) values for the impact of applications, interactions within entry, and exit but not for the impact of awards or eligibility (mixture of example 4 and 5).** Using both equations (37) and (38), this affects how we define the impact of interactions within entry, which also has implications for the definition of $k$.

$$
\begin{aligned}
S^r_{g,t+1} - S^c_{g,t+1} &= \underbrace{(N^r_{g,t} - N^c_{g,t})(P^c_{g,t} - S^c_{g,t})}_{\text{Partial Entry}} \underbrace{-(X^r_{g,t} - X^c_{g,t})S^c_{g,t}}_{\text{Partial Exit}} \\
&+ \underbrace{N^r_{g,t}(P^r_{g,t} - P^c_{g,t})}_{\text{Population Discrepancy}} + \underbrace{(1 - N^r_{g,t} - X^r_{g,t})}_{k^r}(S^r_{g,t} - S^c_{g,t}) \\
&= \underbrace{(B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t}(P^r_{g,t} - S^r_{g,t})}_{\text{Partial Award (c,c,r)}} \\
&+ \underbrace{B^r_{g,t}(A^r_{g,t} - A^c_{g,t})E^r_{g,t}(P^c_{g,t} - S^c_{g,t})}_{\text{Partial Apply (r,r,c)}} \\
&+ \underbrace{B^r_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t})(P^r_{g,t} - S^r_{g,t})}_{\text{Partial Elig (r,c,r)}} \\
&+ \underbrace{((N^r_{g,t} - N^c_{g,t}) - (B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t} - B^r_{g,t}(A^r_{g,t} - A^c_{g,t})E^r_{g,t} - B^r_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t}))(P^c_{g,t} - S^c_{g,t})}_{\text{Interaction (c)}} \\
&\underbrace{-(X^r_{g,t} - X^c_{g,t})S^c_{g,t}}_{\text{Partial Exit (c,c,c,c)}} \\
&+ \underbrace{N^r_{g,t}(P^r_{g,t} - P^c_{g,t})}_{\text{Population Discrepancy}} + \underbrace{(1 - N^r_{g,t} - X^r_{g,t})}_{k^r}(S^r_{g,t} - S^c_{g,t}) \\
&+ (B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t}[(P^c_{g,t} - S^c_{g,t}) - (P^r_{g,t} - S^r_{g,t})] \\
&+ B^r_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t})[(P^c_{g,t} - S^c_{g,t}) - (P^r_{g,t} - S^r_{g,t})] \\
&= \underbrace{(B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t}(P^r_{g,t} - S^r_{g,t})}_{\text{Partial Award (c,c,r)}} \\
&+ \underbrace{B^r_{g,t}(A^r_{g,t} - A^c_{g,t})E^r_{g,t}(P^c_{g,t} - S^c_{g,t})}_{\text{Partial Apply (r,r,c)}} \\
&+ \underbrace{B^r_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t})(P^r_{g,t} - S^r_{g,t})}_{\text{Partial Elig (r,c,r)}} \\
&+ \underbrace{((N^r_{g,t} - N^c_{g,t}) - (B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t} - B^r_{g,t}(A^r_{g,t} - A^c_{g,t})E^r_{g,t} - B^r_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t}))(P^c_{g,t} - S^c_{g,t})}_{\text{Interaction (c)}} \\
&\underbrace{-(X^r_{g,t} - X^c_{g,t})S^c_{g,t}}_{\text{Partial Exit (c,c,c,c)}} \\
&+ \underbrace{[N^r_{g,t} - ((B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t}) - (B^r_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t}))](P^r_{g,t} - P^c_{g,t})}_{\text{Population Discrepancy}} \\
&+ \underbrace{[(1 - N^r_{g,t} - X^r_{g,t}) + ((B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t}) + (B^r_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t}))]}_{k'''}(S^r_{g,t} - S^c_{g,t})
\end{aligned}
$$

**Example 7: In exit, only evaluate impact of death among beneficiaries by fixing SSDI stocks at actual ($r$) values.**

$$
\begin{aligned}
S^r_{g,t+1} - S^c_{g,t+1} = & \underbrace{(N^r_{g,t} - N^c_{g,t})(P^c_{g,t} - S^c_{g,t})}_{\text{Partial Entry}} \underbrace{-(X^r_{g,t} - X^c_{g,t})S^c_{g,t}}_{\text{Partial Exit}} \\
& + \underbrace{N^r_{g,t}(P^r_{g,t} - P^c_{g,t})}_{\text{Population Discrepancy}} + \underbrace{(1 - N^r_{g,t} - X^r_{g,t})}_{k^r}(S^r_{g,t} - S^c_{g,t}) \\
& = \underbrace{(B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t}(P^c_{g,t} - S^c_{g,t})}_{\text{Partial Award (c,c,c)}} \\
& + \underbrace{B^c_{g,t}(A^r_{g,t} - A^c_{g,t})E^c_{g,t}(P^c_{g,t} - S^c_{g,t})}_{\text{Partial Apply (c,c,c)}} \\
& + \underbrace{B^c_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t})(P^c_{g,t} - S^c_{g,t})}_{\text{Partial Elig (c,c,c)}} \\
& + \underbrace{((N^r_{g,t} - N^c_{g,t}) - (B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t} - B^c_{g,t}(A^r_{g,t} - A^c_{g,t})E^c_{g,t} - B^c_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t}))(P^c_{g,t} - S^c_{g,t})}_{\text{Interaction (c)}} \\
& \underbrace{-(XRet^r_{g,t} - XRet^c_{g,t})S^c_{g,t} - (XMort^r_{g,t} - XMort^c_{g,t})S^r_{g,t} - (XRecov^r_{g,t} - XRecov^c_{g,t})S^c_{g,t} - (XOth^r_{g,t} - XOth^c_{g,t})S^c_{g,t}}_{\text{Partial Exit (c,r,c,c)}} \\
& + \underbrace{N^r_{g,t}(P^r_{g,t} - P^c_{g,t})}_{\text{Population Discrepancy}} + \underbrace{(1 - N^r_{g,t} - X^r_{g,t})}_{k^r}(S^r_{g,t} - S^c_{g,t}) \\
& + (XMort^r_{g,t} - XMort^c_{g,t})(S^r_{g,t} - S^c_{g,t}) \\
& = \underbrace{(B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t}(P^c_{g,t} - S^c_{g,t})}_{\text{Partial Award (c,c,c)}} \\
& + \underbrace{B^c_{g,t}(A^r_{g,t} - A^c_{g,t})E^c_{g,t}(P^c_{g,t} - S^c_{g,t})}_{\text{Partial Apply (c,c,c)}} \\
& + \underbrace{B^c_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t})(P^c_{g,t} - S^c_{g,t})}_{\text{Partial Elig (c,c,c)}} \\
& + \underbrace{((N^r_{g,t} - N^c_{g,t}) - (B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t} - B^c_{g,t}(A^r_{g,t} - A^c_{g,t})E^c_{g,t} - B^c_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t}))(P^c_{g,t} - S^c_{g,t})}_{\text{Interaction (c)}} \\
& \underbrace{-(XRet^r_{g,t} - XRet^c_{g,t})S^c_{g,t} - (XMort^r_{g,t} - XMort^c_{g,t})S^r_{g,t} - (XRecov^r_{g,t} - XRecov^c_{g,t})S^c_{g,t} - (XOth^r_{g,t} - XOth^c_{g,t})S^c_{g,t}}_{\text{Partial Exit (c,r,c,c)}} \\
& + \underbrace{N^r_{g,t}(P^r_{g,t} - P^c_{g,t})}_{\text{Population Discrepancy}} + \underbrace{(1 - N^r_{g,t} - XRet^r_{g,t} - XMort^r_{g,t} - XRecov^r_{g,t} - XOth^r_{g,t})}_{k''''}(S^r_{g,t} - S^c_{g,t})
\end{aligned}
$$

Where $X = XRet + XMort + XRecov + XOth$ comes from partitioning SSDI exit rates into the four cause-specific exit rates.

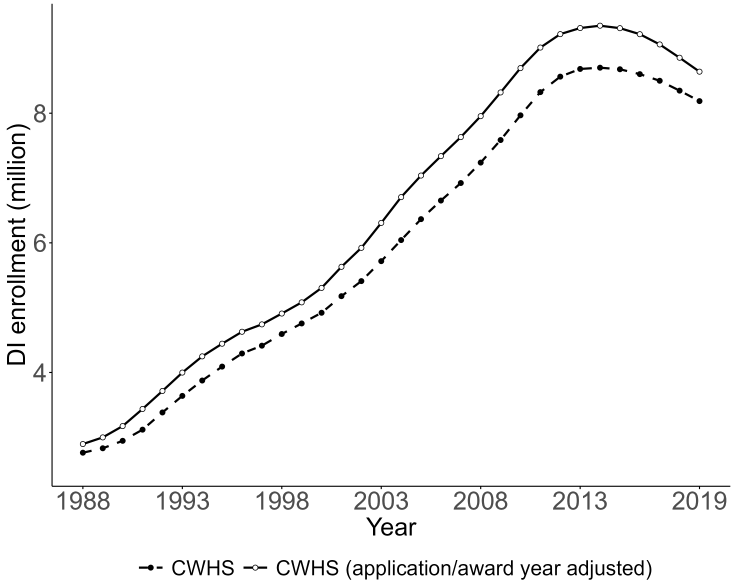
## Ordering of the components of $w^r_{t,g} - w^c_{t,g}$

Following the concept of equation (11), with four covariates (sex, age, earnings bin and birthplace), there are $4! = 24$ arrangements that go into calculating the bounds.

# G Population-level heterogeneity results

## G.1 By sex, age, and skill

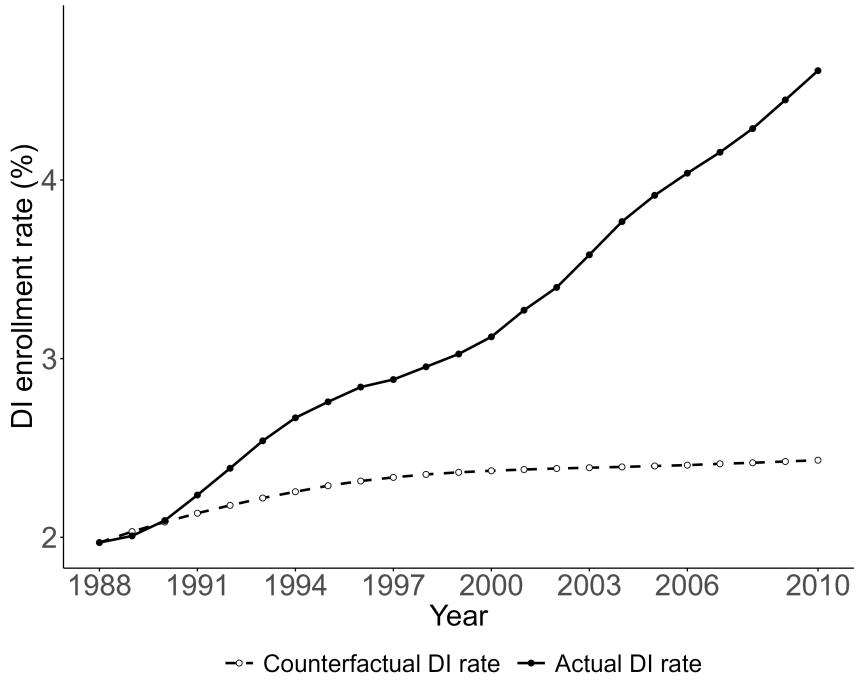
Appendix Figure G.1: Counterfactual vs actual SSDI enrollment level, 1988–2019




Notes: Figure plots actual and simulated SSDI enrollment levels in each year from 1988 to 2019. "CWHS" enrollment level uses the Continuous Work History Sample linked to the Payment History Update System and the Master Beneficiary Record. "CWHS (application/award year adjusted)" enrollment level includes the actual SSDI enrollment plus future awards as described in Section 1. In the decomposition results, we refer to the "CWHS (application/award year adjusted)" divided by population as the "Actual" SSDI rate, as in Appendix Figure G.2, Appendix Table G.1, Appendix Figure G.4, and Appendix Table G.9.

### G.1.1 SSDI increase period (1988–2010)

Appendix Figure G.2: Counterfactual vs actual SSDI enrollment rate, 1988–2010




Notes: Figure plots actual and counterfactual SSDI enrollment rates in each year from 1988 to 2010. Actual SSDI enrollment rate uses the Continuous Work History Sample linked to the Payment History Update System and the Master Beneficiary Record. Counterfactual SSDI enrollment rate uses equations (2), (3), and (4), using 1988 as the base year. The exact annual values are given in Appendix Table G.1.

Appendix Table G.1: Counterfactual vs actual SSDI enrollment rate, 1988–2010

| Year | Actual DI Rate (%) | Counterfactual DI Rate (%) | Total DI change (pp) |
|---|---|---|---|
| 1988 | 1.97 | 1.97 | 0.00 |
| 1989 | 2.01 | 2.03 | -0.02 |
| 1990 | 2.09 | 2.09 | 0.01 |
| 1991 | 2.24 | 2.13 | 0.10 |
| 1992 | 2.39 | 2.18 | 0.21 |
| 1993 | 2.54 | 2.22 | 0.32 |
| 1994 | 2.67 | 2.25 | 0.41 |
| 1995 | 2.76 | 2.29 | 0.47 |
| 1996 | 2.84 | 2.32 | 0.53 |
| 1997 | 2.88 | 2.34 | 0.55 |
| 1998 | 2.95 | 2.35 | 0.60 |
| 1999 | 3.03 | 2.36 | 0.66 |
| 2000 | 3.12 | 2.37 | 0.75 |
| 2001 | 3.27 | 2.38 | 0.89 |
| 2002 | 3.40 | 2.39 | 1.01 |
| 2003 | 3.58 | 2.39 | 1.19 |
| 2004 | 3.77 | 2.39 | 1.37 |
| 2005 | 3.91 | 2.40 | 1.52 |
| 2006 | 4.04 | 2.40 | 1.63 |
| 2007 | 4.16 | 2.41 | 1.74 |
| 2008 | 4.29 | 2.42 | 1.87 |
| 2009 | 4.45 | 2.42 | 2.02 |
| 2010 | 4.61 | 2.43 | 2.18 |

Notes: Table presents actual and counterfactual SSDI enrollment rates in each year from 1988 to 2010. Actual SSDI rate uses the Continuous Work History Sample linked to the Payment History Update System and the Master Beneficiary Record. Counterfactual SSDI rate uses equations (2), (3), and (4), using 1988 as the base year.

Appendix Table G.2: Demographic breakdown of entry share of SSDI enrollment rate, 1988–2010

| Group | Absolute contribution (pp) | Contribution share (%) | Baseline population share (%) | Baseline entrants share (%) | Contribution/ baseline entrants |
|---|---|---|---|---|---|
| Men, Younger, $10K+ | 0.00 | 0.06 | 22.15 | 12.15 | 0.01 |
| Men, Younger, $0-10K | 0.03 | 2.10 | 8.43 | 9.94 | 0.21 |
| Men, Older, $10K+ | 0.21 | 17.90 | 13.01 | 33.67 | 0.53 |
| Men, Older, $0-10K | 0.10 | 8.31 | 4.70 | 6.22 | 1.34 |
| Women, Younger, $10K+ | 0.04 | 3.52 | 13.34 | 4.38 | 0.80 |
| Women, Younger, $0-10K | 0.28 | 23.18 | 15.96 | 7.22 | 3.21 |
| Women, Older, $10K+ | 0.07 | 6.22 | 3.62 | 5.90 | 1.05 |
| Women, Older, $0-10K | 0.48 | 40.34 | 14.50 | 16.61 | 2.43 |
| Missing | -0.02 | -1.64 | 4.30 | 3.92 | -0.42 |
| Total | 1.20 | 100.00 | 100.00 | 100.00 | 1.00 |

Notes: Table presents the "entry" share (from Table 1) of the difference between actual and counterfactual SSDI rates for the increase period (1988–2010), broken down into contribution from demographic subgroups using equation (35). "Contribution share" is "absolute contribution" as a percent of the total change. "Baseline population share" is the share of the US population that the group accounts for in 1988. "Baseline entrants share" is the share of SSDI entrants that the group accounts for in 1988. "Contribution/baseline entrants" presents the ratio of "contribution share" to "baseline entrants share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the change. The demographic groups are characterized by sex, average annual earnings between ages 25-34 (a measure of skill), and whether older or younger than 45 years old.

Appendix Table G.3: Demographic breakdown of application share of SSDI enrollment rate, 1988–2010

| Group | Absolute contribution (pp) | Contribution share (%) | Baseline population share (%) | Baseline entrants share (%) | Contribution/ baseline entrants |
|---|---|---|---|---|---|
| Men, Younger, $10K+ | 0.00 | 0.24 | 22.15 | 12.15 | 0.02 |
| Men, Younger, $0-10K | 0.05 | 6.72 | 8.43 | 9.94 | 0.68 |
| Men, Older, $10K+ | 0.10 | 12.80 | 13.01 | 33.67 | 0.38 |
| Men, Older, $0-10K | 0.02 | 2.74 | 4.70 | 6.22 | 0.44 |
| Women, Younger, $10K+ | 0.04 | 5.52 | 13.34 | 4.38 | 1.26 |
| Women, Younger, $0-10K | 0.28 | 34.74 | 15.96 | 7.22 | 4.81 |
| Women, Older, $10K+ | 0.04 | 4.48 | 3.62 | 5.90 | 0.76 |
| Women, Older, $0-10K | 0.30 | 37.18 | 14.50 | 16.61 | 2.24 |
| Missing | -0.04 | -4.42 | 4.30 | 3.92 | -1.13 |
| Total | 0.80 | 100.00 | 100.00 | 100.00 | 1.00 |

Notes: Table presents the "application" share (from Table 1) of the difference between actual and counterfactual SSDI rates for the increase period (1988–2010), broken down into contribution from demographic subgroups using equation (10). "Contribution share" is "absolute contribution" as a percent of the total change. "Baseline population share" is the share of the US population that the group accounts for in 1988. "Baseline entrants share" is the share of SSDI entrants that the group accounts for in 1988. "Contribution/baseline entrants" presents the ratio of "contribution share" to "baseline entrants share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the change. The demographic groups are characterized by sex, average annual earnings between ages 25-34 (a measure of skill), and whether older or younger than 45 years old.

Appendix Table G.4: Demographic breakdown of award share of SSDI enrollment rate, 1988–2010

| Group | Absolute contribution (pp) | Contribution share (%) | Baseline population share (%) | Baseline entrants share (%) | Contribution/ baseline entrants |
|---|---|---|---|---|---|
| Men, Younger, $10K+ | -0.00 | -0.97 | 22.15 | 12.15 | -0.08 |
| Men, Younger, $0-10K | -0.02 | -12.78 | 8.43 | 9.94 | -1.29 |
| Men, Older, $10K+ | 0.09 | 52.95 | 13.01 | 33.67 | 1.57 |
| Men, Older, $0-10K | 0.03 | 17.65 | 4.70 | 6.22 | 2.84 |
| Women, Younger, $10K+ | -0.00 | -0.22 | 13.34 | 4.38 | -0.05 |
| Women, Younger, $0-10K | -0.02 | -11.71 | 15.96 | 7.22 | -1.62 |
| Women, Older, $10K+ | 0.02 | 11.10 | 3.62 | 5.90 | 1.88 |
| Women, Older, $0-10K | 0.08 | 45.76 | 14.50 | 16.61 | 2.76 |
| Missing | -0.00 | -1.78 | 4.30 | 3.92 | -0.45 |
| Total | 0.17 | 100.00 | 100.00 | 100.00 | 1.00 |

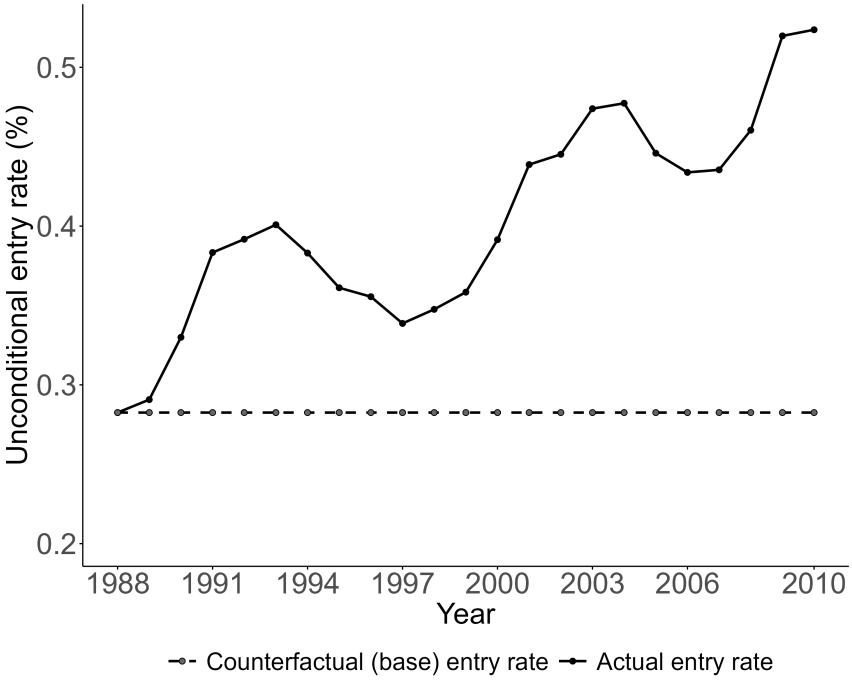
Notes: Table presents the "award" share (from Table 1) of the difference between actual and counterfactual SSDI rates for the increase period (1988–2010), broken down into contribution from demographic subgroups using equation (33). "Contribution share" is "absolute contribution" as a percent of the total change. "Baseline population share" is the share of the US population that the group accounts for in 1988. "Baseline entrants share" is the share of SSDI entrants that the group accounts for in 1988. "Contribution/baseline entrants" presents the ratio of "contribution share" to "baseline entrants share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the change. The demographic groups are characterized by sex, average annual earnings between ages 25-34 (a measure of skill), and whether older or younger than 45 years old.

Appendix Table G.5: Demographic breakdown of eligibility share of SSDI enrollment rate, 1988–2010

| Group | Absolute contribution (pp) | Contribution share (%) | Baseline population share (%) | Baseline entrants share (%) | Contribution/ baseline entrants |
|---|---|---|---|---|---|
| Men, Younger, $10K+ | 0.00 | 1.30 | 22.15 | 12.15 | 0.11 |
| Men, Younger, $0-10K | -0.00 | -0.14 | 8.43 | 9.94 | -0.01 |
| Men, Older, $10K+ | 0.01 | 6.88 | 13.01 | 33.67 | 0.20 |
| Men, Older, $0-10K | 0.03 | 21.30 | 4.70 | 6.22 | 3.43 |
| Women, Younger, $10K+ | 0.00 | 0.23 | 13.34 | 4.38 | 0.05 |
| Women, Younger, $0-10K | 0.03 | 17.94 | 15.96 | 7.22 | 2.48 |
| Women, Older, $10K+ | 0.01 | 7.71 | 3.62 | 5.90 | 1.31 |
| Women, Older, $0-10K | 0.06 | 36.14 | 14.50 | 16.61 | 2.18 |
| Missing | 0.01 | 8.64 | 4.30 | 3.92 | 2.20 |
| Total | 0.15 | 100.00 | 100.00 | 100.00 | 1.00 |

Notes: Table presents the "eligibility" share (from Table 1) of the difference between actual and counterfactual SSDI rates for the increase period (1988–2010), broken down into contribution from demographic subgroups using equation (34). "Contribution share" is "absolute contribution" as a percent of the total change. "Baseline population share" is the share of the US population that the group accounts for in 1988. "Baseline entrants share" is the share of SSDI entrants that the group accounts for in 1988. "Contribution/baseline entrants" presents the ratio of "contribution share" to "baseline entrants share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the change. The demographic groups are characterized by sex, average annual earnings between ages 25-34 (a measure of skill), and whether older or younger than 45 years old.

Appendix Figure G.3: Counterfactual vs actual SSDI entry rate, 1988–2010




Notes: Figure plots actual and counterfactual SSDI entry rate in each year from 1988 to 2010. Actual entry rate uses the Continuous Work History Sample linked to the Payment History Update System and the Master Beneficiary Record. Counterfactual entry rate uses equations (2), (3), and (4), using 1988 as the base year. The exact annual values are given in Appendix Table G.6.

Appendix Table G.6: Counterfactual vs actual SSDI entry rate, 1988–2010

| Year | Actual entry rate (%) | Counterfactual (base) entry rate (%) | Total difference (pp) |
|---|---|---|---|
| 1988 | 0.28 | 0.28 | 0.00 |
| 1989 | 0.29 | 0.28 | 0.01 |
| 1990 | 0.33 | 0.28 | 0.05 |
| 1991 | 0.38 | 0.28 | 0.10 |
| 1992 | 0.39 | 0.28 | 0.11 |
| 1993 | 0.40 | 0.28 | 0.12 |
| 1994 | 0.38 | 0.28 | 0.10 |
| 1995 | 0.36 | 0.28 | 0.08 |
| 1996 | 0.36 | 0.28 | 0.07 |
| 1997 | 0.34 | 0.28 | 0.06 |
| 1998 | 0.35 | 0.28 | 0.07 |
| 1999 | 0.36 | 0.28 | 0.08 |
| 2000 | 0.39 | 0.28 | 0.11 |
| 2001 | 0.44 | 0.28 | 0.16 |
| 2002 | 0.45 | 0.28 | 0.16 |
| 2003 | 0.47 | 0.28 | 0.19 |
| 2004 | 0.48 | 0.28 | 0.19 |
| 2005 | 0.45 | 0.28 | 0.16 |
| 2006 | 0.43 | 0.28 | 0.15 |
| 2007 | 0.44 | 0.28 | 0.15 |
| 2008 | 0.46 | 0.28 | 0.18 |
| 2009 | 0.52 | 0.28 | 0.24 |
| 2010 | 0.52 | 0.28 | 0.24 |
| Total | | | 2.77 |

Notes: Table presents actual and counterfactual SSDI entry rate in each year from 1988 to 2010. Actual SSDI entry rate uses the Continuous Work History Sample linked to the Payment History Update System and the Master Beneficiary Record. Counterfactual SSDI entry rate uses equations (2), (3), and (4), using 1988 as the base year.

Appendix Table G.7: Decomposition of increase in SSDI entry rate, 1988–2010

| | | Absolute contribution (pp) | As a share of total (%) |
|---|---|---|---|
| Entry | | 2.10 | 80.4 |
| | Award | 0.46 | 23.5 |
| | Application | 1.36 | 48.7 |
| | Eligibility | 0.17 | 5.1 |
| | Interaction within entry | 0.10 | 3.1 |
| Across group composition | | 0.68 | 21.0 |
| Within-across interaction | | -0.01 | -1.5 |
| Total | | 2.77 | 100.0 |

Notes: Table presents decomposition of SSDI entry rate increase from 1988 to 2010, using equation (9). The point estimates we report are the ones in which flows (e.g., $B_t$, $A_t$, $E_t$, $X_t$), when fixed, are held fixed at their counterfactual values.

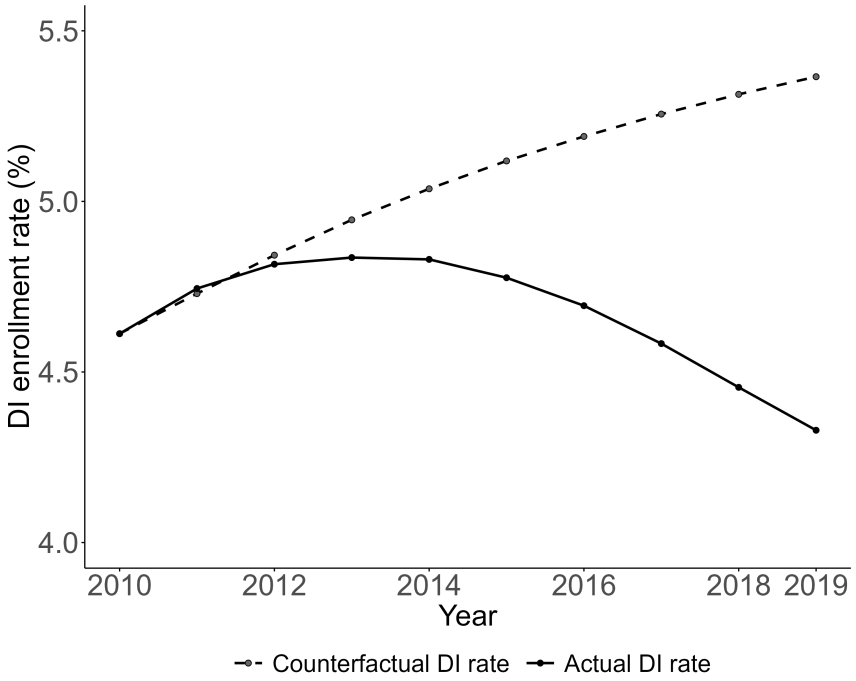
Appendix Table G.8: Demographic breakdown of entry share of SSDI entry rate, 1988–2010

| Group | Absolute contribution (pp) | Contribution share (%) | Baseline entrants share (%) | Contribution/ baseline entrants |
|---|---|---|---|---|
| Men, Younger, $10K+ | 0.02 | 1.1 | 12.1 | 0.1 |
| Men, Younger, $0-10K | 0.10 | 4.9 | 9.9 | 0.5 |
| Men, Older, $10K+ | 0.50 | 23.6 | 33.7 | 0.7 |
| Men, Older, $0-10K | 0.07 | 3.6 | 6.2 | 0.6 |
| Women, Younger, $10K+ | 0.14 | 6.5 | 4.4 | 1.5 |
| Women, Younger, $0-10K | 0.51 | 24.5 | 7.2 | 3.4 |
| Women, Older, $10K+ | 0.15 | 7.2 | 5.9 | 1.2 |
| Women, Older, $0-10K | 0.65 | 30.8 | 16.6 | 1.9 |
| Missing | -0.05 | -2.3 | 3.9 | -0.6 |
| Total | 2.10 | 100.0 | 100.0 | 1.0 |

Notes: Table presents the "entry" share (from Appendix Table G.7) of the difference between actual and counterfactual SSDI entry rates for the increase period (1988–2010), broken down into contribution from demographic subgroups using equation (13). "Contribution share" is "absolute contribution" as a percent of the total change. "Baseline population share" is the share of the US population that the group accounts for in 1988. "Baseline entrants share" is the share of SSDI entrants that the group accounts for in 1988. "Contribution/baseline entrants" presents the ratio of "contribution share" to "baseline entrants share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the change. The demographic groups are characterized by sex, average annual earnings between ages 25-34 (a measure of skill), and whether older or younger than 45 years old.

### G.1.2 SSDI decrease period (2010–2019)

Appendix Figure G.4: Counterfactual vs actual SSDI enrollment rate, 2010–2019




Notes: Figure plots actual and counterfactual SSDI enrollment rates in each year from 2010 to 2019. Actual SSDI rate uses the Continuous Work History Sample linked to the Payment History Update System and the Master Beneficiary Record. Counterfactual SSDI rate uses equations (2), (3), and (4), using 2010 as the base year. The exact annual values are given in Appendix Table G.9.

Appendix Table G.9: Counterfactual vs actual SSDI enrollment rate, 2010–2019

| Year | Actual DI Rate (%) | Counterfactual DI Rate (%) | Total DI change (pp) |
|---|---|---|---|
| 2010 | 4.61 | 4.61 | 0.00 |
| 2011 | 4.74 | 4.73 | 0.01 |
| 2012 | 4.82 | 4.84 | -0.03 |
| 2013 | 4.84 | 4.95 | -0.11 |
| 2014 | 4.83 | 5.04 | -0.21 |
| 2015 | 4.78 | 5.12 | -0.34 |
| 2016 | 4.69 | 5.19 | -0.50 |
| 2017 | 4.58 | 5.26 | -0.67 |
| 2018 | 4.45 | 5.31 | -0.86 |
| 2019 | 4.33 | 5.37 | -1.04 |

Notes: Table presents actual and counterfactual SSDI enrollment rates in each year from 2010 to 2019. Actual SSDI rate uses the Continuous Work History Sample linked to the Payment History Update System and the Master Beneficiary Record. Counterfactual SSDI rate uses equations (2), (3), and (4), using 2010 as the base year.

Appendix Table G.10: Demographic breakdown of entry share of SSDI enrollment rate, 2010–2019

| Group | Absolute contribution (pp) | Contribution share (%) | Baseline population share (%) | Baseline entrants share (%) | Contribution/ baseline entrants |
|---|---|---|---|---|---|
| Older, Men, $10K+ | 0.31 | 31.80 | 17.77 | 31.30 | 1.02 |
| Older, Men, $0-10K | 0.12 | 12.19 | 6.85 | 9.60 | 1.27 |
| Older, Women, $10K+ | 0.11 | 11.47 | 11.29 | 14.93 | 0.77 |
| Older, Women, $0-10K | 0.17 | 16.75 | 13.06 | 19.31 | 0.87 |
| Younger, Men, $10K+ | 0.07 | 6.61 | 16.31 | 6.18 | 1.07 |
| Younger, Men, $0-10K | 0.07 | 6.74 | 7.70 | 5.64 | 1.19 |
| Younger, Women, $10K+ | 0.04 | 4.17 | 12.92 | 4.40 | 0.95 |
| Younger, Women, $0-10K | 0.08 | 8.09 | 10.22 | 6.70 | 1.21 |
| Missing | 0.02 | 2.18 | 3.89 | 1.93 | 1.13 |
| Total | 0.99 | 100.00 | 100.00 | 100.00 | 1.00 |

Notes: Table presents the "entry" share (from Table 2) of the difference between actual and counterfactual SSDI rates for the decrease period (2010–2019), broken down into contribution from demographic subgroups using equation (35). "Contribution share" is "absolute contribution" as a percent of the total change. "Baseline entrants share" is the share of SSDI entrants that the group accounts for in 2010. "Contribution/baseline entrants" presents the ratio of "contribution share" to "baseline entrants share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the change. The demographic groups are characterized by sex, average annual earnings between ages 25-34 (a measure of skill), and whether older or younger than 45 years old.

Appendix Table G.11: Demographic breakdown of application share of SSDI enrollment rate, 2010–2019

| Group | Absolute contribution (pp) | Contribution share (%) | Baseline population share (%) | Baseline entrants share (%) | Contribution/ baseline entrants |
|---|---|---|---|---|---|
| Older, Men, $10K+ | 0.24 | 40.93 | 17.77 | 31.30 | 1.31 |
| Older, Men, $0-10K | 0.07 | 12.77 | 6.85 | 9.60 | 1.33 |
| Older, Women, $10K+ | 0.05 | 8.89 | 11.29 | 14.93 | 0.60 |
| Older, Women, $0-10K | 0.03 | 4.82 | 13.06 | 19.31 | 0.25 |
| Younger, Men, $10K+ | 0.05 | 8.29 | 16.31 | 6.18 | 1.34 |
| Younger, Men, $0-10K | 0.05 | 8.92 | 7.70 | 5.64 | 1.58 |
| Younger, Women, $10K+ | 0.02 | 4.23 | 12.92 | 4.40 | 0.96 |
| Younger, Women, $0-10K | 0.04 | 7.75 | 10.22 | 6.70 | 1.16 |
| Missing | 0.02 | 3.40 | 3.89 | 1.93 | 1.76 |
| Total | 0.58 | 100.00 | 100.00 | 100.00 | 1.00 |

Notes: Table presents the "application" share (from Table 2) of the difference between actual and counterfactual SSDI rates for the decrease period (2010–2019), broken down into contribution from demographic subgroups using equation (10). "Contribution share" is "absolute contribution" as a percent of the total change. "Baseline entrants share" is the share of SSDI entrants that the group accounts for in 2010. "Contribution/baseline entrants" presents the ratio of "contribution share" to "baseline entrants share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the change. The demographic groups are characterized by sex, average annual earnings between ages 25-34 (a measure of skill), and whether older or younger than 45 years old.

Appendix Table G.12: Demographic breakdown of award share of SSDI enrollment rate, 2010–2019

| Group | Absolute contribution (pp) | Contribution share (%) | Baseline population share (%) | Baseline entrants share (%) | Contribution/ baseline entrants |
|---|---|---|---|---|---|
| Older, Men, $10K+ | 0.10 | 22.75 | 17.77 | 31.30 | 0.73 |
| Older, Men, $0-10K | 0.05 | 10.69 | 6.85 | 9.60 | 1.11 |
| Older, Women, $10K+ | 0.07 | 15.71 | 11.29 | 14.93 | 1.05 |
| Older, Women, $0-10K | 0.11 | 25.89 | 13.06 | 19.31 | 1.34 |
| Younger, Men, $10K+ | 0.03 | 6.01 | 16.31 | 6.18 | 0.97 |
| Younger, Men, $0-10K | 0.01 | 3.29 | 7.70 | 5.64 | 0.58 |
| Younger, Women, $10K+ | 0.02 | 5.19 | 12.92 | 4.40 | 1.18 |
| Younger, Women, $0-10K | 0.04 | 9.77 | 10.22 | 6.70 | 1.46 |
| Missing | 0.00 | 0.70 | 3.89 | 1.93 | 0.36 |
| Total | 0.44 | 100.00 | 100.00 | 100.00 | 1.00 |

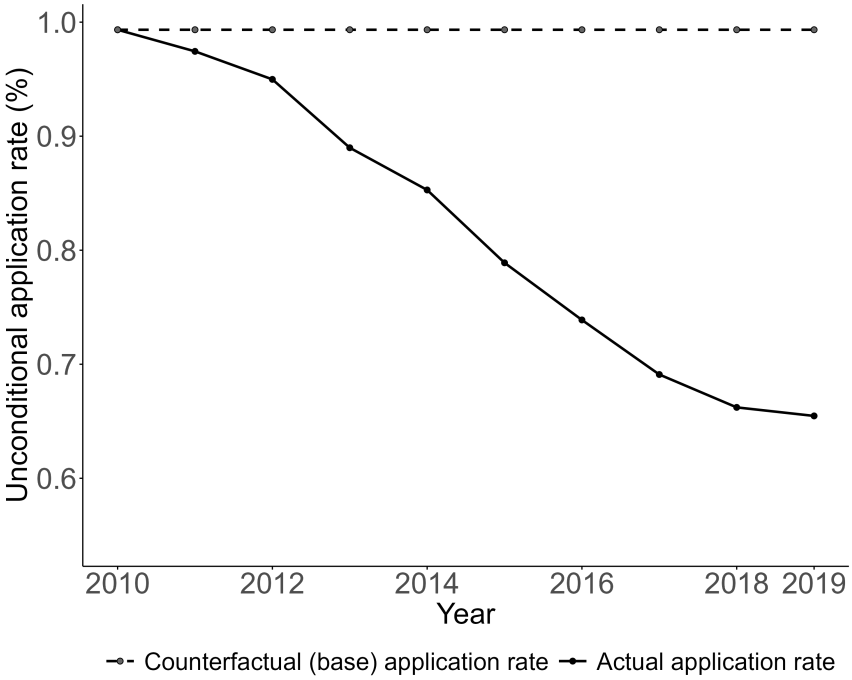
Notes: Table presents the "award" share (from Table 2) of the difference between actual and counterfactual SSDI rates for the decrease period (2010–2019), broken down into contribution from demographic subgroups using equation (33). "Contribution share" is "absolute contribution" as a percent of the total change. "Baseline entrants share" is the share of SSDI entrants that the group accounts for in 2010. "Contribution/baseline entrants" presents the ratio of "contribution share" to "baseline entrants share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the change. The demographic groups are characterized by sex, average annual earnings between ages 25-34 (a measure of skill), and whether older or younger than 45 years old.

Appendix Table G.13: Demographic breakdown of eligibility share of SSDI enrollment rate, 2010–2019

| Group | Absolute contribution (pp) | Contribution share (%) | Baseline population share (%) | Baseline entrants share (%) | Contribution/ baseline entrants |
|---|---|---|---|---|---|
| Older, Men, $10K+ | 0.01 | 7.23 | 17.77 | 31.30 | 0.23 |
| Older, Men, $0-10K | 0.02 | 23.39 | 6.85 | 9.60 | 2.44 |
| Older, Women, $10K+ | 0.00 | 1.15 | 11.29 | 14.93 | 0.08 |
| Older, Women, $0-10K | 0.04 | 51.23 | 13.06 | 19.31 | 2.65 |
| Younger, Men, $10K+ | 0.00 | 1.26 | 16.31 | 6.18 | 0.20 |
| Younger, Men, $0-10K | 0.01 | 9.11 | 7.70 | 5.64 | 1.61 |
| Younger, Women, $10K+ | -0.00 | -0.36 | 12.92 | 4.40 | -0.08 |
| Younger, Women, $0-10K | 0.01 | 6.43 | 10.22 | 6.70 | 0.96 |
| Missing | 0.00 | 0.56 | 3.89 | 1.93 | 0.29 |
| Total | 0.08 | 100.00 | 100.00 | 100.00 | 1.00 |

Notes: Table presents the "eligibility" share (from Table 2) of the difference between actual and counterfactual SSDI rates for the decrease period (2010–2019), broken down into contribution from demographic subgroups using equation (34). "Contribution share" is "absolute contribution" as a percent of the total change. "Baseline entrants share" is the share of SSDI entrants that the group accounts for in 2010. "Contribution/baseline entrants" presents the ratio of "contribution share" to "baseline entrants share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the change. The demographic groups are characterized by sex, average annual earnings between ages 25-34 (a measure of skill), and whether older or younger than 45 years old.

Appendix Figure G.5: Counterfactual vs actual SSDI application rate, 2010–2019




Notes: Figure plots actual and counterfactual SSDI application rate in each year from 2010 to 2019. Actual application rate uses the Continuous Work History Sample linked to the Payment History Update System and the Master Beneficiary Record. Counterfactual application rate uses equations (2), (3), and (4), using 2010 as the base year. The exact annual values are given in Appendix Table G.14.

Appendix Table G.14: Counterfactual vs actual SSDI application rate, 2010–2019

| Year | Actual application rate (%) | Counterfactual (base) application rate (%) | Total difference (pp) |
|---|---|---|---|
| 2010 | 0.99 | 0.99 | 0.00 |
| 2011 | 0.97 | 0.99 | -0.02 |
| 2012 | 0.95 | 0.99 | -0.04 |
| 2013 | 0.89 | 0.99 | -0.10 |
| 2014 | 0.85 | 0.99 | -0.14 |
| 2015 | 0.79 | 0.99 | -0.20 |
| 2016 | 0.74 | 0.99 | -0.25 |
| 2017 | 0.69 | 0.99 | -0.30 |
| 2018 | 0.66 | 0.99 | -0.33 |
| 2019 | 0.65 | 0.99 | -0.34 |
| Total | | | -1.74 |

Table presents actual and counterfactual SSDI application rate in each year from 2010 to 2019. Actual SSDI application rate uses the Continuous Work History Sample linked to the Payment History Update System and the Master Beneficiary Record. Counterfactual SSDI application rate uses equations (2), (3), and (4), using 2010 as the base year.

Appendix Table G.15: Decomposition of decline in SSDI application rate, 2010–2019

| | Absolute contribution (pp) | As a share of total (%) |
|---|---|---|
| Application | -1.51 | 80.8 |
| Eligibility | -0.20 | 16.7 |
| Interaction within apply | 0.05 | -2.3 |
| Across group composition | -0.12 | 7.3 |
| Within-across interaction | 0.05 | -2.5 |
| Total | -1.74 | 100.0 |

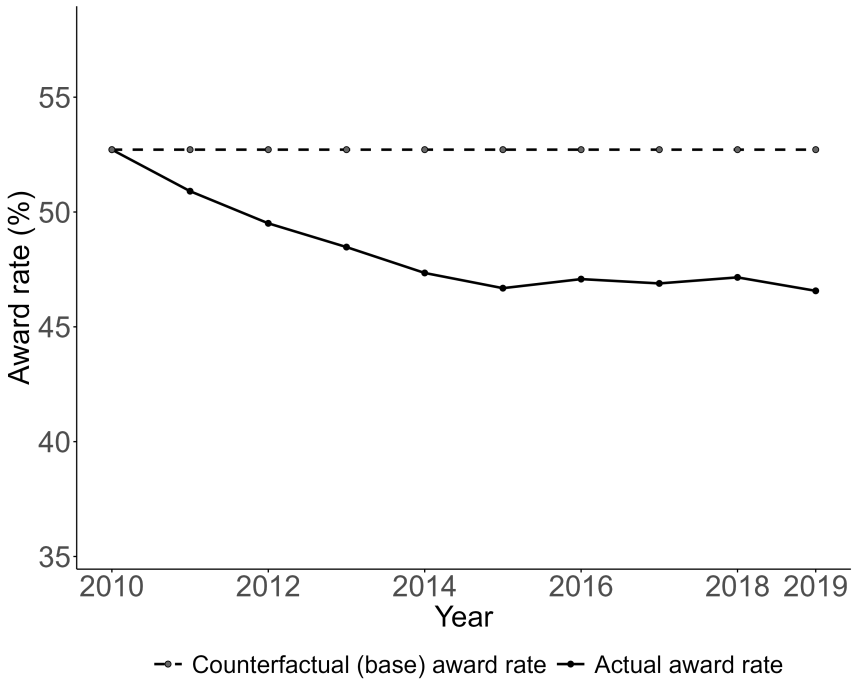
Notes: Table presents decomposition of SSDI application rate decline from 2010 to 2019, using equation (12). The point estimates we report are the ones in which flows (e.g., $B_t$, $A_t$, $E_t$, $X_t$), when fixed, are held fixed at their counterfactual values.

Appendix Table G.16: Demographic breakdown of application share of SSDI application rate, 2010–2019

| Group | Absolute contribution (pp) | Contribution share (%) | Baseline applicants share (%) | Contribution/ baseline applicants |
|---|---|---|---|---|
| Men, Younger, $10K+ | -0.17 | 11.1 | 7.8 | 1.4 |
| Men, Younger, $0-10K | -0.22 | 14.8 | 9.2 | 1.6 |
| Men, Older, $10K+ | -0.46 | 30.3 | 25.3 | 1.2 |
| Men, Older, $0-10K | -0.18 | 11.6 | 9.5 | 1.2 |
| Women, Younger, $10K+ | -0.06 | 4.1 | 5.4 | 0.8 |
| Women, Younger, $0-10K | -0.19 | 12.6 | 11.3 | 1.1 |
| Women, Older, $10K+ | -0.11 | 7.0 | 12.0 | 0.6 |
| Women, Older, $0-10K | -0.08 | 5.6 | 17.7 | 0.3 |
| Missing | -0.04 | 2.9 | 1.8 | 1.6 |
| Total | -1.51 | 100.0 | 100.0 | 1.0 |

Notes: Table presents the "application" share (from Appendix Table G.15) of the difference between actual and counterfactual SSDI application rates for the decline period (2010–2019), broken down into contribution from demographic subgroups using equation (12). "Contribution share" is "absolute contribution" as a percent of the total change. "Baseline applicants share" is the share of SSDI applicants that the group accounts for in 2010. "Contribution/baseline applicants" presents the ratio of "contribution share" to "baseline applicants share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the change. The demographic groups are characterized by sex, average annual earnings between ages 25-34 (a measure of skill), and whether older or younger than 45 years old.

Appendix Figure G.6: Counterfactual vs actual SSDI award rate (conditional on application), 2010–2019




Notes: Figure plots actual and counterfactual SSDI award rate (conditional on application) in each year from 2010 to 2019. Actual award rate uses the Continuous Work History Sample linked to the Payment History Update System and the Master Beneficiary Record. Counterfactual award rate uses equations (2), (3), and (4), using 2010 as the base year. The exact annual values are given in Appendix Table G.17.

Appendix Table G.17: Counterfactual vs actual SSDI award rate (conditional on application), 2010–2019

| Year | Actual award rate (%) | Counterfactual (base) award rate (%) | Total difference (pp) |
|---|---|---|---|
| 2010 | 52.72 | 52.72 | 0.00 |
| 2011 | 50.91 | 52.72 | -1.81 |
| 2012 | 49.51 | 52.72 | -3.21 |
| 2013 | 48.47 | 52.72 | -4.24 |
| 2014 | 47.35 | 52.72 | -5.37 |
| 2015 | 46.68 | 52.72 | -6.03 |
| 2016 | 47.08 | 52.72 | -5.64 |
| 2017 | 46.89 | 52.72 | -5.83 |
| 2018 | 47.15 | 52.72 | -5.56 |
| 2019 | 46.57 | 52.72 | -6.15 |
| Total | | | -43.83 |

Notes: Table presents actual and counterfactual SSDI award rate (conditional on application) in each year from 2010 to 2019. Actual SSDI award rate uses the Continuous Work History Sample linked to the Payment History Update System and the Master Beneficiary Record. Counterfactual SSDI award rate uses equations (2), (3), and (4), using 2010 as the base year.

Appendix Table G.18: Decomposition of decline in SSDI award rate, 2010–2019

| | Absolute contribution (pp) | As a share of total (%) |
|---|---|---|
| Award | -54.15 | 122.5 |
| Across group applicant share | 8.83 | -19.2 |
| Within-across interaction | 1.49 | -3.3 |
| Total | -43.83 | 100.0 |

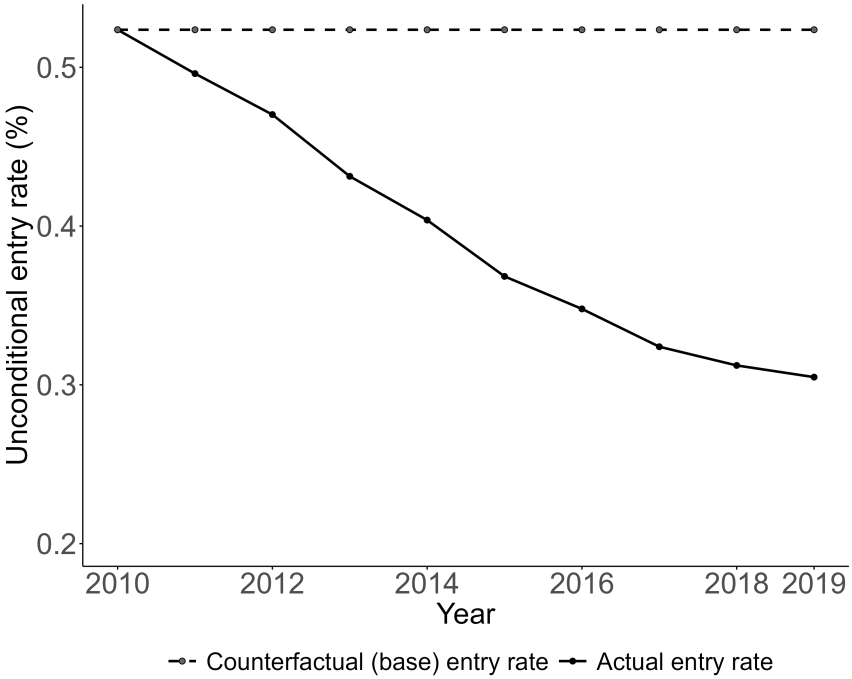
Notes: Table presents decomposition of SSDI award rate decline from 2010 to 2019, using equation (14). The point estimates we report are the ones in which flows (e.g., $B_t$, $A_t$, $E_t$, $X_t$), when fixed, are held fixed at their counterfactual values.

Appendix Table G.19: Demographic breakdown of award share of SSDI award rate, 2010–2019

| Group | Absolute contribution (pp) | Contribution share (%) | Baseline entrants share (%) | Contribution/ baseline entrants |
|---|---|---|---|---|
| Younger, Men, $10K+ | -5.00 | 0.1 | 6.2 | 1.5 |
| Younger, Men, $0-10K | -2.73 | 0.1 | 5.6 | 0.9 |
| Younger, Women, $10K+ | -4.07 | 0.1 | 4.4 | 1.7 |
| Younger, Women, $0-10K | -6.70 | 0.1 | 6.7 | 1.8 |
| Older, Men, $10K+ | -10.52 | 0.2 | 31.3 | 0.6 |
| Older, Men, $0-10K | -5.28 | 0.1 | 9.6 | 1.0 |
| Older, Women, $10K+ | -7.57 | 0.1 | 14.9 | 0.9 |
| Older, Women, $0-10K | -11.99 | 0.2 | 19.3 | 1.1 |
| Missing | -0.30 | 0.0 | 1.9 | 0.3 |
| Total | -54.15 | 1.0 | 100.0 | 1.0 |

Notes: Table presents the "award" share (from Appendix Table G.18) of the difference between actual and counterfactual SSDI award rates for the decline period (2010–2019), broken down into contribution from demographic subgroups using equation (14). "Contribution share" is "absolute contribution" as a percent of the total change. "Baseline entrants share" is the share of SSDI entrants that the group accounts for in 2010. "Contribution/baseline entrants" presents the ratio of "contribution share" to "baseline entrants share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the change. The demographic groups are characterized by sex, average annual earnings between ages 25-34 (a measure of skill), and whether older or younger than 45 years old.

Appendix Figure G.7: Counterfactual vs actual SSDI entry rate, 2010–2019




Notes: Figure plots actual and counterfactual SSDI entry rate in each year from 2010 to 2019. Actual entry rate uses the Continuous Work History Sample linked to the Payment History Update System and the Master Beneficiary Record. Counterfactual entry rate uses equations (2), (3), and (4), using 2010 as the base year. The exact annual values are given in Appendix Table G.20.

Appendix Table G.20: Counterfactual vs actual SSDI entry rate, 2010–2019

| Year | Actual entry rate (%) | Counterfactual (base) entry rate (%) | Total difference (pp) |
|---|---|---|---|
| 2010 | 0.52 | 0.52 | 0.00 |
| 2011 | 0.50 | 0.52 | -0.03 |
| 2012 | 0.47 | 0.52 | -0.05 |
| 2013 | 0.43 | 0.52 | -0.09 |
| 2014 | 0.40 | 0.52 | -0.12 |
| 2015 | 0.37 | 0.52 | -0.16 |
| 2016 | 0.35 | 0.52 | -0.18 |
| 2017 | 0.32 | 0.52 | -0.20 |
| 2018 | 0.31 | 0.52 | -0.21 |
| 2019 | 0.30 | 0.52 | -0.22 |
| Total | | | -1.25 |

Notes: Table presents actual and counterfactual SSDI entry rate in each year from 2010 to 2019. Actual SSDI entry rate uses the Continuous Work History Sample linked to the Payment History Update System and the Master Beneficiary Record. Counterfactual SSDI entry rate uses equations (2), (3), and (4), using 2010 as the base year.

Appendix Table G.21: Decomposition of decrease in SSDI entry rate, 2010–2019

| | | Absolute contribution (pp) | As a share of total (%) |
|---|---|---|---|
| Entry | | -1.25 | 100.1 |
| | Award | -0.54 | 49.7 |
| | Application | -0.75 | 51.6 |
| | Eligibility | -0.09 | 7.9 |
| | Interaction within entry | 0.14 | -9.1 |
| Across group composition | | -0.04 | 2.4 |
| Within-across interaction | | 0.04 | -2.5 |
| Total | | -1.25 | 100.0 |

Notes: Table presents decomposition of SSDI entry rate decline from 2010 to 2019, using equation (9). The point estimates we report are the ones in which flows (e.g., $B_t$, $A_t$, $E_t$, $X_t$), when fixed, are held fixed at their counterfactual values.

Appendix Table G.22: Demographic breakdown of entry share of SSDI entry rate, 2010–2019

| Group | Absolute contribution (pp) | Contribution share (%) | Baseline entrants share (%) | Contribution/ baseline entrants |
|---|---|---|---|---|
| Younger, Men, $10K+ | -0.11 | 8.7 | 6.2 | 1.4 |
| Younger, Men, $0-10K | -0.10 | 8.1 | 5.6 | 1.4 |
| Younger, Women, $10K+ | -0.06 | 4.9 | 4.4 | 1.1 |
| Younger, Women, $0-10K | -0.12 | 9.6 | 6.7 | 1.4 |
| Older, Men, $10K+ | -0.38 | 30.7 | 31.3 | 1.0 |
| Older, Men, $0-10K | -0.14 | 10.9 | 9.6 | 1.1 |
| Older, Women, $10K+ | -0.14 | 10.8 | 14.9 | 0.7 |
| Older, Women, $0-10K | -0.18 | 14.2 | 19.3 | 0.7 |
| Missing | -0.03 | 2.1 | 1.9 | 1.1 |
| Total | -1.25 | 100.0 | 100.0 | 1.0 |

Notes: Table presents the "entry" share (from Appendix Table G.21) of the difference between actual and counterfactual SSDI entry rates for the decline period (2010–2019), broken down into contribution from demographic subgroups using equation (13). "Contribution share" is "absolute contribution" as a percent of the total change. "Baseline entrants share" is the share of SSDI entrants that the group accounts for in 2010. "Contribution/baseline entrants" presents the ratio of "contribution share" to "baseline entrants share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the change. The demographic groups are characterized by sex, average annual earnings between ages 25-34 (a measure of skill), and whether older or younger than 45 years old.

## G.2 By veteran status

### G.2.1 Enrollment decomposition in SSDI decrease period (2010–2019), including veteran status

Appendix Table G.23: Decomposition results for the decrease in SSDI enrollment from 2010 to 2019, including veteran status

| | Absolute contribution (pp) | | | As a share of total (%) | | |
|---|---|---|---|---|---|---|
| | Minimum | Maximum | Used in graph | Minimum | Maximum | Used in graph |
| **Across-group composition** | **0.01** | **-0.06** | **-0.06** | **-0.8** | **5.5** | **5.5** |
| Sex | 0.00 | -0.03 | 0.00 | -0.3 | 2.5 | -0.0 |
| Age | 0.20 | 0.02 | 0.07 | -19.4 | -1.7 | -6.6 |
| Earnings | 0.03 | -0.11 | -0.04 | -3.2 | 10.6 | 3.6 |
| Immigrant status | -0.06 | -0.17 | -0.11 | 5.8 | 16.8 | 10.6 |
| Veteran | 0.04 | -0.00 | -0.00 | -4.2 | 0.2 | 0.0 |
| **Within-group SSDI rates** | **-0.97** | **-1.04** | **-1.04** | **94.5** | **100.8** | **100.8** |
| Entry | -0.93 | -1.01 | -0.99 | 90.2 | 98.0 | 96.2 |
| *Eligibility* | *-0.05* | *-0.09* | *-0.08* | *4.7* | *8.4* | *8.1* |
| *Application* | *-0.48* | *-0.60* | *-0.58* | *46.1* | *57.8* | *56.6* |
| *Award* | *-0.31* | *-0.44* | *-0.43* | *29.7* | *43.0* | *42.0* |
| Exit | -0.02 | -0.04 | -0.04 | 1.5 | 4.2 | 3.8 |
| *Death* | *0.04* | *0.03* | *0.04* | *-3.9* | *-3.4* | *-3.9* |
| *Recovery* | *-0.06* | *-0.08* | *-0.08* | *5.6* | *7.4* | *7.4* |
| *Retirement* | *0.01* | *0.01* | *0.01* | *-1.1* | *-0.7* | *-0.8* |
| *Other* | *-0.01* | *-0.01* | *-0.01* | *0.6* | *1.1* | *1.1* |
| Friction in population | -0.01 | -0.01 | -0.01 | 0.9 | 1.1 | 0.9 |
| Population discrepancies | 0.00 | 0.00 | 0.00 | -0.2 | -0.1 | -0.1 |
| **Within-across interaction** | | | **0.06** | | | **-6.3** |
| Total to be explained | -1.03 | -1.03 | -1.03 | 100.0 | 100.0 | 100.0 |

Notes: Table presents decomposition of the increase in SSDI enrollment rates from 2010 to 2019 into various components. The counterfactual SSDI enrollment rate in 2019 is 5.37 percent while the actual enrollment rate in 2019 is 4.33 percent, creating 1.04 pp of SSDI decline to be decomposed (see Appendix Table G.9). Across-group composition change, within-group change in SSDI enrollment rate, and the interaction between the two are calculated using equation (5). Across-group composition change is decomposed into specific characteristics using equation (11). Within-group SSDI rate is decomposed into entry, exit, friction in population, and population discrepancies using equation (6) and footnotes 13 and 15. Entry is decomposed into partial eligibility (i.e., the part that can only be attributed to eligibility and no other component), partial application, partial award, and the sum of interactions between the three entry components (eligibility, application, and award) using equation (9). Exit is decomposed into death, retirement (defined as aging into the retirement program), recovery (defined as being found no longer eligible due to continuing disability review), and "other" (the residual) using equation (8). "Minimum" means the minimum bound, "Maximum" means the maximum bound, and "Used in graph" means the point estimate used in Figure 3. See Section 2.1 for details on the decomposition method, and Appendix Section F for details on the construction of bounds.

### G.2.2 Unconditional application rate decomposition in SSDI decrease period (2010–2019), including veteran status

Appendix Table G.24: Counterfactual vs actual SSDI application rate including veteran status, 2010–2019

| Year | Actual application rate (%) | Counterfactual (base) application rate (%) | Total difference (pp) |
|---|---|---|---|
| 2010 | 0.99 | 0.99 | 0.00 |
| 2011 | 0.97 | 0.99 | -0.02 |
| 2012 | 0.95 | 0.99 | -0.04 |
| 2013 | 0.89 | 0.99 | -0.10 |
| 2014 | 0.85 | 0.99 | -0.14 |
| 2015 | 0.79 | 0.99 | -0.20 |
| 2016 | 0.74 | 0.99 | -0.25 |
| 2017 | 0.69 | 0.99 | -0.30 |
| 2018 | 0.66 | 0.99 | -0.33 |
| 2019 | 0.65 | 0.99 | -0.34 |
| Total | | | -1.74 |

Notes: Table presents actual and counterfactual SSDI application rate in each year from 2010 to 2019. Actual SSDI application rate uses the Continuous Work History Sample linked to the Payment History Update System and the Master Beneficiary Record. Counterfactual SSDI application rate uses equations (2), (3), and (4), using 2010 as the base year.

Appendix Table G.25: Decomposition of decline in SSDI application rate including veteran status, 2010–2019

| | Absolute contribution (pp) | As a share of total (%) |
|---|---|---|
| Application | -1.52 | 81.4 |
| Eligibility | -0.20 | 16.6 |
| Interaction within apply | 0.04 | -2.2 |
| Across group composition | -0.13 | 7.8 |
| Within-across interaction | 0.07 | -3.6 |
| Total | -1.74 | 100.0 |

Notes: Notes: Table presents decomposition of SSDI application rate decline from 2010 to 2019, using equation (12). The point estimates we report are the ones in which flows (e.g., $B_t$, $A_t$, $E_t$, $X_t$), when fixed, are held fixed at their counterfactual values.

Appendix Table G.26: Demographic breakdown of application share of SSDI application rate including veteran status, 2010–2019

| Group | Absolute contribution (pp) | Contribution share (%) | Baseline applicants share (%) | Contribution/ baseline applicants |
|---|---|---|---|---|
| Veteran, younger male | -0.01 | 0.8 | 2.3 | 0.4 |
| Veteran, younger female | 0.00 | -0.2 | 0.5 | -0.3 |
| Veteran, older male | -0.05 | 3.4 | 4.5 | 0.8 |
| Veteran, older female | -0.00 | 0.1 | 0.6 | 0.2 |
| Non-Veteran | -1.45 | 95.8 | 92.0 | 1.0 |
| Total | -1.52 | 100.0 | 100.0 | 1.0 |

Notes: Table presents the "application" share (from Appendix Table G.15) of the difference between actual and counterfactual SSDI application rates for the decline period (2010–2019), broken down into contribution from demographic subgroups using equation (12). "Contribution share" is "absolute contribution" as a percent of the total change. "Baseline applicants share" is the share of SSDI applicants that the group accounts for in 2010. "Contribution/baseline applicants" presents the ratio of "contribution share" to "baseline applicants share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the change. We identify veteran status by linking EIN from the Detailed Earnings Files to publicly available military EINs. The other demographic groups are characterized by sex, average annual earnings between ages 25-34 (a measure of skill), and whether older or younger than 45 years old.

### G.2.3 Unconditional entry rate decomposition in SSDI decrease period (2010–2019), including veteran status

Appendix Table G.27: Counterfactual vs actual SSDI entry rate including veteran status, 2010–2019

| Year | Actual entry rate (%) | Counterfactual (base) entry rate (%) | Total difference (pp) |
|---|---|---|---|
| 2010 | 0.52 | 0.52 | 0.00 |
| 2011 | 0.50 | 0.52 | -0.03 |
| 2012 | 0.47 | 0.52 | -0.05 |
| 2013 | 0.43 | 0.52 | -0.09 |
| 2014 | 0.40 | 0.52 | -0.12 |
| 2015 | 0.37 | 0.52 | -0.16 |
| 2016 | 0.35 | 0.52 | -0.18 |
| 2017 | 0.32 | 0.52 | -0.20 |
| 2018 | 0.31 | 0.52 | -0.21 |
| 2019 | 0.30 | 0.52 | -0.22 |
| Total | | | -1.25 |

Notes: Table presents actual and counterfactual SSDI entry rate in each year from 2010 to 2019. Actual SSDI entry rate uses the Continuous Work History Sample linked to the Payment History Update System and the Master Beneficiary Record. Counterfactual SSDI entry rate uses equations (2), (3), and (4), using 2010 as the base year.

Appendix Table G.28: Decomposition of decline in SSDI entry rate including veteran status, 2010–2019

| | | Absolute contribution (pp) | As a share of total (%) |
|---|---|---|---|
| Entry | | -1.26 | 100.7 |
| | Award | -0.53 | 49.6 |
| | Application | -0.76 | 52.3 |
| | Eligibility | -0.09 | 7.9 |
| | Interaction within entry | 0.13 | -9.0 |
| Across group composition | | -0.05 | 2.9 |
| Within-across interaction | | 0.05 | -3.6 |
| Total | | -1.25 | 100.0 |

Notes: Table presents decomposition of SSDI entry rate decline from 2010 to 2019, using equation (9). The point estimates we report are the ones in which flows (e.g., $B_t$, $A_t$, $E_t$, $X_t$), when fixed, are held fixed at their counterfactual values.

Appendix Table G.29: Demographic breakdown of entry share of SSDI entry rate including veteran status, 2010–2019

| Group | Absolute contribution (pp) | Contribution share (%) | baseline entrants share (%) | Contribution/ baseline entrants |
|---|---|---|---|---|
| Veteran, younger male | -0.03 | 2.0 | 1.8 | 1.1 |
| Veteran, younger female | -0.00 | 0.3 | 0.3 | 0.9 |
| Veteran, older male | -0.05 | 4.3 | 5.0 | 0.9 |
| Veteran, older female | -0.01 | 0.5 | 0.6 | 0.7 |
| Non-Veteran | -1.17 | 92.9 | 92.3 | 1.0 |
| Total | -1.26 | 100.0 | 100.0 | 1.0 |

Notes: Table presents the "entry" share (from Appendix Table G.21) of the difference between actual and counterfactual SSDI entry rates for the decline period (2010–2019), broken down into contribution from demographic subgroups using equation (13). "Contribution share" is "absolute contribution" as a percent of the total change. "Baseline entrants share" is the share of SSDI entrants that the group accounts for in 2010. "Contribution/baseline entrants" presents the ratio of "contribution share" to "baseline entrants share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the change. We identify veteran status by linking EIN from the Detailed Earnings Files to publicly available military EINs. The other demographic groups are characterized by sex, average annual earnings between ages 25-34 (a measure of skill), and whether older or younger than 45 years old.

## G.3 By pollution abatement

### G.3.1 Unconditional application rate decomposition in SSDI decrease period (2010–2019), including pollution abatement

Appendix Table G.30: Counterfactual vs actual SSDI application rate including pollution abatement, 2010–2019

| Year | Actual application rate (%) | Counterfactual (base) application rate (%) | Total difference (pp) |
|---|---|---|---|
| 2010 | 0.99 | 0.99 | 0.00 |
| 2011 | 0.97 | 0.99 | -0.02 |
| 2012 | 0.95 | 0.99 | -0.04 |
| 2013 | 0.89 | 0.99 | -0.10 |
| 2014 | 0.85 | 0.99 | -0.14 |
| 2015 | 0.79 | 0.99 | -0.20 |
| 2016 | 0.74 | 0.99 | -0.25 |
| 2017 | 0.69 | 0.99 | -0.30 |
| 2018 | 0.66 | 0.99 | -0.33 |
| 2019 | 0.65 | 0.99 | -0.34 |
| Total | | | -1.74 |

Notes: Table presents actual and counterfactual SSDI application rate in each year from 2010 to 2019. Actual SSDI application rate uses the Continuous Work History Sample linked to the Payment History Update System and the Master Beneficiary Record. Counterfactual SSDI application rate uses equations (2), (3), and (4), using 2010 as the base year.

Appendix Table G.31: Decomposition of decline in SSDI application rate including pollution abatement, 2010–2019

| | Absolute contribution (pp) | As a share of total (%) |
|---|---|---|
| Application | -0.78 | 42.1 |
| Eligibility | -0.10 | 7.8 |
| Interaction within apply | 0.02 | -1.2 |
| Across group composition | -0.02 | -0.1 |
| Within-across interaction | 0.03 | -1.8 |
| Total | -0.84 | 46.9 |

Notes: Table presents decomposition of SSDI application rate decline from 2010 to 2019, using equation (12). The point estimates we report are the ones in which flows (e.g., $B_t$, $A_t$, $E_t$, $X_t$), when fixed, are held fixed at their counterfactual values.

Appendix Table G.32: Demographic breakdown of application share of SSDI application rate including pollution abatement, 2010–2019

| Group | Absolute contribution (pp) | Contribution share (%) | Baseline applicants share (%) | Contribution/ baseline applicants |
|---|---|---|---|---|
| More-abatement area | -0.05 | 5.86 | 6.29 | 0.93 |
| Less-abatement area | -0.13 | 16.42 | 18.36 | 0.89 |
| Matched area, no classification | -0.10 | 12.26 | 13.18 | 0.93 |
| Unmatched | -0.51 | 65.47 | 62.17 | 1.05 |
| Total | -0.78 | 100.0 | 100.0 | 1.0 |

Notes: Table presents the "application" share (from Appendix Table G.31) of the difference between actual and counterfactual SSDI application rates for the decline period (2010–2019), broken down into contribution from subgroups based on pollution abatement status in one's place of birth, using equation (12). "Contribution share" is "absolute contribution" as a percent of the total change. "Baseline applicants share" is the share of SSDI applicants that the group accounts for in 2010. "Contribution/baseline applicants" presents the ratio of "contribution share" to "baseline applicants share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the change. We identify abatement and non-abatement areas based on data provided by Isen et al. (2017) linked to place of birth in the Numident. "More-abatement" means that the area was flagged by the 1970 Clean Air Act Amendments to be in "non-attainment" and therefore legally required to reduce pollution levels. "Less-abatement" means that the area was not deemed "non-attainment" and therefore was not required to reduce pollution levels. "Matched area, no classification" means that the individual was matched to a place of birth in the Numident but that their place of birth was not classified as attainment or non-attainment. "Cannot match" means that the individual could not be matched to a place of birth.

### G.3.2 Unconditional entry rate decomposition in SSDI decrease period (2010–2019), including pollution abatement

Appendix Table G.33: Counterfactual vs actual SSDI entry rate including pollution abatement, 2010–2019

| Year | Actual entry rate (%) | Counterfactual (base) entry rate (%) | Total difference (pp) |
|---|---|---|---|
| 2010 | 0.52 | 0.52 | 0.00 |
| 2011 | 0.50 | 0.52 | -0.03 |
| 2012 | 0.47 | 0.52 | -0.05 |
| 2013 | 0.43 | 0.52 | -0.09 |
| 2014 | 0.40 | 0.52 | -0.12 |
| 2015 | 0.37 | 0.52 | -0.16 |
| 2016 | 0.35 | 0.52 | -0.18 |
| 2017 | 0.32 | 0.52 | -0.20 |
| 2018 | 0.31 | 0.52 | -0.21 |
| 2019 | 0.30 | 0.52 | -0.22 |
| Total | | | -1.25 |

Notes: Table presents actual and counterfactual SSDI entry rate in each year from 2010 to 2019. Actual SSDI entry rate uses the Continuous Work History Sample linked to the Payment History Update System and the Master Beneficiary Record. Counterfactual SSDI entry rate uses equations (2), (3), and (4), using 2010 as the base year.

Appendix Table G.34: Decomposition of decline in SSDI entry rate including pollution abatement, 2010–2019

| | | Absolute contribution (pp) | As a share of total (%) |
|---|---|---|---|
| Entry | | -1.27 | 101.8 |
| | Award | -0.54 | 50.0 |
| | Application | -0.78 | 53.7 |
| | Eligibility | -0.10 | 8.7 |
| | Interaction within entry | 0.15 | -10.6 |
| Across group composition | | -0.02 | 0.4 |
| Within-across interaction | | 0.03 | -2.2 |
| Total | | -1.25 | 100.0 |

Notes: Table presents decomposition of SSDI entry rate decline from 2010 to 2019, using equation (9). The point estimates we report are the ones in which flows (e.g., $B_t$, $A_t$, $E_t$, $X_t$), when fixed, are held fixed at their counterfactual values.

Appendix Table G.35: Demographic breakdown of entry share of SSDI entry rate including pollution abatement, 2010–2019

| Group | Absolute contribution (pp) | Contribution share (%) | baseline entrants share (%) | Contribution/ baseline entrants |
|---|---|---|---|---|
| More-abatement area | -0.08 | 6.3 | 5.8 | 1.1 |
| Less-abatement area | -0.22 | 17.0 | 16.5 | 1.0 |
| Matched area, no classification | -0.16 | 12.3 | 11.9 | 1.0 |
| Unmatched | -0.82 | 64.4 | 65.7 | 1.0 |
| Total | -1.27 | 100.0 | 100.0 | 1.0 |

Notes: Table presents the "entry" share (from Appendix Table G.34) of the difference between actual and counterfactual SSDI entry rates for the decline period (2010–2019), broken down into contribution from subgroups based on pollution abatement status in one's place of birth, using equation (13). "Contribution share" is "absolute contribution" as a percent of the total change. "Baseline entrants share" is the share of SSDI entrants that the group accounts for in 2010. "Contribution/baseline entrants" presents the ratio of "contribution share" to "baseline entrants share," indicating whether that subgroup is under- ($<1$) or over- ($>1$) represented in the change. We identify abatement and non-abatement areas based on data provided by Isen et al. (2017) linked to place of birth in the Numident. "More-abatement" means that the area was flagged by the 1970 Clean Air Act Amendments to be in "non-attainment" and therefore legally required to reduce pollution levels. "Less-abatement" means that the area was not deemed "non-attainment" and therefore was not required to reduce pollution levels. "Matched area, no classification" means that the individual was matched to a place of birth in the Numident but that their place of birth was not classified as attainment or non-attainment. "Cannot match" means that the individual could not be matched to a place of birth.

## G.4 By early-career industry

### G.4.1 Unconditional application rate decomposition in SSDI decrease period (2010–2019), by early-career industry

Appendix Table G.36: Counterfactual vs actual SSDI application rate by early-career industry, 2010–2019

| Year | Actual application rate (%) | Counterfactual (base) application rate (%) | Total difference (pp) |
|---|---|---|---|
| 2010 | 0.99 | 0.99 | 0.00 |
| 2011 | 0.97 | 0.99 | -0.02 |
| 2012 | 0.95 | 0.99 | -0.04 |
| 2013 | 0.89 | 0.99 | -0.10 |
| 2014 | 0.85 | 0.99 | -0.14 |
| 2015 | 0.79 | 0.99 | -0.20 |
| 2016 | 0.74 | 0.99 | -0.25 |
| 2017 | 0.69 | 0.99 | -0.30 |
| 2018 | 0.66 | 0.99 | -0.33 |
| 2019 | 0.65 | 0.99 | -0.34 |
| Total | | | -1.74 |

Notes: Table presents actual and counterfactual SSDI application rate in each year from 2010 to 2019. Actual SSDI application rate uses the Continuous Work History Sample linked to the Payment History Update System and the Master Beneficiary Record. Counterfactual SSDI application rate uses equations (2), (3), and (4), using 2010 as the base year.

Appendix Table G.37: Decomposition of decline in SSDI application rate by early-career industry, 2010–2019

| | Absolute contribution (pp) | As a share of total (%) |
|---|---|---|
| Application | -1.58 | 86.2 |
| Eligibility | -0.22 | 18.1 |
| Interaction within apply | 0.05 | -2.3 |
| Across group composition | -0.05 | 1.0 |
| Within-across interaction | 0.06 | -3.0 |
| Total | -1.74 | 100.0 |

Notes: Table presents decomposition of SSDI application rate decline from 2010 to 2019, using equation (12). The point estimates we report are the ones in which flows (e.g., $B_t$, $A_t$, $E_t$, $X_t$), when fixed, are held fixed at their counterfactual values.

Appendix Table G.38: Demographic breakdown of application share of SSDI application rate by early-career industry, 2010–2019

| Group | Absolute contribution (pp) | Contribution share (%) | Baseline applicants share (%) | Contribution/ baseline applicants |
|---|---|---|---|---|
| Ag/mining/utilities/construction | -0.25 | 15.94 | 12.63 | 1.26 |
| Manufacturing | -0.25 | 15.61 | 13.22 | 1.18 |
| Wholesale/retail/transportation | -0.18 | 11.15 | 10.71 | 1.04 |
| White collar/management | -0.19 | 11.96 | 13.53 | 0.88 |
| Education/health care/govt | -0.26 | 16.69 | 19.10 | 0.87 |
| Military | 0.01 | -0.71 | 2.24 | -0.32 |
| No Earning | -0.02 | 1.33 | 1.82 | 0.73 |
| Employed, likely small employer | -0.40 | 25.29 | 24.99 | 1.01 |
| Missing | -0.04 | 2.74 | 1.75 | 1.57 |
| Total | -1.58 | 100.0 | 100.0 | 1.0 |

Notes: Table presents the "application" share (from Appendix Table G.37) of the difference between actual and counterfactual SSDI application rates for the decline period (2010–2019), broken down into contribution from subgroups based on primary industry of work between the ages of 25 and 34 years old, using equation (12). "Contribution share" is "absolute contribution" as a percent of the total change. "Baseline applicants share" is the share of SSDI applicants that the group accounts for in 2010. "Contribution/baseline applicants" presents the ratio of "contribution share" to "baseline applicants share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the change. We identify a person's primary industry using EIN from the Detailed Earnings File linked to NAICS code from the Department of Labor Form 5500. We define primary industry as the industry in which the person worked the most years between 25 and 34 years old. "Employed, likely small employer" means that the applicant had an employer but the employer's EIN is not listed in Form 5500.

### G.4.2 Unconditional entry rate decomposition in SSDI decrease period (2010–2019), by early-career industry

Appendix Table G.39: Counterfactual vs actual SSDI entry rate by early-career industry, 2010–2019

| Year | Actual entry rate (%) | Counterfactual (base) entry rate (%) | Total difference (pp) |
|---|---|---|---|
| 2010 | 0.52 | 0.52 | 0.00 |
| 2011 | 0.50 | 0.52 | -0.03 |
| 2012 | 0.47 | 0.52 | -0.05 |
| 2013 | 0.43 | 0.52 | -0.09 |
| 2014 | 0.40 | 0.52 | -0.12 |
| 2015 | 0.37 | 0.52 | -0.16 |
| 2016 | 0.35 | 0.52 | -0.18 |
| 2017 | 0.32 | 0.52 | -0.20 |
| 2018 | 0.31 | 0.52 | -0.21 |
| 2019 | 0.30 | 0.52 | -0.22 |
| Total | | | -1.25 |

Notes: Table presents actual and counterfactual SSDI entry rate in each year from 2010 to 2019. Actual SSDI entry rate uses the Continuous Work History Sample linked to the Payment History Update System and the Master Beneficiary Record. Counterfactual SSDI entry rate uses equations (2), (3), and (4), using 2010 as the base year.

Appendix Table G.40: Decomposition of decline in SSDI entry rate by early-career industry, 2010–2019

| | | Absolute contribution (pp) | As a share of total (%) |
|---|---|---|---|
| Entry | | -1.29 | 103.9 |
| | Award | -0.54 | 50.1 |
| | Application | -0.80 | 55.8 |
| | Eligibility | -0.10 | 8.7 |
| | Interaction within entry | 0.15 | -10.7 |
| Across group composition | | 0.01 | -1.8 |
| Within-across interaction | | 0.03 | -2.1 |
| Total | | -1.25 | 100.0 |

Notes: Table presents decomposition of SSDI entry rate decline from 2010 to 2019, using equation (9). The point estimates we report are the ones in which flows (e.g., $B_t$, $A_t$, $E_t$, $X_t$), when fixed, are held fixed at their counterfactual values.

Appendix Table G.41: Demographic breakdown of entry share of SSDI entry rate by early-career industry, 2010–2019

| Group | Absolute contribution (pp) | Contribution share (%) | baseline entrants share (%) | Contribution/ baseline entrants |
|---|---|---|---|---|
| Ag/mining/utilities/construction | -0.18 | 13.8 | 11.9 | 1.2 |
| Manufacturing | -0.18 | 14.2 | 13.6 | 1.0 |
| Wholesale/retail/transportation | -0.14 | 10.5 | 9.9 | 1.1 |
| White collar/management | -0.17 | 13.1 | 13.0 | 1.0 |
| Education/health care/govt | -0.21 | 16.6 | 17.6 | 0.9 |
| Military | -0.02 | 1.4 | 2.4 | 0.6 |
| No Earning | -0.03 | 2.3 | 2.0 | 1.2 |
| Employed, likely small employer | -0.34 | 26.0 | 27.6 | 0.9 |
| Missing | -0.03 | 2.0 | 1.9 | 1.1 |
| Total | -1.29 | 100.0 | 100.0 | 1.0 |

Notes: Table presents the "entry" share (from Appendix Table G.40) of the difference between actual and counterfactual SSDI entry rates for the decline period (2010–2019), broken down into contribution from subgroups based on primary industry of work between the ages of 25 and 34 years old, using equation (13). "Contribution share" is "absolute contribution" as a percent of the total change. "Baseline entrants share" is the share of SSDI entrants that the group accounts for in 2010. "Contribution/baseline entrants" presents the ratio of "contribution share" to "baseline entrants share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the change. We identify a person's primary industry using EIN from the Detailed Earnings File linked to NAICS code from the Department of Labor Form 5500. We define primary industry as the industry in which the person worked the most years between 25 and 34 years old. "Employed, likely small employer" means that the applicant had an employer but the employer's EIN is not listed in Form 5500.

# H Applicant-level heterogeneity results

## H.1 SSDI increase period (1988–2010)

Appendix Table H.1: SSDI entry increase by diagnosis

| | | Absolute contribution (pp) | Contribution share (%) |
|---|---|---|---|
| Apply and award in Entry | | 1.84 | 87.5 |
| | Award | 0.38 | 17.9 |
| | Application | 1.36 | 64.8 |
| | Applicant share | -0.19 | -9.1 |
| | Interaction within award and apply | 0.29 | 13.8 |
| Eligibility | | 0.17 | 8.3 |
| Interaction with eligibility | | 0.09 | 4.2 |
| Total | | 2.10 | 100.0 |

Notes: Table presents results from decomposing the Entry row in Appendix Table G.21 using equation (18), by diagnosis. "Award," "Applicant share," "Interaction within award and apply," "Application," and "Eligibility" present the corresponding terms from equation (18).

Appendix Table H.2: Heterogeneity in SSDI entry increase by diagnosis

| Diagnosis | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/baseline |
|---|---|---|---|---|
| Musculoskeletal | 0.81 | 44.0 | 22.7 | 1.9 |
| Mental | 0.52 | 28.4 | 17.2 | 1.7 |
| Cardiovascular | 0.02 | 0.9 | 11.6 | 0.1 |
| Endocrine | 0.16 | 8.6 | 2.0 | 4.3 |
| Other | 0.41 | 22.5 | 36.1 | 0.6 |
| Missing | -0.08 | -4.5 | 10.4 | -0.4 |
| Total | 1.84 | 100.0 | 100.0 | 1.0 |

Table presents results from decomposing the difference between actual and counterfactual SSDI entry rates in 2010 using 1988 as the base year by diagnosis. "Absolute contribution" uses the first terms of equation (16) with diagnoses as the $d$ groups; "contribution share" presents the same number as a percent of the total decline. "Baseline share" presents entrant shares in 1988. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the entry decline. Diagnosis is recorded by the disability examiner assigned to the case.

## H.2 SSDI decrease period (2010–2019)

### H.2.1 Heterogeneity by applicant severity

Appendix Table H.3: Heterogeneity in SSDI application decline by severity

| Severity | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/baseline |
|---|---|---|---|---|
| Least severe | -0.72 | 47.5 | 62.8 | 0.8 |
| Moderately severe | -0.26 | 17.0 | 11.6 | 1.5 |
| Most severe | -0.48 | 31.9 | 22.1 | 1.4 |
| Unknown | -0.05 | 3.6 | 3.6 | 1.0 |
| Total | -1.51 | 100.0 | 100.0 | 1.0 |

Notes: Table presents results from decomposing the Application row in Appendix Table G.15 using equation (15), by severity. "Absolute contribution" reports results from equation (15) with severity levels as the $d$ groups; "contribution share" presents the same number as a percent of the total decline. "Baseline share" presents applicant shares in 2010. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the applicant decline. "Least severe" means that the applicant was rejected at the disability examiner level (but could have been allowed on appeal). "Moderately severe" means that the disability examiner assigned a diary reason of "medical improvement expected" or "medical improvement possible." "Most severe" means that the disability examiner assigned a diary reason of "medical improvement not expected."

Appendix Table H.4: SSDI entry decline by severity

| | | Absolute contribution (pp) | Contribution share |
|---|---|---|---|
| Apply and award in Entry | | -1.19 | 95.2 |
| | Award | -0.36 | 29.1 |
| | Application | -0.75 | 60.3 |
| | Applicant share | -0.19 | 15.0 |
| | Interaction within award and apply | 0.12 | -9.2 |
| Eligibility | | -0.09 | 7.5 |
| Interaction with eligibility | | 0.03 | -2.6 |
| Total | | -1.25 | 100.0 |

Notes: Table presents results from decomposing the Entry row in Appendix Table G.21 using equation (18), by severity. "Award," "Applicant share," "Interaction within award and apply," "Application," and "Eligibility" present the corresponding terms from equation (18).

Appendix Table H.5: Heterogeneity in SSDI entry decline by severity

| Severity | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/baseline |
|---|---|---|---|---|
| Least severe | -0.48 | 40.4 | 36.2 | 1.1 |
| Moderately severe | -0.22 | 18.7 | 20.3 | 0.9 |
| Most severe | -0.43 | 36.3 | 36.5 | 1.0 |
| Unknown | -0.05 | 4.6 | 6.9 | 0.7 |
| Total | -1.19 | 100.0 | 100.0 | 1.0 |

Notes: Table presents results from decomposing the "Apply and award in Entry" row in Appendix Table H.4 into $d$ (severity) groups from the corresponding terms in equation (18) in 2019 using 2010 as the base year. "Absolute contribution" reports results from equation (18) with severity levels as the $d$ groups; "contribution share" presents the same number as a percent of the total decline. "Baseline share" presents entrant shares in 2010. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the entry decline. "Least severe" means that the applicant was rejected at the disability examiner level (but could have been allowed on appeal). "Moderately severe" means that the disability examiner assigned a diary reason of "medical improvement expected" or "medical improvement possible." "Most severe" means that the disability examiner assigned a diary reason of "medical improvement not expected."

### H.2.2 Heterogeneity by applicant diagnosis

Appendix Table H.6: Heterogeneity in SSDI application decline by diagnosis

| Diagnosis | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/baseline |
|---|---|---|---|---|
| Musculoskeletal | -0.34 | 22.6 | 33.0 | 0.7 |
| Mental | -0.36 | 23.9 | 19.7 | 1.2 |
| Cardiovascular | -0.15 | 10.0 | 7.2 | 1.4 |
| Endocrine | -0.15 | 10.1 | 3.8 | 2.6 |
| Other | -0.45 | 29.9 | 32.6 | 0.9 |
| Missing | -0.05 | 3.6 | 3.6 | 1.0 |
| Total | -1.51 | 100.0 | 100.0 | 1.0 |

Notes: Table presents results from decomposing the difference between actual and counterfactual SSDI application rates in 2019 using 2010 as the base year by diagnosis. "Absolute contribution" uses equation (15) with diagnoses as the $d$ groups; "contribution share" presents the same number as a percent of the total change. "Baseline share" presents applicant shares in 2010. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the application decline. Diagnosis is recorded by the disability examiner assigned to the case.

Appendix Table H.7: SSDI award decline by diagnosis

| | Absolute contribution (pp) | Contribution share (%) |
|---|---|---|
| Award | -64.51 | 119.1 |
| Applicant share | 4.76 | -8.8 |
| Interaction | 5.60 | -10.3 |
| Total | -54.15 | 100.0 |

Notes: Table presents results from decomposing the Award row in Appendix Table G.18 using equation (16), by diagnosis. "Award" corresponds to the first term in equation (16), "Applicant share" refers to the second term, and "Interaction" refers to the interaction term.

Appendix Table H.8: Heterogeneity in SSDI award decline by diagnosis

| Diagnosis | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/baseline |
|---|---|---|---|---|
| Musculoskeletal | -23.46 | 36.4 | 32.4 | 1.1 |
| Mental | -19.40 | 30.1 | 17.4 | 1.7 |
| Cardiovascular | -3.21 | 5.0 | 7.5 | 0.7 |
| Endocrine | -3.72 | 5.8 | 3.2 | 1.8 |
| Other | -14.24 | 22.1 | 32.5 | 0.7 |
| Missing | -0.48 | 0.7 | 6.9 | 0.1 |
| Total | -64.51 | 100.0 | 100.0 | 1.0 |

Notes: Table presents results from decomposing the difference between actual and counterfactual SSDI award rates in 2019 using 2010 as the base year by diagnosis. "Absolute contribution" uses the first terms of equation (16) with diagnoses as the $d$ groups; "contribution share" presents the same number as a percent of the total decline. "Baseline share" presents entrant shares in 2010. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the award decline. Diagnosis is recorded by the disability examiner assigned to the case.

Appendix Table H.9: SSDI entry decline by diagnosis

| | | Absolute contribution (pp) | Contribution share |
|---|---|---|---|
| Apply and award in Entry | | -1.19 | 95.2 |
| | Award | -0.64 | 51.3 |
| | Application | -0.75 | 60.3 |
| | Applicant share | 0.05 | -3.8 |
| | Interaction within award and apply | 0.16 | -12.6 |
| Eligibility | | -0.09 | 7.5 |
| Interaction with eligibility | | 0.03 | -2.6 |
| Total | | -1.25 | 100.0 |

Notes: Table presents results from decomposing the Entry row in Appendix Table G.21 using equation (18), by diagnosis. "Award," "Applicant share," "Interaction within award and apply," "Application," and "Eligibility" present the corresponding terms from equation (18).

Appendix Table H.10: Heterogeneity in SSDI entry decline by diagnosis

| Diagnosis | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/baseline |
|---|---|---|---|---|
| Musculoskeletal | -0.33 | 28.0 | 32.4 | 0.9 |
| Mental | -0.31 | 26.4 | 17.4 | 1.5 |
| Cardiovascular | -0.10 | 8.8 | 7.5 | 1.2 |
| Endocrine | -0.09 | 7.2 | 3.2 | 2.3 |
| Other | -0.30 | 25.0 | 32.5 | 0.8 |
| Missing | -0.05 | 4.6 | 6.9 | 0.7 |
| Total | -1.19 | 100.0 | 100.0 | 1.0 |

Notes: Table presents results from decomposing the difference between actual and counterfactual SSDI entry rates in 2019 using 2010 as the base year by diagnosis. "Absolute contribution" uses the first terms of equation (16) with diagnoses as the $d$ groups; "contribution share" presents the same number as a percent of the total decline. "Baseline share" presents entrant shares in 2010. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the entry decline. Diagnosis is recorded by the disability examiner assigned to the case.

### H.2.3 Heterogeneity by applicant decision "step"

Appendix Table H.11: Heterogeneity in SSDI application decline by decision "step"

| Decision step | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/baseline |
|---|---|---|---|---|
| Step 1-2, deny (not severe) | -0.12 | 7.8 | 15.0 | 0.5 |
| Step 3, allow (listings) | -0.28 | 18.6 | 14.8 | 1.2 |
| Step 4, deny (past work) | -0.82 | 54.6 | 18.6 | 2.9 |
| Step 5, allow (any work) | -0.46 | 30.3 | 18.7 | 1.6 |
| Step 5, deny (any work) | 0.26 | -16.9 | 21.7 | -0.8 |
| Other | -0.03 | 2.1 | 7.6 | 0.3 |
| Missing | -0.05 | 3.6 | 3.6 | 1.0 |
| Total | -1.51 | 100.0 | 100.0 | 1.0 |

Notes: Table presents results from decomposing the difference between actual and counterfactual SSDI application rates in 2019 using 2010 as the base year by SSA decision "step." "Absolute contribution" uses equation (15) with decision steps as the $d$ groups; "contribution share" presents the same number as a percent of the total change. "Baseline share" presents the applicant shares in 2010. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the application decline. "Step 1-2 deny" means that the applicant was rejected Steps 1 or 2 of the SSA decision process (working or not severe). "Step 3 allow" means that applicant was allowed at Step 3 (meets medical listings). "Step 4 deny" means that applicant was denied at Step 4 (capacity for past work). "Step 5 allow/deny" means that the applicant was allowed/denied at Step 5 (any work).

### H.2.4 Heterogeneity by state

Appendix Table H.12: Heterogeneity in SSDI application decline by state

| State | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/baseline |
|---|---|---|---|---|
| Alabama | -0.04 | 2.4 | 2.3 | 1.0 |
| Florida | -0.09 | 6.0 | 6.2 | 1.0 |
| Georgia | -0.04 | 2.6 | 3.3 | 0.8 |
| Hawaii | -0.00 | 0.2 | 0.3 | 0.7 |
| Idaho | -0.00 | 0.3 | 0.4 | 0.6 |
| Illinois | -0.07 | 4.9 | 3.5 | 1.4 |
| Indiana | -0.04 | 2.4 | 2.3 | 1.0 |
| Iowa | -0.01 | 0.7 | 0.8 | 0.8 |
| Kansas | -0.01 | 1.0 | 0.8 | 1.1 |
| Kentucky | -0.04 | 2.5 | 2.1 | 1.2 |
| Louisiana | -0.03 | 1.8 | 1.8 | 1.0 |
| Alaska | -0.00 | 0.1 | 0.2 | 0.8 |
| Maine | -0.01 | 0.7 | 0.5 | 1.3 |
| Maryland | -0.02 | 1.2 | 1.6 | 0.7 |
| Massachusetts | -0.03 | 1.8 | 1.8 | 1.0 |
| Michigan | -0.07 | 4.6 | 3.7 | 1.3 |
| Minnesota | -0.02 | 1.2 | 1.3 | 0.9 |
| Mississippi | -0.02 | 1.6 | 1.5 | 1.1 |
| Missouri | -0.04 | 2.9 | 2.5 | 1.2 |
| Montana | -0.01 | 0.5 | 0.3 | 1.4 |
| Nebraska | -0.00 | 0.3 | 0.4 | 0.7 |
| Nevada | -0.01 | 0.7 | 0.7 | 0.9 |
| Arizona | -0.02 | 1.3 | 1.7 | 0.8 |
| New Hampshire | -0.01 | 0.6 | 0.4 | 1.2 |
| New Jersey | -0.02 | 1.2 | 2.0 | 0.6 |
| New Mexico | -0.01 | 0.6 | 0.7 | 0.9 |
| New York | -0.05 | 3.1 | 4.9 | 0.6 |
| North Carolina | -0.06 | 3.7 | 3.5 | 1.1 |
| North Dakota | -0.00 | 0.0 | 0.1 | 0.1 |
| Ohio | -0.08 | 5.0 | 4.0 | 1.3 |
| Oklahoma | -0.02 | 1.0 | 1.5 | 0.7 |
| Oregon | -0.02 | 1.2 | 1.2 | 1.0 |
| Pennsylvania | -0.05 | 3.6 | 4.2 | 0.8 |
| Arkansas | -0.02 | 1.5 | 1.4 | 1.0 |
| Puerto Rico | -0.05 | 3.6 | 1.4 | 2.6 |
| Rhode Island | -0.01 | 0.3 | 0.4 | 0.9 |
| South Carolina | -0.03 | 1.8 | 1.8 | 1.0 |
| South Dakota | -0.00 | 0.2 | 0.2 | 0.9 |
| Tennessee | -0.06 | 3.8 | 2.8 | 1.3 |
| Texas | -0.08 | 5.3 | 7.0 | 0.8 |
| Utah | -0.01 | 0.4 | 0.5 | 0.8 |
| Vermont | -0.00 | 0.2 | 0.2 | 1.2 |
| Virgin Islands | -0.02 | 1.3 | 2.1 | 0.6 |
| California | -0.14 | 9.4 | 8.6 | 1.1 |
| Washington | -0.02 | 1.3 | 1.9 | 0.7 |
| West Virginia | -0.01 | 0.5 | 0.9 | 0.6 |
| Wisconsin | -0.04 | 2.3 | 1.8 | 1.3 |
| Wyoming | -0.00 | 0.1 | 0.2 | 0.5 |
| Colorado | -0.02 | 1.5 | 1.2 | 1.2 |
| Connecticut | -0.01 | 0.7 | 0.9 | 0.8 |
| Delaware | -0.01 | 0.4 | 0.3 | 1.3 |
| District of Columbia | -0.00 | 0.1 | 0.2 | 0.4 |
| Unknown | -0.00 | 0.0 | 0.1 | 0.4 |
| Missing | -0.05 | 3.6 | 3.6 | 1.0 |
| Total | -1.51 | 100.0 | 100.0 | 1.0 |

Notes: Table presents results from decomposing the difference between actual and counterfactual SSDI application rates in 2019 using 2010 as the base year by state. "Absolute contribution" uses equation (15) with states as the $d$ groups; "contribution share" presents the same number as a percent of the total change. "Baseline share" presents applicant shares in 2010. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the application decline. State is observed in the F831 or MEDIB file.

### H.2.5 Heterogeneity by Medicaid expansion

Appendix Table H.13: Heterogeneity in SSDI application decline by Medicaid expansion

| Medicaid expansion | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/baseline |
|---|---|---|---|---|
| After 2019 or never | -0.62 | 41.2 | 40.8 | 1.0 |
| Expansion in 2015-2016 | -0.13 | 8.4 | 8.8 | 1.0 |
| Expansion in 2014 | -0.70 | 46.7 | 46.7 | 1.0 |
| Unknown | -0.00 | 0.0 | 0.1 | 0.4 |
| Missing | -0.05 | 3.6 | 3.6 | 1.0 |
| Total | -1.51 | 100.0 | 100.0 | 1.0 |

Notes: Table presents results from decomposing the difference between actual and counterfactual SSDI application rates in 2019 using 2010 as the base year, by state Medicaid expansion date. "Absolute contribution" uses equation (15) with states as the $d$ groups; "contribution share" presents the same number as a percent of the total change. "Baseline share" presents applicant shares in 2010. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the application decline. State is observed in the F831 or MEDIB file.

### H.2.6 Heterogeneity by applicant urban/rural status

Appendix Table H.14: Heterogeneity in SSDI application decline by population density

| Urban status | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/baseline |
|---|---|---|---|---|
| Big City | -0.99 | 65.8 | 66.1 | 1.0 |
| Big City CZ | -0.13 | 8.9 | 9.6 | 0.9 |
| Small City | -0.19 | 12.3 | 11.2 | 1.1 |
| Rural Area | -0.15 | 9.9 | 9.1 | 1.1 |
| Missing | -0.05 | 3.1 | 4.0 | 0.8 |
| Total | -1.51 | 100.0 | 100.0 | 1.0 |

Notes: Table presents results from decomposing the difference between actual and counterfactual SSDI application rates in 2019 using 2010 as the base year by population density. "Absolute contribution" uses equation (15) with urban status as the $d$ groups; "contribution share" presents the same number as a percent of the total change. "Baseline share" presents applicant shares in 2010. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the application decline. ZIP code is observed in the F831 or MEDIB file, and linked to the USDA Rural-Urban Commuting Area (RUCA) Codes.

### H.2.7 Heterogeneity by ALJ award change

Appendix Table H.15: Heterogeneity in SSDI application decline by ALJ award change

| ALJ change | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/baseline |
|---|---|---|---|---|
| Less affected | -1.15 | 76.3 | 79.9 | 1.0 |
| More affected | -0.20 | 13.4 | 12.7 | 1.1 |
| Other | -0.10 | 6.8 | 3.8 | 1.8 |
| Missing | -0.05 | 3.6 | 3.6 | 1.0 |
| Total | -1.51 | 100.0 | 100.0 | 1.0 |

Notes: Table presents results from decomposing the difference between actual and counterfactual SSDI application rates in 2019 using 2010 as the base year by 2008 award rates at the appeals level. "Absolute contribution" uses equation (15) with award rate group as the $d$ groups; "contribution share" presents the same number as a percent of the total change. "Baseline share" presents applicant shares in 2010. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the application decline. ZIP code is observed in the F831 or MEDIB file. We link ZIP code to appeals award rates calculated using SSA administrative data. "Less affected" means a 2008 appeals award rate of less than 0.4 (and therefore experiences a smaller decline in award rates as a result of the ALJ reform). "More affected" means a 2008 median appeals award rate above 0.4 (and therefore experiences a larger decline in award rates as a result of the ALJ reform).

### H.2.8 Heterogeneity by unemployment rate change

Appendix Table H.16: Heterogeneity in SSDI application decline by unemployment rate change

| Unemp rate change | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/baseline |
|---|---|---|---|---|
| Most improving labor market | -0.67 | 44.1 | 41.0 | 1.1 |
| Moderately improving | -0.50 | 33.4 | 34.4 | 1.0 |
| Least improving labor market | -0.18 | 12.0 | 16.8 | 0.7 |
| Unknown | -0.16 | 10.5 | 7.7 | 1.4 |
| Total | -1.51 | 100.0 | 100.0 | 1.0 |

Notes: Table presents results from decomposing the difference between actual and counterfactual SSDI application rates in 2019 using 2010 as the base year by the change in the unemployment rate between 2010 and 2019. "Absolute contribution" uses equation (15) with unemployment rate change group as the $d$ groups; "contribution share" presents the same number as a percent of the total change. "Baseline share" presents applicant shares in 2010. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the application decline. ZIP code is observed in the F831 or MEDIB file. We link ZIP code to Bureau of Labor Statistics Local Area Unemployment Statistics. "Most improving labor market" means that the change in the unemployment rate from 2010 to 2019 is in the top tercile. "Moderately improving" means in the middle tercile. "Least improving" means the bottom tercile.

Appendix Table H.17: SSDI entry decline by unemployment rate change

| | | Absolute contribution (pp) | Contribution share |
|---|---|---|---|
| Apply and award in Entry | | -1.19 | 95.2 |
| | Award | -0.55 | 44.2 |
| | Application | -0.75 | 60.3 |
| | Applicant share | -0.00 | 0.1 |
| | Interaction within award and apply | 0.12 | -9.5 |
| Eligibility | | -0.09 | 7.5 |
| Interaction with eligibility | | 0.03 | -2.6 |
| Total | | -1.25 | 100.0 |

Notes: Table presents results from decomposing the Entry row in Appendix Table G.21 using equation (18), by unemployment rate change. "Award," "Applicant share," "Interaction within award and apply," "Application," and "Eligibility" present the corresponding terms from equation (18).

Appendix Table H.18: Heterogeneity in SSDI entry decline by unemployment rate change

| Unemp rate change | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/baseline |
|---|---|---|---|---|
| Most improving labor market | -0.50 | 41.7 | 38.7 | 1.1 |
| Moderately improving | -0.39 | 33.0 | 33.2 | 1.0 |
| Least improving labor market | -0.16 | 13.8 | 16.5 | 0.8 |
| Unknown | -0.14 | 11.6 | 11.5 | 1.0 |
| Total | -1.19 | 100.0 | 100.0 | 1.0 |

Notes: Table presents results from decomposing the difference between actual and counterfactual SSDI entry rates in 2019 using 2010 as the base year by the change in the unemployment rate between 2010 and 2019. "Absolute contribution" uses the first terms of equation (16) with unemployment rate change group as the $d$ groups; "contribution share" presents the same number as a percent of the total decline. "Baseline share" presents entrant shares in 2010. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the entry decline. ZIP code is observed in the F831 or MEDIB file. We link ZIP code to Bureau of Labor Statistics Local Area Unemployment Statistics. "Most improving labor market" means that the change in the unemployment rate from 2010 to 2019 is in the top tercile. "Moderately improving" means in the middle tercile. "Least improving" means the bottom tercile.

### H.2.9 Heterogeneity by local minimum wage

Appendix Table H.19: Heterogeneity in SSDI application decline by minimum wage change

| Minimum wage change | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/baseline |
|---|---|---|---|---|
| No increase | -0.50 | 33.1 | 34.8 | 1.0 |
| Small increase | -0.28 | 18.7 | 16.8 | 1.1 |
| Some increase | -0.29 | 19.4 | 21.7 | 0.9 |
| Large increase | -0.07 | 4.6 | 4.0 | 1.2 |
| Unknown | -0.31 | 20.6 | 19.0 | 1.1 |
| Missing | -0.05 | 3.6 | 3.6 | 1.0 |
| Total | -1.51 | 100.0 | 100.0 | 1.0 |

Notes: Table presents results from decomposing the difference between actual and counterfactual SSDI application rates in 2019 using 2010 as the base year by the change in the minimum wage between 2010 and 2016. "Absolute contribution" uses equation (15) with minimum wage change group as the $d$ groups; "contribution share" presents the same number as a percent of the total change. "Baseline share" presents applicant shares in 2010. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the application decline. ZIP code is observed in the F831 or MEDIB file. We link ZIP code to minimum wage changes using data from Vaghul and Zipperer (2022). "No increase" means no change in minimum wage from 2010 to 2016. "Low increase" means an increase of $1 or less between 2010 and 2016. "Some increase" means an increase of between $1 and $2 between 2010 and 2016. "Large increase" means an increase above $2 between 2010 and 2016.

### H.2.10 Heterogeneity by local economic mobility

Appendix Table H.20: Heterogeneity in SSDI application decline by economic mobility

| Economic mobility | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/baseline |
|---|---|---|---|---|
| High mobility | -0.76 | 50.3 | 55.9 | 0.9 |
| Low mobility | -0.59 | 39.1 | 36.6 | 1.1 |
| Other | -0.11 | 7.0 | 3.9 | 1.8 |
| Missing | -0.05 | 3.6 | 3.6 | 1.0 |
| Total | -1.51 | 100.0 | 100.0 | 1.0 |

Notes: Table presents results from decomposing the difference between actual and counterfactual SSDI application rates in 2019 using 2010 as the base year by the Chetty et al. (2014a) county economic mobility classification. "Absolute contribution" uses equation (15) with mobility classification as the $d$ groups; "contribution share" presents the same number as a percent of the total change. "Baseline share" presents applicant shares in 2010. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the application decline. ZIP code is observed in the F831 or MEDIB file. We link ZIP code to counties and then to the Chetty et al. (2014a) county economic mobility classification. "High mobility" means the county is in the top three quartiles of the Chetty et al. (2014a) classification. "Low mobility" means the county is the bottom quartile of the Chetty et al. (2014a) classification.

### H.2.11 Heterogeneity by local opioid crisis severity

Appendix Table H.21: Heterogeneity in SSDI application decline by opioid crisis severity

| Opioid severity | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/baseline |
|---|---|---|---|---|
| Less affected | -0.98 | 65.1 | 68.5 | 1.0 |
| More affected | -0.14 | 9.4 | 8.6 | 1.1 |
| Other | -0.33 | 21.9 | 19.3 | 1.1 |
| Missing | -0.05 | 3.6 | 3.6 | 1.0 |
| Total | -1.51 | 100.0 | 100.0 | 1.0 |

Notes: Table presents results from decomposing the difference between actual and counterfactual SSDI application rates in 2019 using 2010 as the base year by the severity of the opioid crisis. "Absolute contribution" uses equation (15) with opioid crisis severity as the $d$ groups; "contribution share" presents the same number as a percent of the total change. "Baseline share" presents applicant shares in 2010. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the application decline. ZIP code is observed in the F831 or MEDIB file. We link ZIP code to counties and then to the county opioid death rate between 1999 and 2010 from the Centers for Disease Control and Prevention, National Center for Health Statistics (2021). "Less affected" means that the county is in the bottom three quartiles of opioid death rate from 1999 to 2010. "More affected" means that the county is in the top quartile of the opioid death rate from 1999 to 2010.

Appendix Table H.22: SSDI entry decline by opioid crisis severity

| | | Absolute contribution (pp) | Contribution share |
|---|---|---|---|
| Apply and award in Entry | | -1.19 | 95.2 |
| | Award | -0.55 | 44.0 |
| | Application | -0.75 | 60.3 |
| | Applicant share | -0.00 | 0.1 |
| | Interaction within award and apply | 0.12 | -9.3 |
| Eligibility | | -0.09 | 7.5 |
| Interaction with eligibility | | 0.03 | -2.6 |
| Total | | -1.25 | 100.0 |

Notes: Table presents results from decomposing the Entry row in Appendix Table G.21 using equation (18), by opioid crisis severity. "Award," "Applicant share," "Interaction within award and apply," "Application," and "Eligibility" present the corresponding terms from equation (18).

Appendix Table H.23: Heterogeneity in SSDI entry decline by opioid crisis severity

| Opioid severity | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/baseline |
|---|---|---|---|---|
| Less affected | -0.77 | 64.6 | 65.6 | 1.0 |
| More affected | -0.11 | 9.0 | 8.1 | 1.1 |
| Other | -0.26 | 21.9 | 19.4 | 1.1 |
| Missing | -0.05 | 4.6 | 6.9 | 0.7 |
| Total | -1.19 | 100.0 | 100.0 | 1.0 |

Notes: Table presents results from decomposing the difference between actual and counterfactual SSDI entry rates in 2019 using 2010 as the base year by the severity of the opioid crisis. "Absolute contribution" uses the first terms of equation (16) with opioid crisis severity as the $d$ groups; "contribution share" presents the same number as a percent of the total decline. "Baseline share" presents entrant shares in 2010. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the entry decline. ZIP code is observed in the F831 or MEDIB file. We link ZIP code to counties and then to the county opioid death rate between 1999 and 2010 from the Centers for Disease Control and Prevention, National Center for Health Statistics (2021). "Less affected" means that the county is in the bottom three quartiles of opioid death rate from 1999 to 2010. "More affected" means that the county is in the top quartile of the opioid death rate from 1999 to 2010.

### H.2.12 Heterogeneity by local smoking rates

Appendix Table H.24: Heterogeneity in SSDI application decline by smoking rates

| Smoke | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/baseline |
|---|---|---|---|---|
| Lowest quartile | -0.37 | 24.6 | 26.3 | 0.9 |
| Second quartile | -0.36 | 24.1 | 26.6 | 0.9 |
| Third quartile | -0.34 | 22.4 | 21.6 | 1.0 |
| Highest quartile | -0.22 | 14.8 | 14.9 | 1.0 |
| Unknown | -0.16 | 10.5 | 7.0 | 1.5 |
| Missing | -0.05 | 3.6 | 3.6 | 1.0 |
| Total | -1.51 | 100.0 | 100.0 | 1.0 |

Notes: Table presents results from decomposing the difference between actual and counterfactual SSDI application rates in 2019 using 2010 as the base year by county smoking rate from 2010–16. "Absolute contribution" uses equation (15) with county smoking rate as the $d$ groups; "contribution share" presents the same number as a percent of the total change. "Baseline share" presents applicant shares in 2010. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the application decline. ZIP code is observed in the F831 or MEDIB file. We link ZIP code to counties and then to the county smoking rate between 2010 and 2016 from the County Health Rankings & Roadmaps (2025). We group counties by quartile of smoking rate.

### H.2.13 Heterogeneity by applicant pre-application industry

Appendix Table H.25: Heterogeneity in SSDI application decline by pre-application industry

| Recent industry | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/baseline |
|---|---|---|---|---|
| Military | 0.01 | -0.7 | 0.8 | -0.9 |
| Ag/mining/utilities/construction | -0.18 | 12.2 | 7.3 | 1.7 |
| Manufacturing | -0.20 | 13.3 | 7.8 | 1.7 |
| Wholesale/retail/transportation | -0.17 | 11.2 | 10.2 | 1.1 |
| White collar/management | -0.15 | 9.8 | 11.1 | 0.9 |
| Education/health care/govt | -0.25 | 16.3 | 18.2 | 0.9 |
| No Earning | 0.03 | -1.7 | 4.0 | -0.4 |
| Employed, likely small employer | -0.54 | 36.0 | 37.0 | 1.0 |
| Missing | -0.05 | 3.6 | 3.6 | 1.0 |
| Total | -1.51 | 100.0 | 100.0 | 1.0 |

Notes: Table presents results from decomposing the difference between actual and counterfactual SSDI application rates in 2019 using 2010 as the base year by industry in which the applicant worked in the three years before application. "Absolute contribution" uses equation (15) with pre-application industry as the $d$ groups; "contribution share" presents the same number as a percent of the total change. "Baseline share" presents applicant shares in 2010. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the application decline. The EIN of the applicant's employer is observed in the Detailed Earnings File. We link EIN to the Department of Labor Form 5500 to determine NAICS classification. "Employed, likely small employer" means that the applicant had an employer but the employer's EIN is not listed in Form 5500.

### H.2.14 Heterogeneity by applicant pre-application industry remote work

Appendix Table H.26: Heterogeneity in SSDI application decline by pre-application remote work

| Work-from-home rates | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/baseline |
|---|---|---|---|---|
| Low WFH | -0.06 | 3.8 | 3.7 | 1.0 |
| Moderate WFH | -0.34 | 22.6 | 15.5 | 1.5 |
| Substantial WFH | -0.21 | 14.2 | 13.3 | 1.1 |
| High WFH | -0.32 | 21.4 | 21.7 | 1.0 |
| Employed, unknown WFH | -0.54 | 36.0 | 38.2 | 0.9 |
| Unemployed | 0.03 | -1.7 | 4.0 | -0.4 |
| Missing | -0.05 | 3.6 | 3.6 | 1.0 |
| Total | -1.51 | 100.0 | 100.0 | 1.0 |

Notes: Table presents results from decomposing the difference between actual and counterfactual SSDI application rates in 2019 using 2010 as the base year by what share of jobs can be done remotely as classified by Dingel and Neiman (2020). "Absolute contribution" uses equation (15) with pre-application remote work share as the $d$ groups; "contribution share" presents the same number as a percent of the total change. "Baseline share" presents applicant shares in 2010. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the application decline. The EIN of the applicant's employer is observed in the Detailed Earnings File. We link EIN to the Department of Labor Form 5500 to determine NAICS classification, then to the Dingel and Neiman (2020) classification of share of jobs that can be done remotely. "Low WFH" means that fewer than 10% of jobs can be done remotely. "Moderate WFH" means that between 10% and 25% of jobs in the industry can be done remotely. "Substantial WFH" means that between 25% and 50% of jobs in the industry can be done remotely. "High WFH" means that more than 50% of jobs in the industry can be done remotely.

### H.2.15 Heterogeneity by applicant pre-application employment

Appendix Table H.27: Heterogeneity in SSDI application decline by pre-application years of work

| Pre-app employment | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/baseline |
|---|---|---|---|---|
| No earning | 0.03 | -1.9 | 4.1 | -0.5 |
| Earnings in some years | -0.12 | 8.0 | 17.6 | 0.5 |
| Earnings in all years | -1.36 | 90.3 | 74.7 | 1.2 |
| Missing | -0.05 | 3.6 | 3.6 | 1.0 |
| Total | -1.51 | 100.0 | 100.0 | 1.0 |

Notes: Table presents results from decomposing the difference between actual and counterfactual SSDI application rates in 2019 using 2010 as the base year by how many years of the 5 years before application the applicant worked. "Absolute contribution" uses equation (15) with pre-application work years as the $d$ groups; "contribution share" presents the same number as a percent of the total change. "Baseline share" presents applicant shares in 2010. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the application decline. "No earnings" means the applicant did not work in any of the 5 years before application. "Earnings in some years" means the applicant worked for 1-3 years of the 5 years before application. "Earnings in all years" means the applicant worked for 4-5 years of the 5 years before application.

### H.2.16 Heterogeneity by applicant pre-application employment (FTFY)

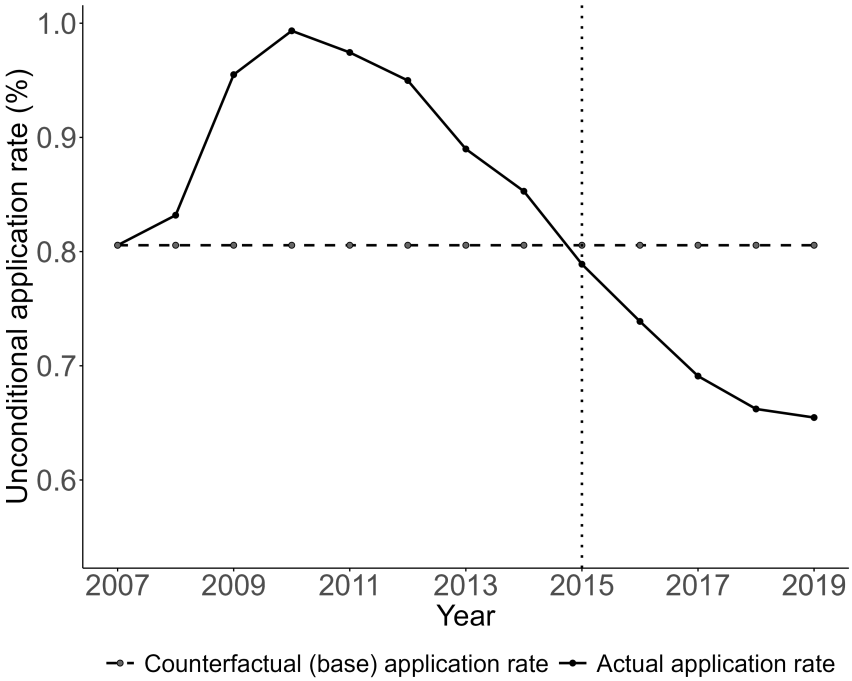
Appendix Table H.28: Heterogeneity in SSDI application decline by pre-application employment

| Pre-app FTFY employment | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/baseline |
|---|---|---|---|---|
| No FTFY earning | -0.32 | 21.4 | 20.6 | 1.0 |
| FTFY earnings in some years | -0.51 | 33.6 | 32.8 | 1.0 |
| FTFY earnings in all years | -0.62 | 41.4 | 43.0 | 1.0 |
| Missing | -0.05 | 3.6 | 3.6 | 1.0 |
| Total | -1.51 | 100.0 | 100.0 | 1.0 |

Notes: Table presents results from decomposing the difference between actual and counterfactual SSDI application rates in 2019 using 2010 as the base year by how many years of the 5 years before application the applicant earned more than $15,000. "Absolute contribution" uses equation (15) with pre-application work years as the $d$ groups; "contribution share" presents the same number as a percent of the total change. "Baseline share" presents applicant shares in 2010. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the application decline. "No earnings" means the applicant did not work in any of the 5 years before application. "Earnings in some years" means the applicant earned more than $15,000 for 1-3 years of the 5 years before application. "Earnings in all years" means the applicant earned more than $15,000 for 4-5 years of the 5 years before application.

## H.3 SSDI in Great Recession (2007–2014) and post-Great Recession period (2015–2019)

Appendix Figure H.1: Counterfactual vs actual SSDI application rate, 2007–2014 and 2015–2019




Notes: Figure plots actual and counterfactual SSDI application rate in each year from 2007 to 2019. Actual application rate uses the Continuous Work History Sample linked to the Payment History Update System and the Master Beneficiary Record. Counterfactual application rate uses equations (2), (3), and (4), using 2007 as the base year. The exact annual values are given in Appendix Table H.29.

Appendix Table H.29: Counterfactual vs actual SSDI application rate, 2007–2014 and 2015–2019

| Period | Year | Actual application rate (%) | Counterfactual (base) application rate (%) | Total difference (pp) |
|---|---|---|---|---|
| Great Recession | | | | |
| | 2007 | 0.81 | 0.81 | 0.00 |
| | 2008 | 0.83 | 0.81 | 0.03 |
| | 2009 | 0.95 | 0.81 | 0.15 |
| | 2010 | 0.99 | 0.81 | 0.19 |
| | 2011 | 0.97 | 0.81 | 0.17 |
| | 2012 | 0.95 | 0.81 | 0.14 |
| | 2013 | 0.89 | 0.81 | 0.08 |
| | 2014 | 0.85 | 0.81 | 0.05 |
| | Total | | | 0.81 |
| Post Great Recession | | | | |
| | 2015 | 0.79 | 0.81 | -0.02 |
| | 2016 | 0.74 | 0.81 | -0.07 |
| | 2017 | 0.69 | 0.81 | -0.11 |
| | 2018 | 0.66 | 0.81 | -0.14 |
| | 2019 | 0.65 | 0.81 | -0.15 |
| | Total | | | -0.49 |

Table presents actual and counterfactual SSDI application rate in each year from 2007 to 2019. Actual SSDI application rate uses the Continuous Work History Sample linked to the Payment History Update System and the Master Beneficiary Record. Counterfactual SSDI application rate uses equations (2), (3), and (4), using 2010 as the base year.

Appendix Table H.30: Decomposition of decline in SSDI application rate, 2007–2014 and 2015–2019

| Period | Variable | Absolute contribution (pp) | As a share of total (%) |
|---|---|---|---|
| Great Recession | | | |
| | Application | 0.89 | 66.1 |
| | Eligibility | -0.09 | -9.0 |
| | Interaction within apply | -0.02 | -1.8 |
| | Across group composition | 0.04 | 3.8 |
| | Within-across interaction | -0.01 | -0.7 |
| | Total | 0.81 | 58.3 |
| Post Great Recession | | | |
| | Application | -0.34 | 17.3 |
| | Eligibility | -0.15 | 24.3 |
| | Interaction within apply | 0.01 | -0.3 |
| | Across group composition | -0.02 | 0.6 |
| | Within-across interaction | 0.01 | -0.2 |
| | Total | -0.49 | 41.7 |

Notes: Table presents decomposition of SSDI application rate decline from 2007 to 2014 (Great Recession) and 2015 to 2019 (post-Great Recession), using equation (12). The point estimates we report are the ones in which flows (e.g., $B_t$, $A_t$, $E_t$, $X_t$), when fixed, are held fixed at their counterfactual values.

Appendix Table H.31: Heterogeneity in SSDI application change by severity during Great Recession (2007–2014) and post-GR (2015–2019)

| Period | Diagnosis | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/baseline |
|---|---|---|---|---|---|
| Great Recession | | | | | |
| | Musculoskeletal | 0.43 | 48.2 | 32.0 | 1.5 |
| | Mental | 0.23 | 25.7 | 18.7 | 1.4 |
| | Cardiovascular | 0.03 | 3.1 | 7.6 | 0.4 |
| | Endocrine | 0.00 | 0.5 | 3.7 | 0.1 |
| | Other | 0.23 | 25.7 | 33.6 | 0.8 |
| | Missing | -0.03 | -3.2 | 4.5 | -0.7 |
| | Total | 0.89 | 100.0 | 100.0 | 1.0 |
| Post Great Recession | | | | | |
| | Musculoskeletal | 0.02 | -5.7 | 32.0 | -0.2 |
| | Mental | -0.07 | 19.9 | 18.7 | 1.1 |
| | Cardiovascular | -0.08 | 22.0 | 7.6 | 2.9 |
| | Endocrine | -0.07 | 20.3 | 3.7 | 5.4 |
| | Other | -0.11 | 31.3 | 33.6 | 0.9 |
| | Missing | -0.04 | 12.2 | 4.5 | 2.7 |
| | Total | -0.34 | 100.0 | 100.0 | 1.0 |

Notes: Table presents results from decomposing the Application row in Appendix Table G.15 using equation (15), by diagnosis. "Absolute contribution" reports results from equation (15) with diagnosis as the $d$ groups; "contribution share" presents the same number as a percent of the total decline. "Baseline share" presents entrant shares in 2007. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the entry decline.

Appendix Table H.32: Heterogeneity in SSDI application change by severity during Great Recession (2007–2014) and post-GR (2015–2019)

| Period | Severity | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/baseline |
|---|---|---|---|---|---|
| Great Recession | | | | | |
| | Least severe | 0.71 | 80.4 | 62.4 | 1.3 |
| | Moderately severe | 0.05 | 5.7 | 12.6 | 0.4 |
| | Most severe | 0.15 | 17.1 | 20.5 | 0.8 |
| | Unknown | -0.03 | -3.2 | 4.5 | -0.7 |
| | Total | 0.89 | 100.0 | 100.0 | 1.0 |
| Post Great Recession | | | | | |
| | Least severe | -0.04 | 12.5 | 62.4 | 0.2 |
| | Moderately severe | -0.12 | 34.0 | 12.6 | 2.7 |
| | Most severe | -0.14 | 41.3 | 20.5 | 2.0 |
| | Unknown | -0.04 | 12.2 | 4.5 | 2.7 |
| | Total | -0.34 | 100.0 | 100.0 | 1.0 |

Notes: Table presents results from decomposing the Application row in Appendix Table G.15 using equation (15), by severity. "Absolute contribution" reports results from equation (15) with severity levels as the $d$ groups; "contribution share" presents the same number as a percent of the total decline. "Baseline share" presents entrant shares in 2007. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the entry decline. "Least severe" means that the applicant was rejected at the disability examiner level (but could have been allowed on appeal). "Moderately severe" means that the disability examiner assigned a diary reason of "medical improvement expected" or "medical improvement possible." "Most severe" means that the disability examiner assigned a diary reason of "medical improvement not expected."

Appendix Table H.33: Heterogeneity in SSDI application change by local unemployment rate change during Great Recession (2007–2014) and post-GR (2015–2019)

| Period | Unemp rate change (07-14) | Absolute contribution (pp) | Contribution share (%) | Baseline share (%) | Contribution/ baseline |
|---|---|---|---|---|---|
| Great Recession | | | | | |
| | Most improving labor market | 0.13 | 14.4 | 18.3 | 0.8 |
| | Moderately improving | 0.34 | 38.0 | 35.7 | 1.1 |
| | Least improving labor market | 0.42 | 47.0 | 37.4 | 1.3 |
| | Unknown | 0.01 | 0.6 | 8.5 | 0.1 |
| | Total | 0.89 | 100.0 | 100.0 | 1.0 |
| Post Great Recession | | | | | |
| | Most improving labor market | -0.08 | 24.0 | 18.3 | 1.3 |
| | Moderately improving | -0.11 | 33.4 | 35.7 | 0.9 |
| | Least improving labor market | -0.07 | 19.0 | 37.4 | 0.5 |
| | Unknown | -0.08 | 23.5 | 8.5 | 2.8 |
| | Total | -0.34 | 100.0 | 100.0 | 1.0 |

Notes: Table presents results from decomposing the Application row in Appendix Table G.15 using equation (15), by local unemployment rate change. "Absolute contribution" reports results from equation (15) with unemployment rate categories as the $d$ groups; "contribution share" presents the same number as a percent of the total decline. "Baseline share" presents entrant shares in 2007. "Contribution/baseline" presents the ratio of "contribution share" to "baseline share," indicating whether that subgroup is under- (<1) or over- (>1) represented in the entry decline. ZIP code is observed in the F831 or MEDIB file. We link ZIP code to Bureau of Labor Statistics Local Area Unemployment Statistics. "Most improving labor market" means that the change in the unemployment rate from 2010 to 2019 is in the top tercile. "Moderately improving" means in the middle tercile. "Least improving" means the bottom tercile.